III Unsolved problems

1. A two bus system is shown in fig E2.1. Determine the voltage at the end of third iteration by G-S method. The elements of bus impedance matrix are $Y_{11} = Y_{22} = 4.5 \angle -86°$ p.u and $Y_{21} = Y_{12} = 1.8 \angle 110°$ p.u.

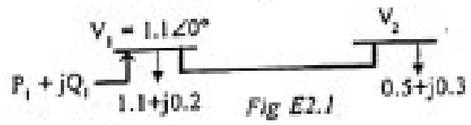
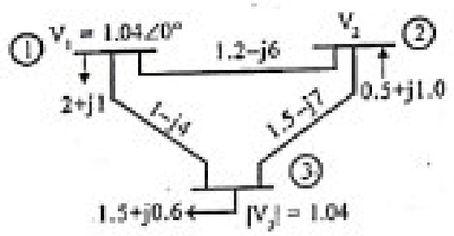


Fig E2.1

2. A three bus system is shown in fig E2.2. Determine the bus-2 voltage at the end of second iteration by G-S method.



Note : All elements are in p.u. admittances.

Fig E2.2

3. The load flow data for a three bus system are shown in table E3.1 and table E3.2. The reactive power limit of bus-2 is $0.01 \le Q_2 \le 0.25$. Calculate the bus voltages at the end of second iteration by G-S method.

Table E3.1

Bus code	Impedance
1-2	0.07 + j0.2
1-3	0.01 + j0.05
2-3	0.02 + j0.15

Table E3.2

Buscode	P	Q	V	Remark
1	·	·	1.06∠0°	Slack
2	0.2	·	1.05	PV
3	0.6	0.25	·	PQ

4. The load flow data for a 4-bus systems are given in table E4.1 and E4.2. Determine the voltages at the end of first G-S iteration.

Table E4.1

Bus code	Impedance
1-2	0.01 + j0.09
1-3	0.05 + j0.20
2-3	0.02 + j0.14
2-4	0.02 + j0.14
3-4	0.01 + j0.06

Table E4.2

Buscode	P	Q	V	Remark
1	·	·	1.06∠0°	Slack
2	0.25	0.1	·	PQ
3	0.5	0.3	·	PQ
4	0.4	0.05	·	PQ

SYMMETRICAL FAULT ANALYSIS



3.1 TYPES OF FAULT

A fault in a circuit is any failure which interferes with the normal flow of current. The faults occur in power system due to insulation failure of equipments, flash over of lines initiated by a lightning stroke, due to permanent damage to conductors and towers or due to accidental faulty operations. The faults can be broadly classified into shunt faults (short circuits) and series faults (open conductors).

The shunt type of faults involves short circuit between conductor and ground or short circuit between two or more conductors. The shunt faults are characterized by increase in current and fall in voltage and frequency. The shunt faults can be classified as shown below.

1. Line-to-ground fault
2. Line-to-line fault
3. Double line-to-ground fault
4. Three phase fault

The series fault may occur with one or two broken conductors which creates open circuits. It also happens in circuits controlled by fuses or breakers which do not open all the three phases, i.e., one or two phases of the circuit may open and the other phases may be closed. The series faults are characterized by increase in voltage and frequency and fall in current in the faulted phase. The series faults may be classified as one open conductor fault and two open conductor fault.

In the faults mentioned above the 3-phase fault is a symmetrical fault and all other faults are unsymmetrical faults. The symmetrical fault conditions are analyzed on per phase basis using thevenin's theorem or using bus impedance matrix. The unsymmetrical faults are analyzed using symmetrical components.

From the experience in the operation of power system it can be shown that 70 to 80% of the faults are single line-to-ground faults, and 5% of the faults are three phase faults. The relative frequency of occurrence of various types of faults in the power systems in the order of decreasing severity as follows,

3-phase faults	-	5%
Double line-to-ground faults	-	10%
Line-to-line fault	-	15%
Single line-to-ground fault	-	70%

3.2 FAULT CALCULATIONS

When fault occur in a part of power system, heavy current flows in that part of circuit which may cause permanent damage to the equipments. Hence the faulty part should be isolated from the healthy part immediately on the occurrence of a fault. This can be achieved by providing protective relays and circuit breakers. The protective relays sense the faulty conditions and send signals to circuit breakers to open the circuit under faulty condition.

The current which flows in different parts of a power system immediately after the occurrence of a fault differs from the current flowing after few cycles (after a small time) from the occurrence of the fault. The selection of circuit breakers depends on the current flowing immediately after the fault occurs and the current which the breakers must interrupt. (The interrupting current is the current flowing after few cycles). The estimation of these currents for various types of faults at various locations in the system are called fault calculations. The data obtained from fault calculations are also used to determine the settings of the relay which control the circuit breakers.

3.3 TRANSIENTS DUE TO SHORT CIRCUITS

Most of the components of the power system have inductive property which give rise to transients when there is a sudden change in current. Faults on the power system are accompanied by sudden change in current which give rise to transient conditions in power system. The transient states of transmission line and synchronous generator are discussed in this section.

Transients due to a short circuit in transmission line

An unloaded transmission line can be represented by an R-L circuit excited by a sinusoidal source as shown in fig 3.1. (In fig 3.1., the capacitance of the transmission line is neglected).

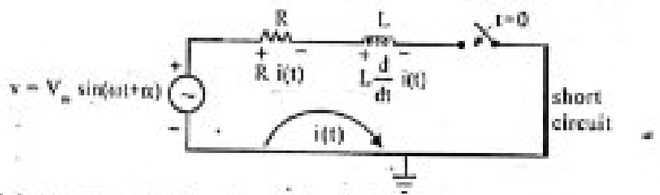


Fig 3.1 : Representation of an unloaded transmission line under short circuit

Let i(t) = Current in transmission line under short circuit condition
Z = Impedance of the transmission line.

Here $Z = R + j\omega L = \sqrt{R^2+\omega^2L^2}\ \angle \tan^{-1}\frac{\omega L}{R} = |Z|\angle\theta$

where $|Z| = \sqrt{R^2+\omega^2L^2}$ and $\theta = \tan^{-1}\frac{\omega L}{R}$

The fault condition can be simulated by closing the switch at t = 0. When the switch is closed at t = 0, fault current (short circuit current) flows in the circuit. The differential equation governing the circuit can be obtained using kirchoff's voltage law as shown in equ(3.1).

$$R\,i(t) + L\frac{d\,i(t)}{dt} = V_m \sin(\omega t + \alpha) \qquad \text{.....(3.1)}$$

The equ(3.1) can be solved using any technique available in circuit theory to find i(t). On solving the equation (3.1), the short circuit current i(t) is given by the equation (3.2).

$$i(t) = \underbrace{\frac{V_m}{|Z|}\sin(\omega t+\alpha-\theta)}_{\text{Symmetrical short circuit current or steady state current } (i_s)} + \underbrace{\frac{V_m}{|Z|}\sin(\theta-\alpha)\,e^{-\frac{R}{L}t}}_{\text{DC off-set current or transient current } (i_t)} \qquad \text{.....(3.2)}$$

The plot of i(t), i_s, i_t and v with respect to t are shown in fig 3.2.

From equ(3.2) it is observed that the short circuit current has two components and they are sinusoidal steady state component and unidirectional transient component. The steady state current is called symmetrical short circuit current and the transient component is called DC off-set current.

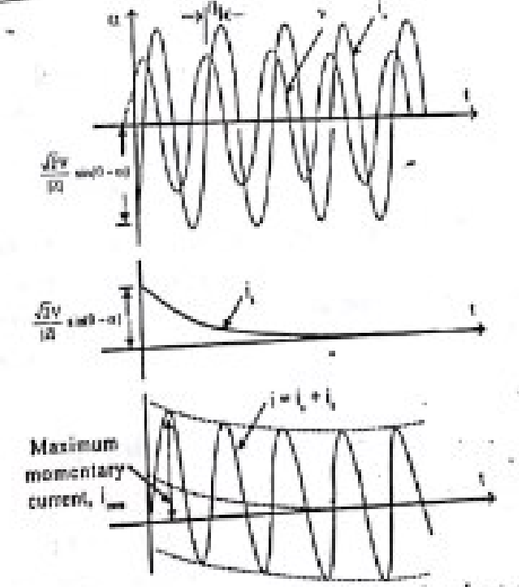


Fig 3.2 : Waveform of currents in a transmission line under short circuit condition

In the short circuit current i(t), the value corresponding to the first peak is called the maximum momentary short circuit current (i_{max}). In equ(3.2) the first peak value is obtained when $\sin(\omega t + \alpha - \theta) = 1$.

$$\therefore i_{max} = \frac{V_m}{|Z|} + \frac{V_m}{|Z|}\sin(\theta-\alpha)\,e^{-\frac{R}{L}t} \qquad \text{.....(3.3)}$$

If the decay of transient current in the interval between t = 0 and time at which first peak occurs is neglected then equ(3.5) can be written as shown below (i.e., $e^{-\frac{R}{L}t} \approx e^{0} = 1$).

$$i_{mm} = \frac{V_m}{|Z|} + \frac{V_m}{|Z|}\sin(\theta - \alpha) \qquad \text{.....(3.4)}$$

In transmission lines the resistance is very low when compared to reactance and so $\theta \approx 90°$ (i.e., $\tan^{-1}(\omega L/R) \approx \tan^{-1}\infty = 90°$).

Hence by taking θ as 90°, the equ(3.4) can be written as,

$$i_{mm} = \frac{V_m}{|Z|} + \frac{V_m}{|Z|}\sin(90 - \alpha)$$

$$\therefore i_{mm} = \frac{V_m}{|Z|} + \frac{V_m}{|Z|}\cos\alpha \qquad \text{.....(3.5)}$$

From equ(3.5), we can say that, i_{mm} has maximum possible value when α = 0. This implies that the effect of short circuit will be severe if the fault occurs when the voltage wave is going through zero.

When α = 0, the equ(3.5) can be written as

$$i_{mm\,(max\ possible)} = \frac{V_m}{|Z|} + \frac{V_m}{|Z|} = 2\frac{V_m}{|Z|} \qquad \text{.....(3.6)}$$

From equ(3.2) we get the maximum value of symmetrical short circuit current as $V_m/|Z|$. From equ(3.6) we can observe that the maximum possible value of maximum momentary short circuit is double the value of maximum symmetrical short circuit current. If such a condition exists in a transmission line then this effect is called doubling effect.

A safer choice of momentary current rating of circuit breaker can be maximum possible value of maximum momentary short circuit current. The interrupting current rating of the circuit breaker can be obtained by multiplying the symmetrical short circuit current by a suitable factor. The multiplication by a constant is neccessary to account for the DC-off set current at the time of interruption.

Transients due to a short circuit in 3-phase alternator

Consider a three phase alternator running on no-load. If a 3-phase fault occurs at the terminals of the alternator then a heavy short circuit current flows in the armature circuit. The oscillogram of the short circuit current after removing the DC-off set current is shown in fig 3.3.

At the time of a short circuit in a 3-phase machine, the voltage wave of the three phases will have different phases. Therefore the DC-offset current will be different in each phase of a 3-phase machine and so it is accounted separately on an empirical basis. The ac component of the short circuit current is called symmetrical short circuit current, whose waveform is shown in fig 3.3.

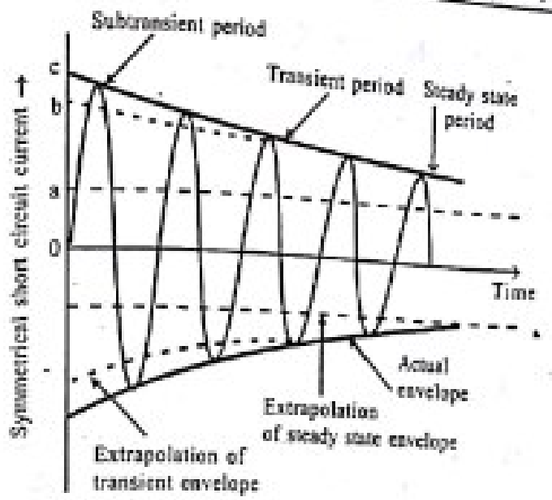


Fig 3.3 : Symmetrical short circuit armature current in an alternator

The symmetrical short circuit current shown in fig 3.3. can be divided into three regions called subtransient, transient and steady state region. (In a practical situation the circuit breaker will open the generator circuit before the steady state fault condition is established).

Under steady state short circuit conditions, the armature reaction of a synchronous generator produces a demagnetizing flux. This effect is represented as a reactance called armature reaction reactance X_a. The sum of leakage reactance, X_l and X_a is called synchronous reactance, X_s. In case of salient pole machines the synchronous reactance is called direct axis reactance and denoted by X_d. On neglecting the armature resistance the steady state short circuit model of an alternator will be as shown in fig 3.4(a).

At the instant of short circuit the DC-off set current appears in all the three phases of stator. This dc-offset current can induce currents in rotor field winding and damper winding by transformer action. The increase in field current and damper winding current will set up flux in a direction to augument the main flux. This effect can be represented by two reactances

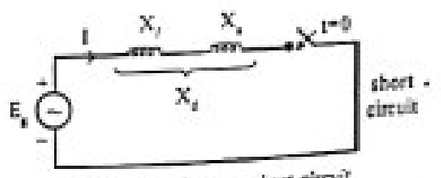


Fig a : Steady state short circuit model of an alternator

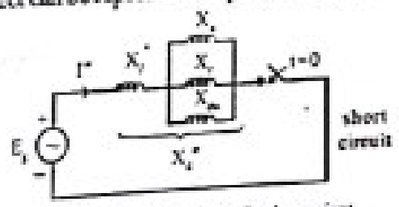


Fig b : Circuit model of subtransient state of an alternator

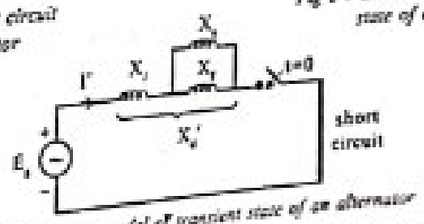


Fig c : Circuit model of transient state of an alternator

Fig 3.4 : Equivalent circuit of an alternator under subtransient, transient and steady state conditions

in parallel with X_a as shown in fig 3.4(b). Here X_f represents the flux created by induced current in the field winding and X_{dw} represents the flux created by induced currents in the damper winding. The combined effect of all the three reactances is to reduce the total reactance of the machine and so the short circuit current is very high in this state which is called as sub-transient state. The total reactance under this condition is called sub-transient reactance and denoted by X_d''

$$\therefore X_d'' = X_l + \frac{1}{\frac{1}{X_a} + \frac{1}{X_f} + \frac{1}{X_{dw}}} \quad \text{.....(3.7)}$$

The induced currents in the damper winding disappears after few cycles from the instant of fault. Because the time constant of the damper winding is smaller than the field winding. This effect is equivalent to open circuited X_{dw} and this state is called transient state. The transient state of the alternator is shown in fig 3.4(c). The total reactance in transient state is called transient reactance and it is denoted by X_d'.

$$\therefore X_d' = X_l + \frac{1}{\frac{1}{X_a} + \frac{1}{X_f}} \quad \text{.....(3.8)}$$

The transient state will exist for few cycles and then steady state conditions are achieved. Because, the effect of field winding current will also die-out in a short time depending on its time constant. This effect is equivalent to open circuited X_f and this state is referred to as steady state. In steady state the total reactance is given by sum of X_l and X_a.

$$\therefore X_d = X_l + X_a \quad \text{.....(3.9)}$$

From equations (3.7), (3.8) and (3.9) we can say that the subtransient reactance of the machine is smallest and steady state reactance of the machine is highest among the reactances. Therefore, $X_d'' < X_d' < X_d$.

The various reactances of the synchronous machine can be estimated from the oscillogram shown in fig 3.3. The envelope of the current wave during transient period can be extrapolated backwards in time to meet the y-axis at point-b. Similarly the envelope of the current wave during steady state period can be extrapolated backwards in time to meet the y-axis at point a.

Let $|I|$ = RMS value of steady state current.
$|I'|$ = RMS value of transient current excluding dc component.
$|I''|$ = RMS value of subtransient current excluding dc component.

From the waveform of fig 3.3, we get,

$$|I| = \frac{oa}{\sqrt{2}} \quad ; \quad |I'| = \frac{ob}{\sqrt{2}} \quad ; \quad |I''| = \frac{oc}{\sqrt{2}}$$

$$\therefore X_d'' = \frac{|E_g|}{|I''|} = \frac{|E_g|}{oa/\sqrt{2}} \quad \text{.....(3.10)}$$

$$X_d' = \frac{|E_g|}{|I'|} = \frac{|E_g|}{ob/\sqrt{2}} \quad \text{.....(3.11)}$$

$$X_d = \frac{|E_g|}{|I|} = \frac{|E_g|}{oc/\sqrt{2}} \quad \text{.....(3.12)}$$

The momentary current rating of the circuit breakers used for generators and synchronous motors are determined using subtransient reactances. The interrupting capacity of the circuit breaker are determined using subtransient reactance for generators and transient reactance for motors.

3.4 INTERNAL VOLTAGES OF LOADED SYNCHRONOUS MACHINES UNDER TRANSIENT CONDITIONS

Consider a generator connected to a bus with voltage V_t. Let I_L be the current delivered by the generator. The circuit model of a synchronous generator operating under steady state conditions supplying a load current I_L is shown in fig 3.5. Here E_g is the induced emf under loaded condition and X_d is the direct axis synchronous reactance of the machine.

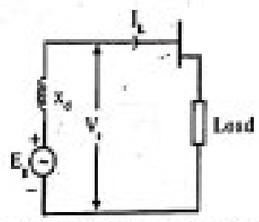


Fig 3.5 : Equivalent circuit of a loaded generator under steady state condition

$$\text{Now, } E_g = V_t + j I_L X_d \quad \text{.....(3.13)}$$

Let a short circuit occur at the terminals of the generator while delivering the load current I_L. Now in order to study the subtransient state the E_g and X_d should be replaced by E_g'' and X_d'' as shown in fig 3.6. In order to study the transient state the E_g and X_d of fig 3.5 should be replaced by E_g' and X_d' as shown in fig 3.6.

Here, E_g'' = Subtransient internal voltage.

E_g' = Transient internal voltage.

In fig (3.6) and (3.7) the prefault current is I_L when the switch is not closed. The fault condition can be represented by closing the switch. In fig (3.6) when the switch is closed

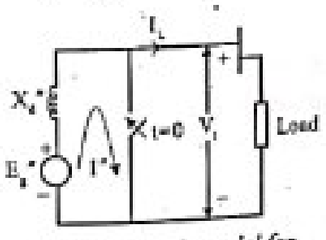


Fig 3.6 : Circuit model for computing subtransient current

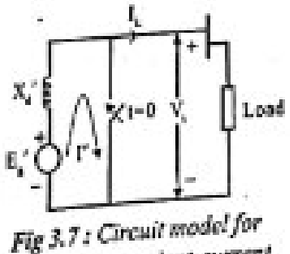


Fig 3.7 : Circuit model for computing transient current

subtransient fault current will flow in the fault path. In fig (3.7) when the switch is closed, transient fault current will flow in the fault path.

With reference to fig (3.6) and (3.7) if prefault load current is I_L then we can write,

$$E_g'' = V_t + jI_L X_d'' \quad \text{....(3.14)}$$

$$E_g' = V_t + jI_L X_d' \quad \text{....(3.15)}$$

It is important to note that the value of E_g'' and E_g' are functions of prefault load current. Therefore E_g'' and E_g' have to be estimated for each value of load current.

The synchronous motors have internal emfs and reactances similar to that of a generator except that the current direction is reversed. Hence for short circuit studies the circuit models similar to that of fig (3.5) to (3.7) can be used for synchronous motor with reversed direction of current. Therefore for synchronous motor,

$$E_g'' = V_t - jI_L X_d'' \quad \text{....(3.16)}$$

$$E_g' = V_t - jI_L X_d' \quad \text{....(3.17)}$$

3.5 SYMMETRICAL THREE PHASE FAULTS

When a fault occurs in a power system, the network, the current flowing is determined by the internal emfs of the machines in the system, by their internal impedances and by impedances in the network between the machines and the fault. When the fault current is equal in all the phases, the fault is called symmetrical fault. The fault current will be symmetrical only in three phase faults in which all the three phases are shorted to ground.

The symmetrical fault can be analysed on per phase basis using reactance diagram or by using per unit reactance diagram. The symmetrical fault analysis has to be performed separately for sub-transient, transient and steady state conditions of the fault, because the reactances and internal emfs of the synchronous machines will be different in each state.

In symmetrical fault analysis, the reactance diagram of the power system is formed using the information in single line diagram. For calculation of sub-transient fault current, the synchronous machine is represented by its sub-transient internal emf in series with sub-transient reactance. Similarly for calculation of transient fault current, the synchronous machine is represented by its transient internal emf in series with transient reactance. For steady state fault analysis, steady state internal emfs and reactances are used for synchronous machines.

Once the per unit reactance diagram of the power system is formed for a particular state (i.e. either sub-transient or transient or steady state) of fault condition, then the currents and voltages in the various parts of the system can be determined by any one of the following methods.

(i) Using Kirchoff's voltage and current law.
(ii) Using thevenin's theorem.
(iii) By forming bus impedance matrix.

Symmetrical fault current estimation using kirchoff's laws

The following procedure can be followed to directly calculate the voltages and currents during symmetrical fault condition in a power system, using Kirchoff's laws.

1. Choose appropriate base values and determine the prefault condition reactance diagram of the given power system. The prefault condition reactance diagram is separately formed for sub-transient, transient and steady state condition of the fault.
2. Calculate the internal emfs of synchronous machines and the prefault voltage at the fault point using prefault current (load current).

Note: If the power system is unloaded (i.e., if there is no prefault current then prefault voltage at the fault point is 1 p.u. Also the internal emfs for sub-transient and transient state are same as steady state induced emf.

3. Draw the fault condition reactance diagram of the system. This diagram is same as prefault reactance diagram except that the fault is represented by a short circuit or by the specified fault impedance. The currents in this reactance diagram are fault condition currents.
4. Calculate the p.u. value of fault currents in the various parts of the system and in the fault.
5. The actual values of the fault currents are obtained by multiplying the p.u. values by the respective base currents.

Note : When transformers are used in the power system, the base currents will be different for various sections of power system.

Symmetrical fault current estimation using thevenin's theorem

The following procedure can be followed to calculate the voltages and currents during symmetrical fault using thevenin's theorem.

1. Choose appropriate base values and determine the prefault condition reactance diagram of the given power system.
2. Calculate the prefault voltage at the fault point using the prefault current (load current). If the system is unloaded, then the prefault voltage is 1 p.u. The prefault voltage at the fault point is the thevenin's voltage.
3. Determine the thevenin's impedance of the system at the fault point. (This is given by looking back impedance at the fault point). To determine the thevenin's impedance, replace all the sources by zero value source and then reduce the resultant network to single equivalent impedance.
4. Draw the thevenin's equivalent at the fault point, F as shown in fig 3.8a. Here the fault can be represented by a short circuit or by fault impedance as shown in fig 3.8b or fig 3.8c.

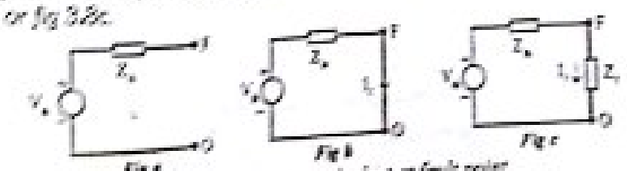


Fig a Fig b Fig c

Fig 3.8 : Thevenin's equivalent at fault point

Now the p.u. value fault current I_f is given by

$$\text{p.u. value of fault current, } I_f = \frac{V_{th}}{Z_0} \qquad \text{.....(3.18)}$$

The actual value of fault current is obtained by multiplying the p.u. value with base current.

5. *The fault currents in other parts of the network are determined from the knowledge of change in current due to fault and prefault current. The fault current (i.e., postfault current) in any part of the system is given by sum of prefault current and change in current due to fault.*

6. *The change in current due to fault can be estimated by connecting the thevenin's source with reversed polarity (i.e., negative of thevenin's voltage source) at the fault. Replace all other sources by zero value sources. Now the currents in various parts of the system are the change in currents due to fault. Calculate these currents by any conventional technique.*

Note : *In a practical power system the normal working voltage will be very close to 1 p.u. Hence the prefault voltage can be assumed as 1 p.u. for simplified calculations. Also the short circuit current will be very much higher than load current (usually 10 to 20 times the load current) and it is purely imaginary. The load current is almost real and negligible. Therefore the prefault current can be neglected for simplified calculations.*

3.6 BUS IMPEDANCE MATRIX IN FAULT CALCULATIONS

The bus impedance matrix can be used to estimate the fault at any point of the system. Usually this method is useful for large system.

Consider the following matrix equation relating the bus voltages, currents injected to the buses and bus impedance matrix for a n-bus system.

$$Z_{bus}\, I = V \qquad \text{.....(3.19)}$$

$$\begin{bmatrix} Z_{11} & Z_{12} & \dots & Z_{1k} & \dots & Z_{1n} \\ Z_{21} & Z_{22} & \dots & Z_{2k} & \dots & Z_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ Z_{k1} & Z_{k2} & \dots & Z_{kk} & \dots & Z_{kn} \\ \vdots & \vdots & & \vdots & & \vdots \\ Z_{n1} & Z_{n2} & \dots & Z_{nk} & \dots & Z_{nn} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_k \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_k \\ \vdots \\ V_n \end{bmatrix} \qquad \text{.....(3.20)}$$

where $I_1, I_2, \dots, I_n$ are currents injected to buses 1, 2 ,....,n respectively.
and $V_1, V_2, \dots, V_n$ are voltages at buses 1, 2 ,...., n respectively.

Let a three-phase fault occur in bus-k.

The prefault voltage at bus-k be V_{pf}. Let the prefault condition be represented by a source of value V_{pf} as shown in fig 3.9a. The fault condition can be represented by a source of $-V_{pf}$ in series with prefault voltage source as shown in fig 3.9b, so that the total bus voltage in bus-k is zero. Due to this fault all other bus voltages will change. Let the change in bus voltage be $\Delta V_1, \Delta V_2, \dots, \Delta V_n$.

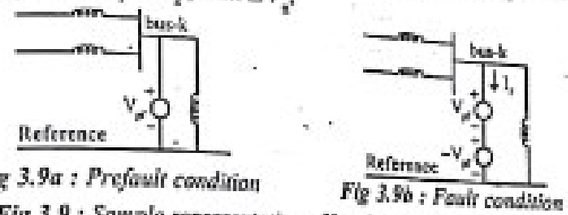


Fig 3.9a : Prefault condition *Fig 3.9b : Fault condition*

Fig 3.9 : Sample representation of bus-k in a n-bus system

Let us assume that the prefault current is zero. Therefore the prefault voltages of all the buses including bus-k will be 1 p.u. When a three phase fault occurs in bus-k the fault current, I_f will flow away from the bus-k and so the current injected to bus-k is $-I_f$. Now if we replace all other sources by zero value sources then the current injected to all other buses will be zero. Under this condition the only source in the network is $-V_{pf}$ and current injected to bus-k is $-I_f$.

Also the voltages in all other buses are only the change in voltages. This condition can be represented by the following matrix equation.

$$\begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \vdots \\ -V_{pf} \\ \vdots \\ \Delta V_n \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \dots & Z_{1k} & \dots & Z_{1n} \\ Z_{21} & Z_{22} & \dots & Z_{2k} & \dots & Z_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ Z_{k1} & Z_{k2} & \dots & Z_{kk} & \dots & Z_{kn} \\ \vdots & \vdots & & \vdots & & \vdots \\ Z_{n1} & Z_{n2} & \dots & Z_{nk} & \dots & Z_{nn} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -I_f \\ \vdots \\ 0 \end{bmatrix} \qquad \text{.....(3.21)}$$

From the matrix equation (3.21) we get,

$$\begin{aligned} \Delta V_1 &= -I_f Z_{1k} \\ \Delta V_2 &= -I_f Z_{2k} \\ &\vdots \\ -V_{pf} &= -I_f Z_{kk} \\ &\vdots \\ \Delta V_n &= -I_f Z_{kn} \end{aligned} \qquad \text{.....(3.22)}$$

$\therefore$ The fault current in bus-k, $I_f = \dfrac{V_{pf}}{Z_{kk}}$

In general the change in bus-q voltage due to a three phase fault in bus-k is given by,

$$\Delta V_q = -I_f Z_{qk} \quad \text{.....(3.23)}$$

The voltage at a bus during a fault in bus-k is given by sum of prefault bus voltage and change in bus voltage. Since the system is unloaded system the prefault voltage at all buses be V_{pf} = 1 p.u., (i.e., same as bus-k voltage).

$$\therefore V_1 = V_{pf} - I_f Z_{1k} = 1\angle 0^\circ - I_f Z_{1k}$$

$$V_2 = V_{pf} - I_f Z_{2k} = 1\angle 0^\circ - I_f Z_{2k}$$

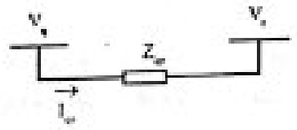
$$\vdots$$

$$V_k = V_{pf} - V_{pf} = 0$$

$$\vdots$$

$$V_n = V_{pf} - I_f Z_{nk} = 1\angle 0^\circ - I_f Z_{nk}$$

The postfault voltages at the buses can be determined as shown above. Then the fault current flowing through the lines can be estimated from the knowledge of line impedances. A sample calculation for fault current in the line-qr is shown below.



$$I_{qr} = \frac{V_q - V_r}{Z_{qr}} = (V_q - V_r)\, y_{qr} \quad \text{.....(3.24)}$$

3.7 SELECTION OF CIRCUIT BREAKERS

The circuit breakers are protective devices which are used in power system to automatically open the faulty part of the system in the event of a fault. In normal working condition they can be used as a switch. Hence the two functions of circuit breakers are

(i) to act as switch for normal load conditions.

(ii) to automatically isolate or open the faulty part in the event of a fault.

The circuit breakers are normally used in power system at places where the power level is very high. They are used in high voltage transmission lines, substations, generating stations and for heavy loads in industries.

Since the circuit breakers are employed in places where the power level is high, whenever its contacts open it has to interrupt heavy currents both during load conditions and faulty conditions. Since the power system is predominantly inductive in nature, the interruption of current when the circuit breaker open its contact is associated with large voltages induced across its contacts which inturn results in sparking at the contacts. Hence in circuit breakers the amount of current it has to interrupt is an important criteria.

The circuit breaker for a particular application (or load) is selected based on the following ratings.

1. Normal working power level specified as rated interrupting current or rated interrupting kVA.
2. The fault level specified as either the rated short circuit interrupting current or rated short circuit interrupting MVA.
3. Momentary current rating.
4. Normal working voltage.
5. Speed of circuit breaker.

The speed of circuit breaker is the time between the occurrence of the fault to the extinction of the arc (when the contact opens). It is normally specified in cycles of power frequency. [1 cycle for 50Hz power frequency is 1/50 = 0.02 msec]. The standard speed of circuit breakers are 8, 5, 3 or 1½ cycles.

The momentary current rating is the maximum current that may flow through a circuit breaker for a short duration. It is the current that may flow during subtransient period of fault condition. In fault analysis the subtransient fault current calculated using subtransient circuit model is the symmetrical subtransient current. It is then multiplied by a factor of 1.6 to get the maximum momentary current during fault. [The factor 1.6 accounts for dc-offset current during subtransient period]. The circuit breaker is chosen such that its momentary current rating is less than the calculated value.

Usually the circuit breaker will open its contacts in the transient period and so the short circuit interrupting current rating depends on transient period currents. In fault analysis the transient fault current calculated using transient circuit model is the symmetrical transient fault current. It is then multiplied by a factor 1.0 to1.5 to get the maximum interrupting current. [The factor 1.0 to 1.5 accounts for dc-offset current during transient period]. The circuit breaker is chosen such that its short circuit interrupting current rating is less than the calculated value. The multiplying factor to find interrupting current depends on the speed of circuit breaker.

The multiplying factor for various speeds of circuit breaker are shown in table 3.1.

Table 3.1 : Multiplying factor to find the short circuit interrupting current

Speed of circuit breaker	Multiplying factor
8 cycles or more	1.0
5 cycles	1.1
3 cycles	1.2
2 cycles	1.4
1½ cycles	1.5

The short circuit interrupting MVA can be estimated from the knowledge of prefault voltage and short-circuit interrupting current as shown below.

Short circuit interrupting MVA $= \sqrt{3}\,|V_{pfL}|\,|I_{FL}|$ ....(3.25)

where $|V_{pfL}|$ = Magnitude of prefault line voltage at the fault point in kV.

$|I_{FL}|$ = Magnitude of line value of short circuit interrupting current at the fault in kA.

or Short circuit interrupting MVA $= |V_{pfL,pu}| \times |I_{FL,pu}| \times MVA_b$ ....(3.26)

where $|V_{pfL,pu}|$ = Magnitude of prefault voltage at the fault point in p.u.

$|I_{FL,pu}|$ = Magnitude of short circuit interrupting current at the fault in p.u.

Note : *Here the short circuit interrupting MVA is a three phase power rating.*

The equations (3.25) and (3.26) can be used to compute the interrupting kVA in normal working condition if we use normal working voltage in kV and normal (load condition) interrupting current instead of fault condition voltage and currents.

EXAMPLE 3.1

A synchronous generator and motor are rated for 30,000 kVA, 13.2 kV and both have subtransient reactance of 20%. The line connecting them has a reactance of 10% on the base of machine ratings. The motor is drawing 20,000 kW at 0.8 pf leading. The terminal voltage of the motor is 12.8 kV. When a symmetrical three-phase fault occurs at motor terminals, find the subtransient current in generator, motor and at the fault point.

SOLUTION

The single line diagram and the circuit model to compute subtransient fault current are shown in fig 3,1.1. and 3.1.2. respectively.

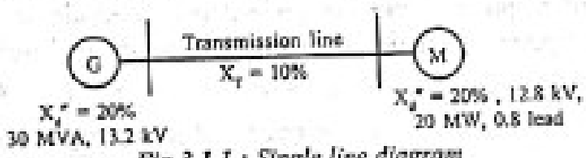


Fig 3.1.1 : Single line diagram

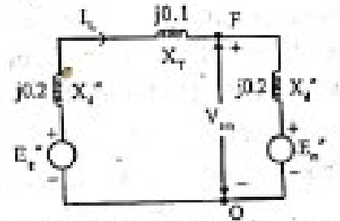


Fig 3.1.2 : Circuit model to compute subtransient fault current

In the fig 3.1.2, the generator and motor are represented by a source in series with subtransient reactance. The value of the sources are subtransient internal voltages.

The base values are, $MVA_b = 30\,MVA$; $kV_b = 13.2\,kV$

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}kV_b} = \dfrac{MVA_b \times 1000}{\sqrt{3} \times kV_b} = \dfrac{30 \times 1000}{\sqrt{3} \times 13.2} = 1312.16\,A$

Actual value of prefault voltage at fault point, $V_{fm} = 12.8\,kV$

p.u. value of prefault voltage at fault point, $V_{fm} = \dfrac{\text{Actual value}}{\text{Base value}} = \dfrac{12.8}{13.2} = 0.9697$ p.u.

Actual value of real power of the load, $P_m = 20$ MW, 0.8 lead

p.u. value of real power of the load, $P_m = \dfrac{\text{Actual value}}{\text{Base value}} = \dfrac{20}{30} = 0.6667$ p.u.

When voltage, current and power are expressed in p.u., then in 3-phase circuits

$P = VI\cos\phi$

where $\cos\phi$ = power factor of the load

$\therefore$ p.u. value of magnitude of load current, $|I| = \dfrac{P_m}{V_{fm}\cos\phi} = \dfrac{0.6667}{0.9697 \times 0.8} = 0.8594$ p.u.

Alternatively the load current can be calculated as shown below

$$\text{Load kVA} = \frac{\text{Read Power}}{\text{pf}} = \frac{20}{0.8} = 25\text{kVA}$$

$$\text{p.u value of load kVA, } S_m = \frac{\text{Load kVA}}{\text{Base kVA}} = \frac{25}{30} = 0.8333\text{p.u.}$$

$$\text{p.u value of load current, } |I| = \frac{S_m}{V_{fm}} = \frac{0.8333}{0.9697} = 0.8595\text{ p.u.}$$

Let us take the terminal voltage of motor V_{fm} as reference vector and so the load current will lead the terminal voltage of motor with an angle $\cos^{-1} 0.8$.

$\therefore V_{fm} = 0.9697\,\angle 0°$ p.u.

and $I_L = 0.8594\,\angle\cos^{-1} 0.8 = 0.8594\,\angle 36.9°$ p.u.

Method-1

Prefault condition

The voltages and currents in the various elements of the system just before the fault are shown in fig (3.1.3). In this circuit V_{fm} and I_L are known values and using these values the subtransient internal voltages E_g'' and E_m'' can be calculated by Kirchoff's Voltage Law (KVL) as shown below.

By applying KVL in the circuit of fig (3.1.3) we get,

$$\begin{aligned} E_g'' &= j0.2\,I_L + j0.1\,I_L + V_{fm} \\ &= j0.3\,I_L + V_{fm} \\ &= 0.3\,\angle 90° \times 0.8594\,\angle 36.9° + 0.9697\,\angle 0° \\ &= 0.2578\,\angle 126.9° + 0.9697\,\angle 0° \\ &= -0.1548 + j0.2062 + 0.9697 \\ &= 0.8149 + j0.2062 = 0.8406\,\angle 14.2°\text{ p.u.} \end{aligned}$$

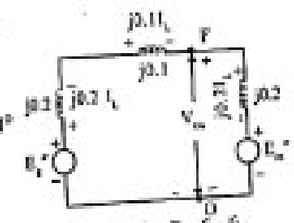


Fig 3.1.3 : Prefault current and voltages

Using KVL in the circuit of fig 3.1.3 we get,

$$\begin{aligned} E_m'' + j0.2\,I_L &= V_{fm} \\ \therefore E_m'' &= V_{fm} - j0.2\,I_L = 0.9697\,\angle 0° - (0.2\,\angle 90° \times 0.8594\,\angle 36.9°) \\ &= 0.9697\,\angle 0° - (0.1719\,\angle 126.9°) = 0.9697 - (-0.1032 + j0.1375) \\ &= 1.0729 - j0.1375 = 1.0817\,\angle -7.3°\text{ p.u.} \end{aligned}$$

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subtransient state of
fig 3.1.4. The circuit
wo parallel circuits
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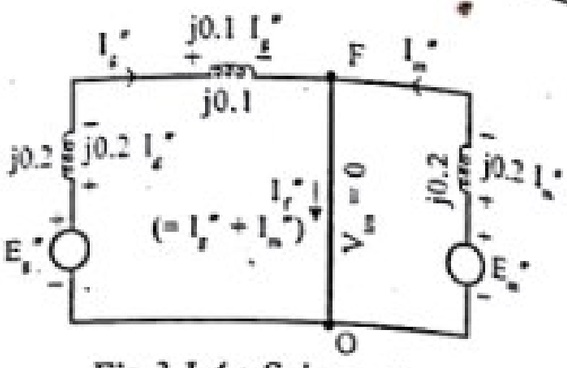


Fig 3.1.4 : Subtransient current and voltages

of fig 3.1.4. we get,

$= E_g''$ or $j0.3\, I_s'' = E_s''$

$$\frac{E_g''}{0.3} = \frac{0.8406\ \angle 14.2°}{0.3\ \angle 90°} = 2.802\ \angle -75.8°\ \text{p.u.}$$

of fig 3.1.4. we get, $j0.2\, I_m'' = E_m''$

$$\frac{E_m''}{0.2} = \frac{1.0817\ \angle -7.3°}{0.2\ \angle 90°} = 5.4085\ \angle -97.3°\ \text{p.u.}$$

$= I_g'' + I_m''$

$= 2.802\ \angle -75.8° + 5.4085\ \angle -97.3°$

$= 0.687 - j2.716 - 0.687 - j5.365$

$= -j8.081 = 8.081\angle -90°$ p.u.

current can be obtained by multiplying the p.u. values of

$I_g'' = 2.802\ \angle -75.8° \times 1312.16$

$= 3676.67\ \angle -75.8°\ \text{A} = 3.67667\ \angle -75.8°\ \text{kA}$

$I_m'' = 5.4085\ \angle -97.3° \times 1312.16$

$= 7096.8\ \angle -97.3°\ \text{A} = 7.0968\ \angle -97.3°\text{kA}$

$I_f'' = 8.081\ \angle -90° \times 1312.16$

$= 10603.56\ \angle -90°\ \text{A} = 10.60356\ \angle -90°\text{kA}$

Method-2 : Using thevenin's theorem

To find fault current

Since the prefault voltage at the fault point is known can be represented by the thevenin's equivalent circuit.

Consider the prefault circuit model shown in fig 3.1. the fault point is V_{tm}. The thevenin's impedance is given by from the fault point. It is obtained by replacing all the vo sources (i.e., by short circuit) as shown in fig 3.1.6 and then re to single equivalent impedance as shown below.

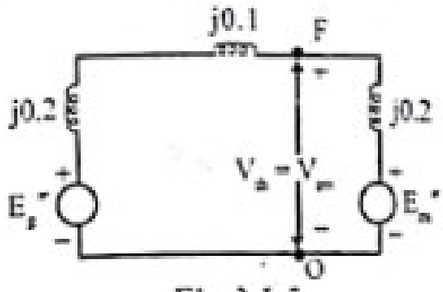


Fig 3.1.5

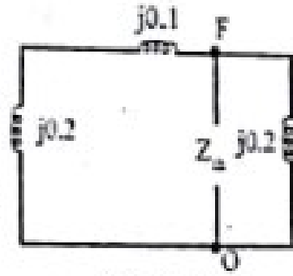


Fig 3.1.6

Thevenin's equivalent impedance, $Z_{th} = \frac{(j0.1 + j0.2)\, j0.2}{(j0.1 + j0.2) + j0}$

The thevenin's equivalent of the circuit with respect 3.1.7. Now short circuiting the terminals of the thevenin's e fig 3.1.8. is equivalent to fault condition. The current flow fault current.

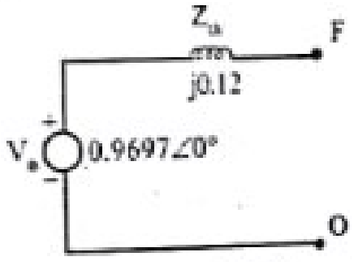


Fig 3.1.7 : Prefault thevenin's equivalent at fault point



Fig 3.1.8 : Th under fa

Current in the fault, $I_f'' = \frac{V_{th}}{Z_{th}} = \frac{0.9697\angle 0°}{j0.12} = \frac{0.9697\angle 0}{0.12\angle 90}$

To find the change in current due to fault

The change in current due to fault can be calculate by connecting the thevenin's generator with reversed polari at the fault point as shown in fig 3.1.9. (Here all other sourc are replaced by zero value sources, i.e., the voltage sourc are replaced by short circuit).

Now, $I_1 = \frac{V_g}{jX_d'' + jX_t} = \frac{1.0\angle 0^\circ}{0.2\angle 90^\circ} = 5\angle -90^\circ$ p.u

$I_2 = \frac{V_m}{jX_m} = \frac{1.0\angle 0^\circ}{0.2\angle 90^\circ} = 5\angle -90^\circ$ p.u

To find the subtransient fault current in motor and generator

The subtransient fault currents of motor and generator are given by the sum of prefault current and the change in current due to fault (current delivered by thevenin's generator). Here the prefault current is the load current I_L.

$\therefore I_g'' = I_1 - I_L = 5\angle -90^\circ - 1.654\angle -36.9^\circ = -j5 - 1.323 + j0.992$

$= 1.677 - j4.008 = 4.345\angle -67.3^\circ$ p.u

$\therefore I_m'' = I_2 - I_L = 5\angle -90^\circ - (1.654\angle -36.9^\circ) = -j5 - 1.323 + j0.992$

$= -1.677 - j4.008 = 4.345\angle -112.7^\circ$ p.u

Note : It can be observed that the currents obtained by both the method are same.

RESULT

Subtransient fault current in generator, $I_g'' = 4.345\angle -67.3^\circ$ p.u or $4.345\angle -67.3^\circ$ kA

Subtransient fault current in motor, $I_m'' = 4.345\angle -112.7^\circ$ p.u or $4.345\angle -112.7^\circ$ kA

Subtransient current at the fault, $I_f'' = 8.69\angle -90^\circ$ p.u or $8.69\angle -90^\circ$ kA

EXAMPLE 2.2

A generator is connected through a transformer to a synchronous motor. The subtransient reactance of generator and motor are 0.25 and 0.35 respectively. The leakage reactance of the transformer is 0.1 p.u. All the reactances are calculated on a common base. A three phase fault occurs at the terminals of the motor when the terminal voltage of the generator is 0.9 p.u. The output current of generator is 1 p.u and 0.8 pf leading. Find the subtransient current in p.u. in the fault, generator and motor. Use the terminal voltage of generator as reference vector.

SOLUTION

The single line diagram and the circuit model to compute subtransient fault current are shown in fig 2.2.1 and 2.2.2 respectively.



Fig 2.2.1 : Single line diagram

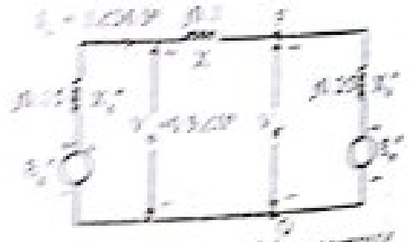


Fig 2.2.2 : Circuit model to compute subtransient fault current

Here V_g = Terminal voltage of generator.

V_f = Prefault voltage at the fault point.

In fig 2.2.2 the generator and motor are represented by a source in series with subtransient reactance. The value of the source are subtransient internal voltage.

The terminal voltage of generator is reference vector and as the load current will lead the generator terminal voltage by an angle $\cos^{-1} 0.8 = 36.9^\circ$.

$\therefore$ Load current in p.u, $I_L = 1\angle \cos^{-1} 0.8 = 1\angle -36.9^\circ$ p.u

Method-1

Prefault condition

The voltages and currents in the various elements of the system just before the fault occurs is shown in fig 2.2.3. In this circuit V_g and I_L are source values. Using these value the subtransient internal voltages E_g' and E_m' can be calculated by Kirchoff's voltage law (KVL) as shown below.

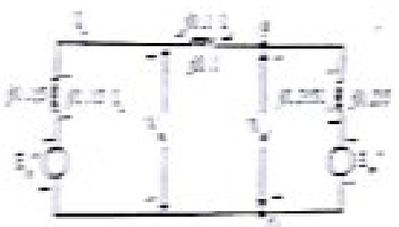


Fig 2.2.3 : Prefault current and voltages

By applying KVL in the circuit of fig 2.2.3, we get

$E_g' = j0.25 I_L + V_g$

$= 0.25\angle 90^\circ \times 1\angle -36.9^\circ + 0.9\angle 0^\circ$

$= 0.25\angle 126.9^\circ + 0.9\angle 0^\circ = -0.15 + j0.2 + 0.9$

$= 0.8499 + j0.2 = 0.87\angle 13.2^\circ$ p.u

Using KVL in the circuit of fig 2.2.3 we get

$E_m' - j0.35 I_L - j0.1 I_L = V_g$

$\therefore E_m' = V_g + j0.45 I_L$

$= 0.9\angle 0^\circ + 0.45\angle 90^\circ \times 1\angle -36.9^\circ = 0.9\angle 0^\circ + 0.45\angle 53.1^\circ$

$= 0.9 + j0.27 + j0.36 = 1.17\angle 53.1^\circ$

$= 1.026 - \angle 47.1^\circ$ p.u

Fault condition

The voltages and currents in the various elements of the system during subtransient state of the fault condition are shown in fig 2.2.4. The circuit of fig 2.2.4 can be treated as two parallel sources feeding current in the fault point.



Fig 2.2.4 : Subtransient current and voltages

Using KVL in the circuit of fig 3.2.4. we get,

$$j0.15\,I_g'' + 0.1\,I_g'' = E_g'' \quad \text{or} \quad j0.25\,I_g'' = E_g''$$

$$\therefore \text{Subtransient fault current in generator } I_g'' = \frac{E_g''}{j0.25} = \frac{0.8187\angle 84^\circ}{0.25\angle 90^\circ} = 3.2748\angle -81.6^\circ \text{ p.u.}$$

Using KVL in the circuit of fig 3.2.4. we get,

$$j0.35\,I_m'' = E_m''$$

$$\therefore \text{Subtransient fault current in motor, } I_m'' = \frac{E_m''}{j0.35} = \frac{1.2243\angle -17.1^\circ}{0.35\angle 90^\circ} = 3.498\angle -107.1^\circ \text{ p.u.}$$

The current in the fault during subtransient state $I_f'' = I_g'' + I_m''$

$$= 3.2748\angle -81.6^\circ + 3.498\angle -107.1^\circ$$
$$= 0.4784 - j3.2397 - 1.0286 - j3.3434$$
$$= -0.5502 - j6.5831 = 6.606\angle -94.8^\circ \text{ p.u.}$$

Method-2 : Using thevenin's theorem

To find fault current

The prefault voltage at the fault point can be estimated from the prefault circuit model. Then the thevenin's equivalent of the circuit is determined and from which the fault current is estimated.

Consider the prefault circuit model shown in fig 3.2.5. The thevenin's voltage at the fault point is V_{pf}. The thevenin's impedance is given by looking back impedance from the fault point. It is obtained by replacing all the voltage sources by zero value sources (i.e., by short circuit) as shown in fig 3.2.6 and then reducing the resultant network to single equivalent impedance as shown below.

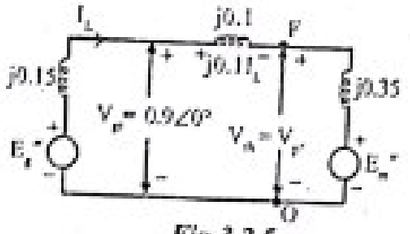


Fig 3.2.5

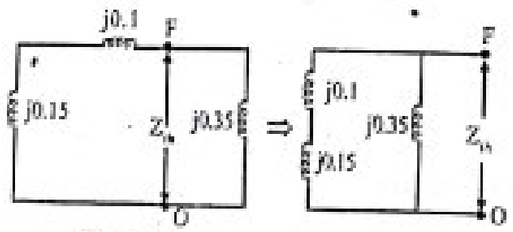


Fig 3.2.6

In fig 3.2.5. using KVL we get,

$$V_{pf} = j0.1\,I_L + V_{th}$$

Thevenin's voltage, $V_{th} = V_{pf} - j0.1\,I_L$

$$= 0.9\angle 0^\circ - (0.1\angle 90^\circ \times 1\angle 36.9^\circ)$$

$$= 0.9\angle 0^\circ - (0.1\angle 126.9^\circ)$$
$$= 0.9 - (-0.06 + j0.08)$$
$$= 0.96 - j0.08 = 0.9633\angle -4.8^\circ \text{ p.u.}$$

Thevenin's equivalent impedance, $Z_{th} = \frac{(j0.15 + j0.1)\,j0.35}{(j0.15 + j0.1) + j0.35} = j0.1458$ p.u.

The thevenin's equivalent of the circuit with respect to fault point is shown in fig 3.2.7. Now short circuiting the terminals of the thevenin's equivalent circuit as shown in fig 3.2.8. is equivalent to fault condition. The current flowing through the short is the fault current.

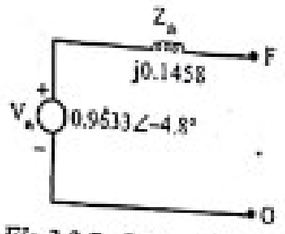


Fig 3.2.7 : Prefault thevenin's equivalent at fault point

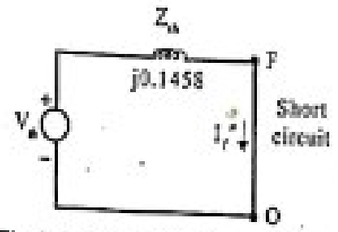


Fig 3.2.8 : Thevenin's equivalent under fault condition

Current in the fault, $I_f'' = \frac{V_{th}}{Z_{th}} = \frac{0.9633\angle -4.8^\circ}{j0.1458} = \frac{0.9633\angle -4.8^\circ}{0.1458\angle 90^\circ} = 6.606\angle -94.8^\circ$ p.u.

To find the change in current due to fault

The change in current due to fault can be calculated by connecting the thevenin's generator with reversed polarity at the fault point as shown in fig 3.2.9. (Here all other sources are replaced by zero value sources, i.e., the voltage sources are replaced by short circuit).

$$\text{Now, } I_1 = \frac{V_{th}}{j0.15 + j0.1} = \frac{0.9633\angle -4.8^\circ}{0.25\angle 90^\circ} = 3.8532\angle -94.8^\circ$$

$$I_2 = \frac{V_{th}}{j0.35} = \frac{0.9633\angle -4.8^\circ}{0.35\angle 90^\circ} = 2.7523\angle -94.8^\circ \text{ p.u.}$$

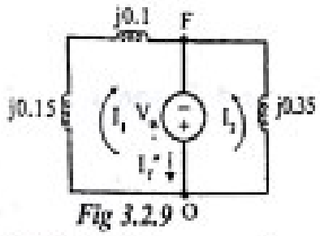


Fig 3.2.9

To find the subtransient fault current in motor and generator

The subtransient fault currents of motor and generator are given by the sum of prefault current and change in current due to fault (current delivered by thevenin's generator). Here the prefault current is the load current, I_L.

$$\therefore I_g'' = I_1 + I_L = 3.8532\angle -94.8^\circ + 1\angle 36.9^\circ = -0.3224 - j3.8397 + 0.7997 + j0.6004$$
$$= 0.4773 - j3.2393 = 3.274\angle -81.6^\circ \text{ p.u.}$$

$$\therefore I_m'' = I_2 - I_L = 2.7523\angle -94.8^\circ - 1\angle 36.9^\circ = -0.2303 - j2.7426 - (0.7997 + j0.6004)$$
$$= -1.03 - j3.343 = 3.498\angle -107.1^\circ \text{ p.u.}$$

Note : It can be observed that the currents calculated by both the methods are same.

RESULT

Subtransient fault current in generator, $I_g'' = 3.274 \angle -81.6°$ p.u.

Subtransient fault current in motor, $I_m'' = 3.098 \angle -107.1°$ p.u.

Subtransient current in the fault, $I_f'' = 6.606 \angle -94.8°$ p.u.

EXAMPLE 3.3

A 3-phase, 5 MVA, 6.6 kV alternator with a reactance of 8% is connected to a feeder of series impedance of 0.12 + j0.48 ohms/phase per km. The transformer is rated at 3 MVA, 6.6 kV/33 kV and has a reactance of 5%. Determine the fault current supplied by the generator operating under no-load with a voltage of 6.9 kV, when a 3-phase symmetrical fault occurs at a point 15 km along the feeder.

SOLUTION

The single line diagram of the power system and the prefault reactance diagram are shown in fig 3.3.1 and fig 3.3.2 respectively. Let the point F be the point where the fault occurs.

Let us choose generator rating as base values.

$\therefore MVA_b = 5$ MVA and $kV_b = 6.6$ kV

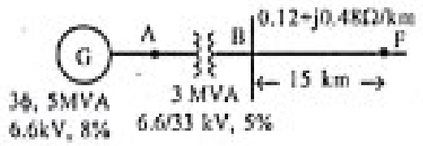


Fig 3.3.1 : Single line diagram

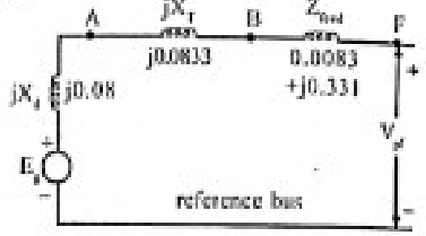


Fig 3.3.2 : Prefault reactance diagram

To find E_g and V_{pf}

Here the generator is not delivering any load current and so the induced emf of the generator will be same as operating voltage.

Actual value of induced emf, $E_g = 6.9$ kV

p.u. value of induced emf, $E_g = \dfrac{\text{Actual value}}{\text{Base value}} = \dfrac{6.9}{6.6} = 1.0455$ p.u.

The prefault voltage, V_{pf} at the fault point F can be obtained using the transformer ratio as shown below.

Actual value of prefault voltage, $V_{pf} = 6.9 \times \dfrac{33}{6.6} = 34.5$ kV

The base kV referred to HT side of transformer $= 6.6 \times \dfrac{33}{6.6} = 33$ kV

$\therefore$ The p.u. value of prefault voltage, $V_{pf} = \dfrac{\text{Actual value}}{\text{Base value}} = \dfrac{34.5}{33} = 1.0455$ p.u.

Note : The open circuit p.u. value of voltage is same at every point in a series path irrespective of their actual voltages.

To find generator reactance

Since the rating of the generator is chosen as base value, the p.u. reactance of the generator will be same as the specified value.

$\therefore$ p.u. reactance of the generator, $X_g = 8\% = 0.08$ p.u.

To find transformer reactance

The transformer reactance can be computed with respect to new base using the following equation.

$$X_{pu,new} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

Here $X_{pu,old} = 5\% = 0.05$ p.u.

$\therefore$ p.u. reactance of transformer, $X_T = 0.05 \times \left(\dfrac{6.6}{6.6}\right)^2 \times \dfrac{5}{3} = 0.0833$ p.u.

To find feeder reactance

The base impedance, $Z_b = \dfrac{(kV_b)^2}{MVA_b} = \dfrac{33^2}{5} = 217.8\ \Omega$/phase

Actual impedance of the feeder for a length of 15 km $\Big\}$, Z_{feed} = impedance / km × length

$= (0.12 + j0.48) \times 15 = 1.8 + j7.2\ \Omega$/phase

$\therefore$ p.u. value of the impedance of the feeder, $Z_{feed} = \dfrac{\text{Actual impedance}}{\text{Base impedance}} = \dfrac{1.8 + j7.2}{217.8}$

$= 0.0083 + j0.0331$ p.u.

To find fault current

The prefault reactance diagram of fig 3.3.2 can be represented by thevenin's equivalent circuit shown in fig 3.3.3. Here the thevenin's voltage is the prefault voltage at fault point F and the thevenin's impedance is the sum of three impedances in series.

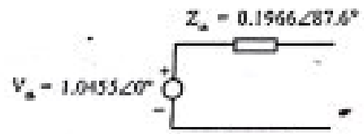


Fig 3.3.3.

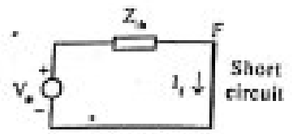


Fig 3.3.4.

$$\therefore Z_{th} = jX_d + jX_T + Z_{feed}$$

$$= j0.08 + j0.0833 + 0.0083 + j0.0331$$

$$= 0.0083 + j0.1964 \text{ p.u.} = 0.1966 \angle 87.6° \text{ p.u.}$$

$$= 0.1966 \angle 87.6° \text{ p.u.}$$

The fault in the feeder can be represented by a short circuit as shown in fig 3.3.4. Now the current I_f through the short circuit is the fault current.

$$\therefore \text{p.u. value of fault current, } I_f = \frac{V_{th}}{Z_{th}} = \frac{1.0455\angle 0°}{0.1966\angle 87.6°} = 5.3179\angle -87.6° \text{ p.u.}$$

Note : *The solution can also be attempted by neglecting the resistance of the feeder. In this case*

$$Z_{th} = j0.1964 = 0.1964\angle 90° \text{ p.u. and } I_f = 5.3233\angle -90° \text{ p.u.}$$

It is important to note that the error in neglecting the resistive component is negligible.

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{MVA_b \times 1000}{\sqrt{3}\,kV_b} = \frac{5 \times 1000}{\sqrt{3} \times 33} = 87.4773 \text{ A.}$$

$\therefore$ Actual value of fault current, I_f = p.u. value of $I_f \times I_b$

$$= 5.3179\angle -87.6° \times 87.4773$$

$$= 465.2 \angle -87.6 \text{ amps.}$$

RESULT

Fault current = $5.3179\angle -87.6$ p.u. or $465.2\angle -87.6$ A.

EXAMPLE 3.4

For the radial network shown in fig 3.4.1 a 3-phase fault occurs at point -F. Determine the fault current.

SOLUTION

The prefault impedance diagram of the given system is shown in fig 3.4.2

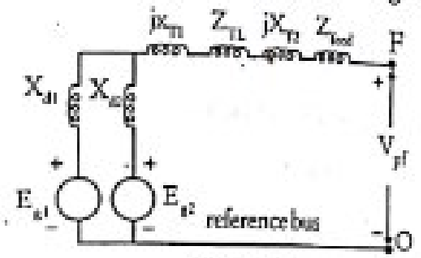


Fig 3.4.2

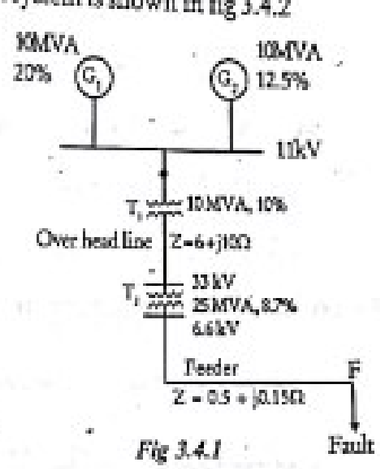


Fig 3.4.1

X_{d1}, X_{d2} - Reactances of generator 1&2.
X_{T1} - Reactance of transformer T1.
Z_{TL} - Impedance of transmission line.
X_{T2} - Reactance of transformer T2.
Z_{feed} - Impedance of feeder.

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 10MVA \quad ; \quad kV_b = 11kV.$$

There is no load on the system and so the induced emf of the generator can be assumed as 1 p.u. Therefore the prefault voltage V_{pf} at the fault point is also 1 p.u.

To find the generator reactances

Since the generator ratings are chosen as base values, the p.u. reactance of the generators will remain same.

p.u. reactance of generator-1, X_{d1} = 20 % = 0.2 p.u.
p.u. reactance of generator-2, X_{d2} = 12.5 % = 0.125 p.u.

To find the reactance of T_1

The base values referred to LT side of transformer is same as chosen base and so its reactance is same as specified value.

p.u. reactance of transformer-T_1, X_{T1} = 10% = 0.1 p.u.

To find the p.u. impedance of overhead line

$$\text{Base kV on HT side of transformer-T}_1 = 11 \times \frac{33}{11} = 33 \text{ kV}$$

$$\text{Base impedance, } Z_b = \frac{kV_b^2}{MVA_b} \times \frac{33^2}{10} = 108.9\,\Omega/\text{phase}$$

Given that the actual impedance of overhead line = $6 + j10\ \Omega$

$\therefore$ p.u. impedance overhead line, $Z_{TL} = \frac{\text{Actual impedance}}{\text{Base impedance}} = \frac{6 + j10}{108.9} = 0.0551 + j0.0918$ p.u.

To find the reactance of T_2

The new p.u. reactance of the transformer T_2 can be computed using the following formula

$$X_{pu,new} = X_{p.u,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

Here $X_{pu,old} = 8.7\% = 0.087$, $MVA_{b,old} = 25$ MVA, $kV_{b,old} = 33$ kV.

$MVA_{b,new} = 10$ MVA, $kV_{b,new} = 33$ kV.

$\therefore$ p.u. reactance of transformer T_2, $X_{T2} = 0.087 \times \left(\frac{33}{33}\right)^2 \times \frac{10}{25} = 0.0348$ p.u

To find the p.u. impedance of the feeder.

Base kV on the LT side of transformer, $T_2 = 33 \times \frac{6.6}{33} = 6.6$ kV

Base impedance, $Z_b = \frac{kV_b^2}{MVA_b} = \frac{6.6^2}{10} = 4.356\Omega$/phase

Given that the actual impedance of feeder = $0.5 + j0.15\ \Omega$/phase

$\therefore$ p.u. impedance of the feeder, $Z_{feed} = \frac{\text{Actual impedance}}{\text{Base impedance}} = \frac{0.5 + j0.15}{4.356} = 0.1148 + j0.0344$ p.u.

To find thevenin's equivalent at fault point

The thevenin's equivalent network of the prefault impedance diagram shown in fig 3.4.2 can be obtained as shown below.

The thevenin's voltage at the fault point is the prefault voltage and so $V_{th} = 1\angle 0°$ p.u.

The thevenin's impedance can be obtained by reducing the following network (fig 3.4.3) to single equivalent impedance.

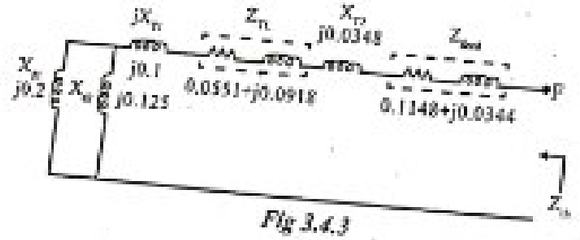


Fig 3.4.3

$$Z_{th} = \frac{j0.2 \times j0.125}{j0.2 + j0.125} + (j0.1 + 0.0551 + j0.0918 + j0.0348 + 0.1148 + j0.0344)$$

$$= j0.0769 + 0.1699 + j0.261 = 0.1699 + j0.3379 = 0.3782\angle 63.3°\ \text{p.u.}$$

The prefault thevenin's equivalent circuit is shown in fig 3.4.4. In this the fault is represented by a short circuit as shown in fig 3.4.5. The current through the short circuit is the fault current I_f

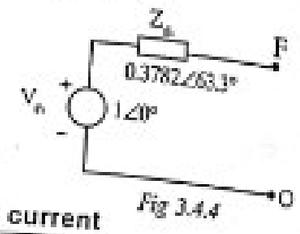


Fig 3.4.4

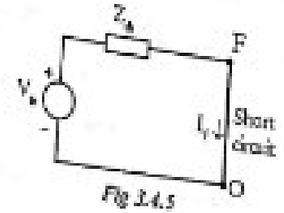


Fig 3.4.5

To find fault current

The p.u. value of fault current, $I_f = \frac{V_{th}}{Z_{th}} = \frac{1\angle 0°}{0.3782\angle 63.3°} = 2.6441\angle -63.3°$ p.u.

The base Current, $I_b = \frac{kVA_b}{\sqrt{3}kV_b} = \frac{10 \times 1000}{\sqrt{3} \times 6.6} = 874.77$A

The actual value of fault current, I_f = p.u. value of fault current × Base current

$= 2.6441\angle -63.3° \times 874.77 = 2313\angle -63.3°$ A

$= 2.313\angle -63.3°$ kA

RESULT

The fault current, $I_f = 2.6441\angle -63.3°$ p.u. or $2.313\angle -63.3°$ kA

EXAMPLE 3.5

A 1250 kVA, 5000V generator with $X_d'' = 0.08$ p.u. supplies a purely resistive load of 1000 kW at rated voltage. The load is connected directly across the terminals of the generator. If all the three phases of the load are short circuited simultaneously find initial symmetrical short circuit current in the generator.

SOLUTION

The single line diagram and the prefault impedance diagram of the system are shown

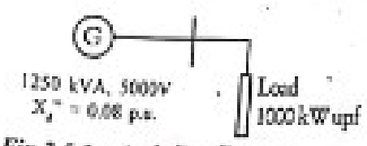


Fig 3.5.1 : single line diagram

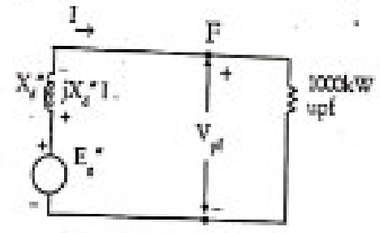


Fig 3.5.2 : Prefault impedance diagram

Let the base values be generator ratings

$$\therefore kVA_b = 1250kVA \quad ; \quad kV_b = 5000V = 5kV$$

$$\therefore \text{p.u. value of load power, } P = \frac{\text{Actual value of load power}}{\text{Base kVA}} = \frac{1000}{1250} = 0.8 \text{ p.u.}$$

p.u. value of load voltage $V = 1\angle 0°$ p.u (Because the load is supplied at rated voltage)

When power, voltage and current are expressed in p.u. values then in 3-phase circuits,

$$P = |V|\,|I| \cos\phi$$

Here $\cos\phi = 1$, because the load is resistive.

$$\therefore \text{p.u. value of magnitude of load current, } |I| = \frac{P}{|V|} = \frac{0.8}{1.0} = 0.8 \text{ p.u.}$$

$$\therefore I = |I|\angle \cos^{-1} 1 = 0.8\angle 0° \text{ p.u}$$

Here the prefault voltage, V_{pf} is same as load voltage.

$$\therefore V_{pf} = 1.0\angle 0° \text{ p.u.}$$

The subtransient internal emf E_g'' during fault can be estimated from the circuit of fig 3.5.2, using kirchoff's law as shown below.

$$E_g'' = jX_d'' I + V_{pf} = j0.08 \times 0.8\angle 0° + 1.0\angle 0°$$

$$= 0.08\angle 90° \times 0.8\angle 0° + 1.0\angle 0° = 0.064\angle 90° + 1.0\angle 0° = 1 + j0.064$$

$$= 1.002\angle 3.7° \text{ p.u.} = 1.002\angle 3.7° \text{ p.u.}$$

The fault condition impedance diagram of the system is shown in fig 3.5.3.

From fig 3.5.3, using KVL we get,

$$jX_d'' I_g'' = E_g''$$

$$I_g'' = \frac{E_g''}{jX_d''} = \frac{1.002\angle 3.7°}{j0.08}$$

$$= \frac{1.002\angle 3.7°}{0.08\angle 90°} = 12.525\angle -86.3° \text{ p.u.}$$

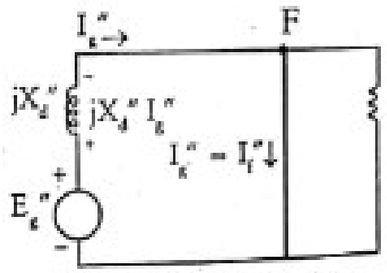


Fig 3.5.3 : Fault condition impedance diagram

Here the subtransient current I_g'' supplied by the generator is same as current through the fault I_f''.

$$\therefore I_f'' = I_g'' = 12.525\angle -86.3° \text{ p.u.}$$

$$\text{Base Current, } I_b = \frac{kVA_b}{\sqrt{3}kV_b} = \frac{1250}{\sqrt{3} \times 5} = 144.34 \text{ amps.}$$

Actual value of fault current I_f = p.u. value of fault current × Base current

$$= 12.525\angle -86.3 \times 144.34 = 1807.86\angle -86.3° \text{ A}$$

$$= 1.80786\angle -86.3° \text{ kA}$$

RESULT

The fault current, $I_f = 12.525\angle -86.3°$ p.u. or $1.80786\angle -86.3°$ kA.

EXAMPLE 3.6

Two synchronous motors are connected to the bus of a large system through a transmission line as shown in fig 3.6.1.

The ratings of the various components are,

Motor each : 1MVA, 440V, 0.1p.u transient reactance

Line : 0.05 ohm reactance

Large system : Short circuit MVA at its bus at 440V is 8.

When the motors are operated at 400V, calculate the short circuit current (symmetrical) fed into a three phase fault at motor bus.

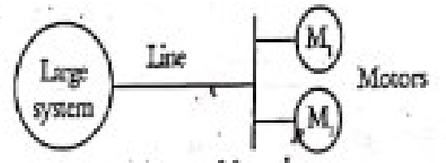


Fig : 3.6.1

SOLUTION

The large system feeding the line can be considered as a source of constant voltage. Also the bus in the large system is an infinite bus and so its voltage and frequency will not change due to a fault in the motor bus. Since the short circuit rating of the infinite bus in specified at 440V, we assume its voltage rating as 440V.

The prefault reactance diagram of the given system is shown in fig 3.6.2.

Let us choose motor ratings as base values.

$$\therefore MVA_b = 1MVA \quad ; \quad kV_b = 0.44 \text{ kV}$$

$$\text{Base impedance, } Z_b = \frac{kV_b^2}{MVA_b} = \frac{0.44^2}{1} = 0.1936.$$

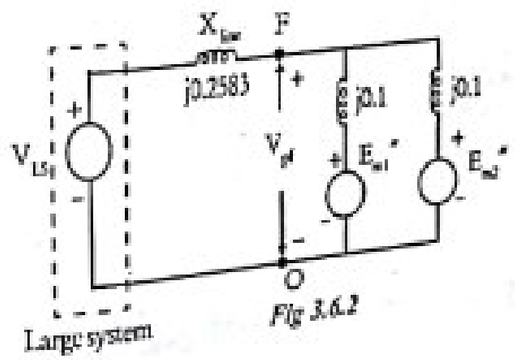


Fig 3.6.2

p.u. reactance of transmission line = $\frac{\text{Actual reactance}}{\text{Base impedance}} = \frac{0.05}{0.1936} = 0.2583$ p.u.

The motor is operated at 400V. Hence the voltage at motor bus is 400V. The prefault voltage is the voltage at motor bus.

Actual value of prefault voltage = 400V.

p.u. value of prefault voltage $= \frac{400}{440} = 0.9091$ p.u

p.u. value of the voltage of infinite bus, $V_{LS} = \frac{440}{440} = 1$ p.u.

The fault condition of the system is shown fig 3.6.3. The fault current can be estimated using thevenin's theorem. The thevenin's voltage at the fault point is the prefault voltage V_f. The thevenin's impedance is obtained by reducing the network shown in fig 3.6.4 to single equivalent impedance.

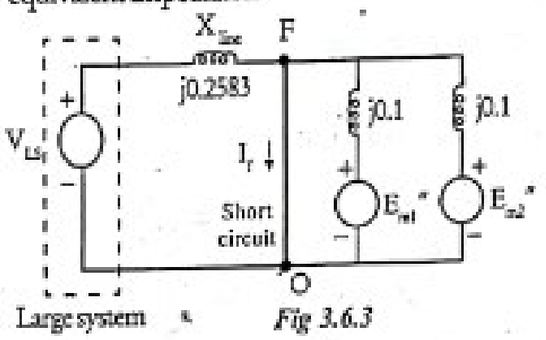


Fig 3.6.3

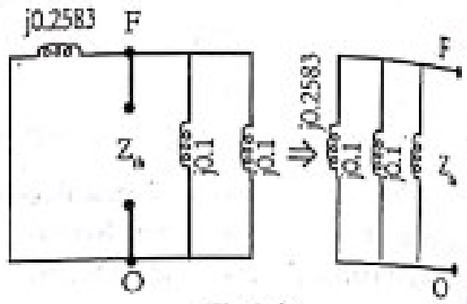


Fig 3.6.4

$$Z_{th} = \frac{1}{\frac{1}{j0.2583} + \frac{1}{j0.1} + \frac{1}{j0.1}} = j0.0419 \text{ p.u.}$$

The prefault thevenin's equivalent of the system is shown in fig 3.6.5. The fault is represented by short circuit as shown in fig 3.6.6.

The p.u. value of fault current, $I_f = \frac{V_{th}}{Z_{th}} = \frac{0.9091}{j0.0419} = j21.6969$ p.u. $= 21.6969\angle -90°$ p.u.

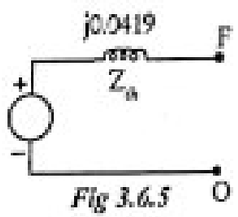


Fig 3.6.5

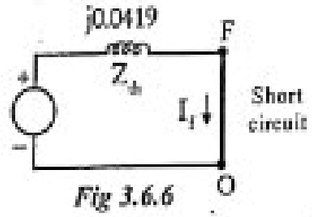


Fig 3.6.6

Base current, $I_b = \frac{kVA_b}{\sqrt{3}kV_b} = \frac{1 \times 1000}{\sqrt{3} \times 0.44} = 1312.16A = 1.3122kA$

Actual value of fault current, I_f = p.u. value of fault current × Base current

$= 21.6969\angle -90° \times 1.3122 = 28.4707\angle -90°$ kA.

RESULT

Fault current = $21.6969\angle -90°$ p.u. or $28.4707\angle -90°$ kA.

EXAMPLE 3.7

A generator is connected through a circuit breaker to a transformer. The ratings of the generator are 100 MVA, 18kV, $X_d'' = 19\%$, $X_d' = 26\%$ and $X_d = 130\%$. The transformer ratings are 100 MVA, 240/18kV, Y-Δ, X = 10% with 18kV on Δ side. If a 3-phase short circuit occurs on the high tension side of a transformer at rated voltage and no load, find (a) initial symmetrical rms current in the transformer winding on the high tension side. (b) The initial symmetrical rms current in the line on the low tension side.

SOLUTION

The single line diagram and the prefault reactance diagram of the given power system are shown in fig 3.7.1 and 3.7.2 respectively.

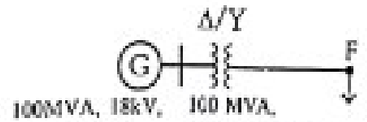


Fig 3.7.1

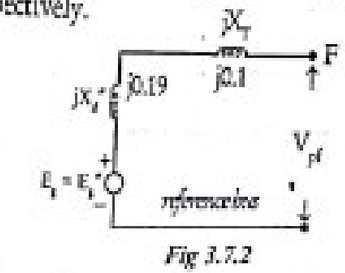


Fig 3.7.2

Let us choose the generator rating as base values

$\therefore MVA_b = 100$ MVA ; $kV_b = 18$kV

Subtransient reactance of generator, $X_d'' = 19\% = 0.19$ p.u.

Leakage reactance of transformer, $X_T = 10\% = 0.1$ p.u.

The generator is running on no-load and so the subtransient internal emf is same as induced emf. Since the generator is running on no-load, $E_g = 1\angle 0°$ p.u. Also the prefault voltage, $V_{pf} = E_g = 1\angle 0°$ p.u.

The fault condition reactance diagram is shown in fig 3.7.3. The fault is represented by a short circuit. The current through the fault is the initial symmetrical rms value of fault current. (Here it is same as subtransient fault current after eliminating dc-offset current).

The p.u. value of initial symmetrical fault current $|I_f''| = \frac{E_g}{j0.19 + j0.1} = \frac{1\angle 0°}{0.29\angle 90°}$

$= 3.4483\angle -90°$

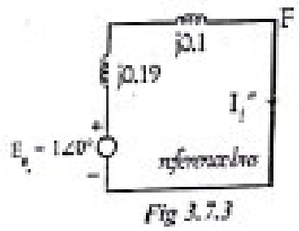


Fig 3.7.3

Now, $|I_f''| = 3.4483$ p.u.

Base kV on HT side of transformer $= 18 \times \frac{240}{18} = 240$ kV

Base current on HT side of transformer $= \frac{kVA_b}{\sqrt{3}kV_b} = \frac{100 \times 1000}{\sqrt{3} \times 240} = 240.56A$

The HT winding of the transformer is star connected and so the fault current in the transformer winding is same as the fault current in the line.

∴ p.u. value of initial symmetrical fault current on the HT winding of transformer $= |I_f''| = 3.4483$ p.u.

Actual value of initial symmetrical fault current on the HT winding of transformer = p.u. value × Base current

$= 3.4483 \times 240.56 = 829.5$ A $= 0.8295$ kA

The LT winding of transformer is delta connected but the transmission line is considered as star circuit.

∴ Base current on LT side of tranformer $= \frac{kVA_b}{\sqrt{3}\, kV_b} = \frac{100 \times 1000}{\sqrt{3} \times 18} = 3207.5$ A

∴ p.u. value of initial symmetrical fault current on the LT line $= |I_f''| = 3.4483$ p.u.

Actual value of initial symmetrical fault current on the LT line = p.u. value × Base current

$= 3.4483 \times 3207.5 = 11060.4$ A $= 11.0604$ kA.

Actual value of initial symmetrical fault current on the LT winding of tranformer $= \frac{11.0604}{\sqrt{3}} = 6.3857$ kA

RESULT

Initial symmetrical fault current on the HTwinding of transformer = 3.4483 p.u.or0.8295kA.

Initial symmetrical fault current on the LT line of transformer = 3.4483 p.u.or 11.0604 kA.

EXAMPLE 3.8

The bus impedance matrix of four bus system with values in p.u is given by,

$$Z_{bus} = j\begin{bmatrix} 0.15 & 0.08 & 0.04 & 0.07 \\ 0.08 & 0.15 & 0.06 & 0.09 \\ 0.04 & 0.06 & 0.13 & 0.05 \\ 0.07 & 0.09 & 0.05 & 0.12 \end{bmatrix}$$

In this system generators are connected to buses 1 and 2 and their subtransient reactances were included when finding Z_{bus}. If prefault current is neglected, find subtransient current in

p.u. in the fault for a 3 phase fault on bus -4. Assume prefault voltage as 1 p.u. If the subtransient reactance of generator in bus - 2 is 0.2 p.u. find the subtransient fault current supplied by generator.

SOLUTION

Let I_f'' be the subtransient current in the fault on bus - 4.

Now, $I_f'' = \frac{V_{pf}}{Z_{44}}$

where V_{pf} = prefault voltage at bus - 4 = $1\angle 0°$ p.u.

$\therefore I_f'' = \frac{1\angle 0°}{j0.12} = -j8.333 = 8.333\angle -90°$ p.u.

The voltage at bus -2 when there is a 3-phase fault in bus - 4 is given by

$V_2 = V_{pf} - I_f'' Z_{24}$

$\therefore V_2 = 1\angle 0° - 8.333\angle -90° \times j0.09 = 1 - 8.333\angle -90° \times 0.09\angle 90°$

$= 1 - 0.74997 = 0.25003 \approx 0.25\angle 0°$ p.u.

Since there is no current in the system prior to the fault all the buses will be at same potential prior to fault. Also there won't be any potential drop in the synchronous reactance of generator, because it does not deliver any current prior to fault. Hence the induced emf (or subtransient internal emf) of the generator in bus-2 is also 1 p.u. The generator in bus-2 can be represented as shown in fig 3.8.1

With reference to fig 3.8.1,

The subtransient fault current delivered by generator at bus - 2 $= I_{g2}'' = \frac{E_{g2}'' - V_2}{jX_{d2}''}$

$= \frac{1\angle 0° - 0.25\angle 0°}{j0.2} = \frac{1 - 0.25}{0.2\angle 90°}$

$= 3.75\angle -90°$ p.u.

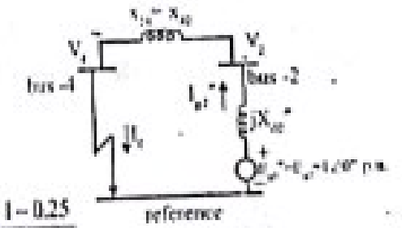


Fig 3.8.1

Note : $= I_f'' = I_{g1}'' + I_{g2}''$

RESULT

The subtransient fault current in bus-4 = $I_f'' = 8.333\angle -90°$ p.u.

The voltage at bus - 2 when there is a 3 phase fault in bus-4 = $V_2 = 0.75\angle 0°$ p.u.

The subtransient fault current delivered by generator - 2 when there is a 3 - phase fault in bus - 4 $= I_{g2}'' = 3.75\angle -90°$ p.u.

EXAMPLE 3.9

The bus impedance matrix of four bus system with values in p.u. is given by

$$Z_{bus} = j\begin{bmatrix} 0.1488 & 0.0651 & 0.0864 & 0.0978 \\ 0.0651 & 0.1554 & 0.0799 & 0.0967 \\ 0.0864 & 0.0798 & 0.1341 & 0.1058 \\ 0.0978 & 0.0967 & 0.1058 & 0.1566 \end{bmatrix}$$

If a 3-phase fault occurs at bus-1 when there is no-load, find the subtransient current in the fault and voltages at all buses. Also find the subtransient current supplied by the generator connected to bus-2 by taking the subtransient reactance of generator as j0.2 p.u. The prefault voltage is 1 p.u.

SOLUTION

Let I_f'' be the subtransient fault current injected to (or drawn from) bus-1 due to a 3-phase fault.

$$\text{Now, } I_f'' = \frac{V_{pf}}{Z_{11}}$$

where V_{pf} = prefault voltage = $1\angle 0°$ p.u.

$$\therefore I_f'' = \frac{1\angle 0°}{j0.1488} = \frac{1\angle 0°}{0.1488\angle 90°} = 6.7204\angle -90° \text{ p.u.}$$

Let V_1, V_2, V_3 and V_4 be the voltages at buses 1,2,3 and 4 respectively in the fault condition. The voltage in bus-q due to a 3-phase fault in bus - k is given by

$$V_q = V_{pf} - I_f'' Z_{qk}$$

$$\therefore V_1 = V_{pf} - I_f'' Z_{11} = V_{pf} - V_{pf} = 0$$

$$V_2 = V_{pf} - I_f'' Z_{21} = 1\angle 0° - 6.7204\angle -90° \times 0.0651\angle 90° = 1 - 0.4375 = 0.5625 \text{ p.u.}$$

$$V_3 = V_{pf} - I_f'' Z_{31} = 1\angle 0° - 6.7204\angle -90° \times 0.0864\angle 90° = 1 - 0.5806 = 0.4194 \text{ p.u.}$$

$$V_4 = V_{pf} - I_f'' Z_{41} = 1\angle 0° - 6.7204\angle -90° \times 0.0978\angle 90° = 1 - 0.6573 = 0.3427 \text{ p.u.}$$

Since there is no current in the system prior to the fault all the buses will be at same potential, prior to the fault. Also there won't be any potential drop in the synchronous reactance of generator, because it does not deliver any current prior to fault. Hence the induced emf (or subtransient internal emf) of the generator in bus-2 is also 1p.u. The generator in bus-2 can be represented as shown in fig 3.9.1

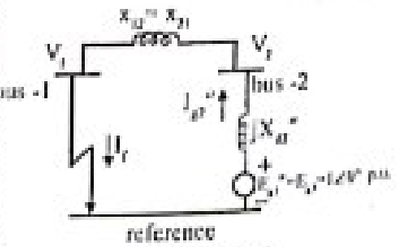


Fig 3.9.1

With reference to fig 3.9.1,

The subtransient fault current delivered by generator at bus-2

$$I_{g2}'' = \frac{E_{g2}'' - V_2}{jX_{d2}''}$$

$$= \frac{1\angle 0° - 0.5625\angle 0°}{j0.2} = \frac{1 - 0.5625}{0.2\angle 90°}$$

$$= \frac{0.4375}{0.2}\angle -90° = 2.1875\angle -90° \text{ p.u.}$$

RESULT

The subtransient fault current in bus-1 = $I_f'' = 6.7204\angle -90°$ p.u.

The subtransient fault current supplied by generator at bus-2 = $I_g'' = 2.1875\angle -90°$ p.u.

The bus voltages when there is a 3-phase fault in bus-1 are

$V_1 = 0$; $V_2 = 0.5625$ p.u. ; $V_3 = 0.4194$ p.u. ; $V_4 = 0.3427$ p.u.

EXAMPLE 3.10

A generator connected through a five cycle circuit breaker to a transformer, is rated at 100 MVA, 18 kV with reactances $X_d'' = 20\%$, $X_d' = 25\%$ and $X_d = 110\%$. It is operated on no-load and at rated voltage. When a 3 -phase fault occurs between the breaker and the transformer, find.

a) Short circuit current in circuit breaker
b) The initial symmetrical rms current in the circuit breaker.
c) The maximum possible dc component of the short circuit current in the breaker.
d) The current to be interrupted by the breaker.
e) The interrupting MVA.

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 100 \text{ MVA}, kV_b = 18 \text{ kV}.$$

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}kV_b} = \frac{100 \times 1000}{\sqrt{3} \times 18} = 3207.5 \text{A}$$

a) To find short circuit currents in circuit breaker

The circuit breaker clears the faults in 5 cycles and so we can assume that the circuit opens during transient period of the fault. Therefore the fault currents that may flow through the circuit breaker are maximum momentary short circuit current, subtransient fault current and transient fault current. If the fault exists even after 5 cycles (i.e, if the fault is not cleared by

EXAMPLE 3.9

The bus impedance matrix of four bus system with values in p.u. is given by

$$Z_{bus} = j\begin{bmatrix} 0.1488 & 0.0651 & 0.0864 & 0.0978 \\ 0.0651 & 0.1554 & 0.0799 & 0.0967 \\ 0.0864 & 0.0798 & 0.1341 & 0.1058 \\ 0.0978 & 0.0967 & 0.1058 & 0.1566 \end{bmatrix}$$

If a 3-phase fault occurs at bus-1 when there is no-load, find the subtransient current in the fault and voltages at all buses. Also find the subtransient current supplied by the generator connected to bus-2 by taking the subtransient reactance of generator as j0.2 p.u. The prefault voltage is 1 p.u.

SOLUTION

Let I_f'' be the subtransient fault current injected to (or drawn from) bus-1 due to a 3-phase fault.

$$\text{Now, } I_f'' = \frac{V_{pf}}{Z_{11}}$$

where V_{pf} = prefault voltage = $1\angle 0°$ p.u.

$$\therefore I_f'' = \frac{1\angle 0°}{j0.1488} = \frac{1\angle 0°}{0.1488\angle 90°} = 6.7204\angle -90° \text{ p.u.}$$

Let V_1, V_2, V_3 and V_4 be the voltages at buses 1,2,3 and 4 respectively in the fault condition. The voltage in bus-q due to a 3-phase fault in bus - k is given by

$$V_q = V_{pf} - I_f'' Z_{qk}$$

$$\therefore V_1 = V_{pf} - I_f'' Z_{11} = V_{pf} - V_{pf} = 0$$

$$V_2 = V_{pf} - I_f'' Z_{21} = 1\angle 0° - 6.7204\angle -90° \times 0.0651\angle 90° = 1 - 0.4375 = 0.5625 \text{ p.u.}$$

$$V_3 = V_{pf} - I_f'' Z_{31} = 1\angle 0° - 6.7204\angle -90° \times 0.0864\angle 90° = 1 - 0.5806 = 0.4194 \text{ p.u.}$$

$$V_4 = V_{pf} - I_f'' Z_{41} = 1\angle 0° - 6.7204\angle -90° \times 0.0978\angle 90° = 1 - 0.6573 = 0.3427 \text{ p.u.}$$

Since there is no current in the system prior to the fault all the buses will be at same potential, prior to the fault. Also there won't be any potential drop in the synchronous reactance of generator, because it does not deliver any current prior to fault. Hence the induced emf (or subtransient internal emf) of the generator in bus-2 is also 1p.u. The generator in bus-2 can be represented as shown in fig 3.9.1

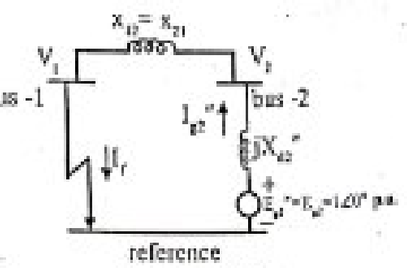


Fig 3.9.1

With reference to fig 3.9.1,

The subtransient fault current delivered by generator at bus-2

$$I_{g2}'' = \frac{E_{g2}'' - V_2}{jX_{d2}''}$$

$$= \frac{1\angle 0° - 0.5625\angle 0°}{j0.2} = \frac{1 - 0.5625}{0.2\angle 90°}$$

$$= \frac{0.4375}{0.2}\angle -90° = 2.1875\angle -90° \text{ p.u.}$$

RESULT

The subtransient fault current in bus-1 = $I_1'' = 6.7204\angle -90°$ p.u.

The subtransient fault current supplied by generator at bus-2 = $I_{g2}'' = 2.1875\angle -90°$ p.u.

The bus voltages when there is a 3-phase fault in bus-1 are

$V_1 = 0$; $V_2 = 0.5625$ p.u. ; $V_3 = 0.4194$ p.u. ; $V_4 = 0.3427$ p.u.

EXAMPLE 3.10

A generator connected through a five cycle circuit breaker to a transformer, is rated at 100 MVA, 18 kV with reactances $X_d'' = 20\%$, $X_d' = 25\%$ and $X_d = 110\%$. It is operated on no-load and at rated voltage. When a 3 -phase fault occurs between the breaker and the transformer, find.

a) Short circuit current in circuit breaker
b) The initial symmetrical rms current in the circuit breaker.
c) The maximum possible dc component of the short circuit current in the breaker.
d) The current to be interrupted by the breaker.
e) The interrupting MVA.

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 100 \text{ MVA}, kV_b = 18 \text{ kV}.$$

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}kV_b} = \frac{100 \times 1000}{\sqrt{3} \times 18} = 3207.5 \text{A}$$

a) To find short circuit currents in circuit breaker

The circuit breaker clears the faults in 5 cycles and so we can assume that the circuit opens during transient period of the fault. Therefore the fault currents that may flow through the circuit breaker are maximum momentary short circuit current, subtransient fault current and transient fault current. If the fault exists even after 5 cycles (i.e, if the fault is not cleared by

opening the circuit in 5 cycles) then steady state fault current will also flow through the circuit breaker.

The generator is running on no-load and so the internal emfs during subtransient and transient state are same as steady state induced emf. The generator is operated at rated voltage and so $E_g = E_g' = E_g'' = 1$p.u.

Given that, $X_d'' = 20\% = 0.2$ p.u, $X_d' = 25\% = 0.25$ p.u. and $X_d = 110\% = 1.1$ p.u.

Subtransient fault current in generator excluding dc - offset current $\Big\} I_g'' = \dfrac{E_g''}{jX_d''} = \dfrac{1\angle 0°}{j0.2} = \dfrac{1\angle 0°}{0.2\angle 90°} = 5\angle -90°$ p.u.

Transient fault current in generator excluding dc - offset current $\Big\} I_g' = \dfrac{E_g'}{jX_d'} = \dfrac{1\angle 0°}{j0.25} = \dfrac{1\angle 0°}{0.25\angle 90°} = 4\angle -90°$ p.u.

Steady state fault current in generator, $I_g = \dfrac{E_g}{jX_d} = \dfrac{1\angle 0°}{j1.1} = \dfrac{1\angle 0°}{1.1\angle 90°} = 0.9091\angle -90°$ p.u.

The maximum momentary short circuit current is obtained by multiplying $|I_g''|$ by 1.6, to account for dc-offset current.

$\therefore$ Maximum momentary short circuit current $= 1.6|I_g''| = 1.6 \times 5 = 8$ p.u.

The actual values of fault currents can be obtained by multiplying the p.u. current by base current.

Actual value of subtransient fault current $= |I_g''| \times I_b = 5 \times 3207.5 = 16037.5 = 16.0375$ kA

Actual value of transient fault current $= |I_g'| \times I_b = 4 \times 3207.5 = 12830$ A $= 12.83$ kA

Actual value of steady state fault current $\Big\} = |I_g| \times I_b = 0.9091 \times 3207.5 = 2915.9$ A $= 2.9159$ kA.

Actual value of maximum momentary short circuit current $\Big\} = 8 \times I_b = 8 \times 3207.5 = 25660$ A $= 25.66$ kA.

b) To find initial symmetrical rms current.

The initial symmetrical rms current is the subtransient fault current excluding dc-offset current, i.e., it is same as I_g''.

$\therefore$ Initial symmetrical rms current $= 5\angle -90°$ p.u. or 16.0375 kA.

c) To find maximum possible dc component

The maximum possible dc component of the short circuit current in the breaker is the difference between maximum momentary current and initial symmetrical short circuit current.

$\therefore$ Maximum possible dc component of the short circuit current $= 25.66 - 16.0375$
$= 9.6225$ kA.

d) To find the interrupting current

The interrupting current of the circuit breaker can be estimated from the transient current by multiplying it by a suitable constant to account for dc-offset current at the time of interruption. For 5 cycles circuit breaker the multiplying factor is 1.1.

$$\text{p.u value of short circuit interrupting current} = \left(\begin{matrix}\text{p.u. value of}\\ \text{transient fault current}\end{matrix}\right) \times 1.1$$

$$= 4 \times 1.1 = 4.4 \text{ p.u}$$

Actual value short circuit interrupting current = p.u. value × base current

$$= 4.4 \times 3207.5 = 14113 \text{ A} = 14.113 \text{ kA.}$$

e) To find interrupting MVA

$$\text{The short circuit interrupting MVA} = \left(\begin{matrix}\text{p.u. value of}\\ \text{prefault voltage}\end{matrix}\right) \times \left(\begin{matrix}\text{p.u value of}\\ \text{interrupting current}\end{matrix}\right) \times \text{Base MVA}$$

$$= 1.0 \times 4.4 \times 100 = 440 \text{ MVA}$$

3.8 SHORT QUESTIONS AND ANSWERS

1. *What is meant by a fault?*

A fault in a circuit is any failure which interferes with the normal flow of current. The faults are associated with abnormal change in current, voltage and frequency of the power system. The faults may cause damage to the equipments if it is allowed to persist for a long time. Hence every part of a system has been protected by means of relays and circuit breakers to sense the faults and to isolate the faulty part from the healthy part in the event of fault.

2. *Why faults occur in a power system?*

The faults occur in a power system due to insulation failure of equipments, flashover of lines initiated by a lightning stroke, due to permanent damage to conductors and towers or due to accidental faulty operations.

3. *How the faults are classified ?*

In one method of classification, the faults are classified as shunt and series faults. The shunt faults are due to short circuits in conductors and the series faults are due to open conductors.

In another method of classification, the faults are classified into symmetrical and unsymmetrical faults. In symmetrical faults the fault currents are equal in all the phases and can be analysed on per phase basis. In unsymmetrical faults the fault currents are unbalanced and so they can be analysed only using symmetrical components.

4. *List the various types of shunt and series faults.*

The various types of shunt faults are

i) Line-to-ground fault
ii) Line-to-line fault
iii) Double line-to-ground fault
iv) Three phase fault.

The various types of series faults are

i) One open conductor fault
ii) Two open conductor fault

5. *List the symmetrical and unsymmetrical faults*

The three phase fault is the only symmetrical fault. All other types of faults are unsymmetrical faults.

The various unsymmetrical faults are

i) Line-to-ground fault
ii) Line-to-Line fault
iii) Double line-to-ground fault
iv) One or two open conductor faults.

6. *What is symmetrical and unsymmetrical fault?*

The fault is called symmetrical fault if the fault current is equal in all the phases. The fault is called unsymmetrical fault if the fault current is not equal in all the phases.

7. *Name any two methods of reducing short-circuit current.*

The following are the two methods of reducing the short circuit current

i) By providing neutral reactance
ii) By introducing a large value of shunt reactance between buses.

8. *Name the main differences in representation of power system for load flow and short circuit studies.*

i) For load flow studies both the resistances and reactances are considered whereas for fault analysis the resistances are neglected.
ii) For load flow studies the bus admittance matrix is useful whereas for short circuit studies bus impedance matrix is used.
iii) The load flow study is performed to determine the exact voltages and currents whereas in short circuit studies the voltages can be safely assumed as 1 pu and the prefault current can be neglected.

9. *For a fault at a given location, rank the various faults in the order of severity.*

In a power system relatively the most severe fault is three phase fault and less severe fault is open conductor fault.

The various faults in the order of decreasing severity are given below.

i) 3-phase fault
ii) Double line-to-ground fault
iii) Line-to-line fault
iv) Single line-to ground fault
v) Open conductor faults.

10. *Write the relative frequency of occurrence of various types of faults.*

Type of fault	Relative frequency of occurrence
i) 3-phase fault	5%
ii) Double line-to ground fault	10%
iii) Line-to-line fault	15%
iv) Single line-to-ground fault	70%

11. *What is meant by fault calculations?*

The fault condition of a power system can be divided into substransient, transient and steady state periods. The currents in the various parts of the system and in the fault are different in these periods. The estimation of these currents for various types of faults at various locations in the system are commonly referred to as fault calculations.

12. *What is the need for short circuit studies or fault analysis ?*

The short circuit studies are essential in order to design or develop the protective schemes for various parts of the system. The protective schemes consists of current & voltage sensing devices, protective relays and circuit breakers. The selection (or proper choice) of these devices mainly depends on various currents that may flow in the fault conditions.

13. *What is the reason for transients during short circuits?*

The faults or short circuits are associated with sudden change in currents. Most of the components of the power system have inductive property which opposes any sudden change in currents and so the faults (short circuit) are associated with transients.

14. *Draw the waveform of a short circuit current on a transmission line.*

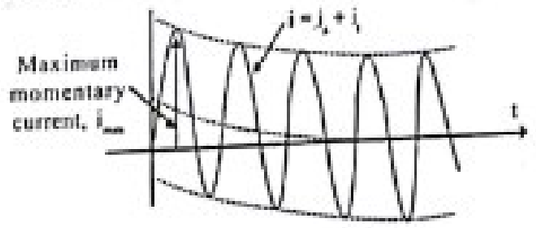


Fig Q14 : Waveform of a short circuit current on a transmission line

15. *What is meant by doubling effect?*

If a symmetrical fault occurs when the voltage wave is going through zero then the maximum momentary short circuit current will be double the value of maximum symmetrical short circuit current. This effect is called doubling effect.

16. *Define DC off-set current.*

The unidirectional transient component of short circuit current is called DC off-set current.

17. *Draw the oscillogram of short circuit current when an unloaded generator is subjected to symmetrical fault, clearly marking subtransient, transient and steady state regions*

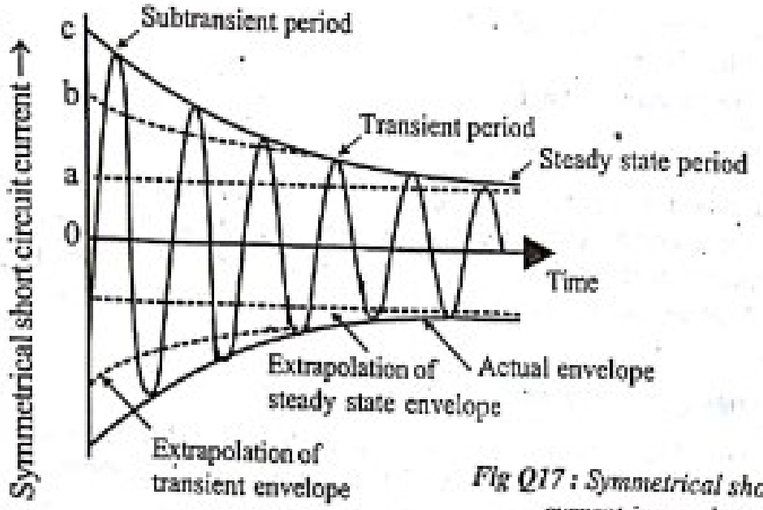


Fig Q17 : Symmetrical short circuit armature current in synchronous machine

18. *What is synchronous reactance?*

The synchronous reactance is the ratio of induced emf and the steady state rms current (i.e it is the reactance of a synchronous machine under steady state condition). It is the sum of leakage reactance and the reactance representing armature reaction.

19. *Define subtransient reactance.*

The subtransient reactance is the ratio of induced emf on no-load and the subtransient symmetrical rms current, (i.e, it is the reactance of a synchronous machine under subtransient condition).

20. *Define transient reactance.*

The transient reactance is the ratio of induced emf on no-load and the transient symmetrical rms current. (i.e, it is the reactance of a synchronous machine under transient condition).

21. *How will you determine the synchronous, transient and subtransient reactances from the oscillogram of the short circuit current.*

The oscillogram can be divided into three regions namely subtransient, transient and steady state regions. The envelopes of these regions can be extrapolated backwards to meet the y-axis as shown in fig Q21.

Subtransient symmetrical rms current, $I'' = \frac{oc}{\sqrt{2}}$

Transient symmetrical rms current, $I' = \frac{ob}{\sqrt{2}}$

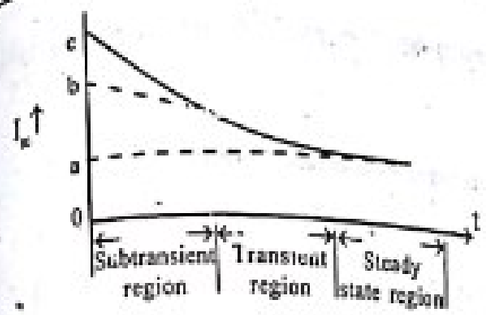


Fig Q21 : Envelope of the oscillogram of symmetrical short circuit current in an alternator.

Steady state symmetrical rms current, $I = \frac{oa}{\sqrt{2}}$

Subtransient reactance, $X''_d = \frac{E_g}{I''}$

Transient reactance, $X'_d = \frac{E_g}{I'}$

Synchronous reactance, $X_d = \frac{E_g}{I}$

22. *What is the significance of subtransient reactance in short circuit studies?*

The subtransient reactance can be used to estimate the initial value of fault current immediately on the occurence of the fault. The maximum momentary short circuit current rating of the circuit breaker used for protection or fault clearing should be less than this initial fault current.

23. *What is the significance of transient reactance in short circuit studies?*

The transient reactance is used to estimate the transient state fault current. Most of the circuit breakers open their contacts only during this period. Therefore for a circuit breaker used for fault clearing (or protection), its interrupting short circuit current rating should be less than the transient fault current.

24. *Distinguish between the transient and subtransient reactances of a synchronous machine.*

The transient reactance is the ratio of induced emf and transient current. But the subtransient reactance is the ratio of induced emf and subtransient current.

25. *Why the armature current decreases when the flux diminishes?*

The voltage generated by the air-gap flux determines the current which will flow through the resistance and leakage reactance of the armature winding. When the flux diminishes, the generated emf decreases and so the current decreases.

26. *Write down the equation determining fault current in a generator when its reactance are known?*

$$|I| = \frac{|E_g|}{X_d} \qquad |I'| = \frac{|E_g|}{X'_d}$$

Where, $|I|$ = Steady state symmetrical fault current
$|I'|$ = Transient symmetrical fault current
X_d = Direct axis synchronous reactance
X'_d = Direct axis transient reactance
$|E_g|$ = RMS voltage from one terminal to neutral at no load

27. *Write the equation to find the subtransient current?*

$$|I''| = \frac{|E_g|}{X''_d}$$

$|I''|$ = Subtransient symmetrical fault current
$|E_g|$ = RMS voltage from one terminal to neutral at no load
X''_d = Direct axis subtransient reactance

28. *Given one application of subtransient reactance?*

Subtransient reactance of generators and motors are used to determine the initial current flowing on the occurrence of the short circuits.

29. *Write the equation to determine subtransient internal voltage of the generator?*

$$E''_g = V_t + j I_L X''_d$$

E''_g = Subtransient internal voltage ; I_L = Load current
V_t = Terminal voltage ; X''_d = Direct axis subtransient reactance

30. *Write the equation to determine transient internal voltage of the generator?*

$$E'_g = V_t + j I_L X'_d$$

E'_g = Transient internal voltage ; I_L = Load current
V_t = Terminal voltage ; X'_d = Direct axis transient reactance

31. *Write equation for subtransient internal voltage and transient internal voltage of the motor.*

$$E''_m = V_t - j I_L X''_d$$
$$E'_m = V_t - j I_L X'_d$$

E''_g = Subtransient internal voltage ; I_L = Load current
E'_g = Transient internal voltage ; X'_d = Direct axis transient reactance
V_t = Terminal voltage ; X''_d = Direct axis subtransient reactance

32. *A motor is drawing 10,000 kW at 0.85 pf leading and a terminal voltage of 12 kV. Determine the load current.*

$$\text{Load current, } I_L = \frac{10{,}000}{\sqrt{3} \times 0.85 \times 12} \angle \cos^{-1} 0.85 = 566 \angle 31.8° \text{ amps}$$

33. *In fig.Q33 find the fault current if the prefault voltage at the fault point is 0.97 p.u.*

Solution

$$Z_{th} = j\frac{(0.15 + 0.2) \times 0.15}{(0.15 + 0.2) + 0.15} = j0.105 \text{ p.u.}$$

$$\text{Fault current, } I_f = \frac{V_{pf}}{Z_{th}} = \frac{0.97}{j0.105} = -j9.238 \text{ p.u}$$

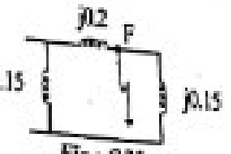


Fig : Q33

34. *The generator emf is 1 p.u. and the subtransient reactance is 20%. Find the subtransient current.*

Solution

Given that $E_g = 1$ p.u ; $X''_d = j0.2$

$$\therefore \text{Subtransient current } I'' = \frac{E_g}{X''_d} = \frac{1}{j0.2} = -j5 \text{ p.u.} = 5\angle -90° \text{ p.u.}$$

35. *Generator emf is 1 p.u and the transient reactance is 25%. Find the transient current.*

Solution

Given that $E_g = 1$ p.u ; $X'_d = j0.25$

$$\therefore \text{Transient current, } I' = \frac{E_g}{X'_d} = \frac{1}{j0.25} = -j4 \text{ p.u.} = 4\angle -90° \text{ p.u.}$$

36. *E_g = 1p.u and X_d = 40%. Find the short circuit current.*

Solution

$$\text{Short circuit current } I = \frac{E_g}{X_d} = \frac{1}{j0.4} = -j2.5 \text{ p.u.} = 2.5\angle -90° \text{ p.u.}$$

37. *When a synchronous machine is suddenly short circuit, what is the maximum possible instantaneous current in terms of symmetrical short circuit current?*

The initial short circuit current will have its maximum possible value if the fault occurs when the voltage wave is going through zero. This maximum possible value will be double that of maximum symmetrical short circuit current.

38. *How symmetrical faults are analyzed?*

The symmetrical faults are analysed using per unit reactance diagram of the power system. Once the reactance diagram of the power system is formed then the fault is simulated by short circuit (or by connecting the fault impedance at the fault point). The currents and voltages at various parts of the system can be estimated by any

one of the following method.

i) Using kirchoff's laws

ii) Using thevenin's theorem

iii) by forming bus impedance matrix.

39. *Write the equation to find the fault current in bus-k and change in voltages in other buses due to a 3-phase fault in bus-k using bus impedance matrix.*

The fault current in bus-k, $I_f = \frac{V_{pf}}{Z_{kk}}$

Where V_{pf} = prefault voltage in bus-k, (normally 1 p.u.)

$$\left.\begin{matrix}\text{The change in bus-q voltage} \\ \text{due to a 3-phase fault in bus-k}\end{matrix}\right\} \Delta V_q = -I_f Z_{qk} \quad ; \quad \text{for } q = 1, 2 \ldots n$$

40. *What is a circuit breaker?*

The circuit breaker is a protective device which is used in power system to automatically open the faulty part of the system in the event of a fault. In normal working condition if can be used as a switch.

41. *What are the two main functions of circuit breaker?*

The two main functions of circuit breaker are

i) To act as switch for normal load condition

ii) To automatically isolate or open the faulty part in the event of a fault.

42. *What are the main factors to be considered to select a circuit breaker?*

The choice of a circuit breaker for particular application depends on the following ratings of the circuit breaker

i) Normal working power level specified as rated interrupting current or rated interrupting MVA.

ii) The fault level specified as either rated short circuit interrupting current or rated short circuit interrupting MVA.

iii) Momentary current rating

iv) Normal working voltage

v) Speed of circuit breaker.

43. *Why the circuit breaker interrupting current is asymmetrical?*

The interrupting current of circuit breaker is the sum of symmetrical short circuit current and dc-offset current. The presence of dc-offset current makes the interrupting current as asymmetric.

44. *Define the speed of circuit breaker. How it is specified?*

The speed of circuit breaker is defined as the time between the occurrence of the fault to the extinction of the arc (when the contact opens).

The speed is normally specified in cycles of power frequency (1 cycle for 50Hz power frequency is $^1/_{50}$ =0.02 ms). The standard speed of circuit breakers are 8,5,3 or $1^1/_2$ cycles.

45. *What is momentary current rating of circuit breaker? How it is estimated?*

The momentary current rating is the maximum current that may flow through a circuit breaker for a short duration. It is estimated by multiplying the symmetrical subtransient fault current by a factor of 1.6 (The factor 1.6 accounts for dc-offset current during subtransient period).

46. *What is interrupting short circuit current rating of circuit breaker? How it is estimated?*

The interrupting short circuit current rating of the circuit breaker is the maximum current that may flow through it when its contacts open due to a fault.

It is estimated by multiplying the transient short circuit current by a factor 1.0 to 1.5. The value of the factor depends on the speed of the breaker. (The factor 1.0 to 1.5 accounts for dc-offset current during transient period.)

47. *Define short circuit interrupting MVA of a circuit breaker.*

The short circuit interrupting MVA of a circuit breaker is the volt-amperes (power) flowing through it at the moment of opening its contacts due to a fault.

It is estimated by the following equations

Short circuit interrupting MVA $= \sqrt{3}|V_{pL}||I_{fL}|$ or

Short circuit interrupting MVA $= |V_{pfL_pu}||I_{fL_pu}| \times MVA_b$

where, $|V_{pfL}|$ = Magnitude of line voltage at the fault point in kV.

$|I_{fL}|$ = Magnitude of line value of short circuit interrupting current at the fault in kA.

$|V_{pfL_pu}|$ = Magnitude of prefault voltage at the fault point in p.u.

$|I_{fL_pu}|$ = Magnitude of short circuit interrupting current at the fault in p.u.

48. *Find the momentary current through the circuit breaker if the initial symmetrical short circuit current through it is 5270.9 A*

Solution

The momentary current = 1.6× Initial symmetrical short circuit current

= 1.6 ×5270.9 A = 8433.44 A

= 8.433 kA

49. *The thevenin's impedance and voltage at a fault point is 0.576∠84°pu and 1∠0°pu respectively. Determine the short circuit MVA for a base of 30 MVA, 11 kV.*

Solution

Fault current in pu, $I_f = \frac{V_{th}}{Z_{th}} = \frac{1\angle 0^\circ}{0.576\angle 84^\circ} = 1.736\angle -84^\circ$ pu.

Base current, $I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{30\times 1000}{\sqrt{3}\times 11} = 1574.6$ A

Line value of the magnitude of fault current in amperes $= |I_{fL}| = |I_{fL}$ in p.u.$| \times I_b$.

$= 1.736 \times 1574.6 = 2733.5\text{A} = 2.7335$ kA.

Short circuit MVA $= \sqrt{3}\times|V_{pfL}|\times|I_{fL}| = \sqrt{3}\times 11\times 2.7835 = 52.08$ MVA.

50. *Determine the interrupting current of 3-cycles circuit breaker employed in a generator rated at 20 MVA, 33kV. Take X_d'=25%, E_g=1pu.*

Solution

Transient symmetrical rms current, $I'' = \frac{E_g}{X_d''} = \frac{1}{j0.25} = -j4\text{pu} = 4\angle -90^\circ$ pu.

Interrupting current in pu $= 1.2 \times |I''| = 1.2 \times 4 = 4.8$ p.u

(i.e, for 3 -cycles breaker the multiplication factor is 1.2)

Base current, $I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{20\times 10^3}{\sqrt{3}\times 33} = 350$ A

$\therefore$ Interrupting current in kA $= \frac{4.8\times 350}{1000} = 1.68$ kA

3.9 EXERCISES

I. State whether the following statements are TRUE/FALSE

1. The fault current is symmetrical only in 3-phase faults.
2. In power systems 70 to 80% faults are single line-to-ground faults.
3. The series faults are associated with heavy current flow.
4. The selection of circuit breaker does not depends on fault level.
5. The estimation of various fault currents are commonly referred as fault calculation.
6. The transients during a short circuit are due to capacitance of the components of power system.

7. The initial value of short circuit current is very low if the fault occurs when the voltage wave is going through zero.
8. The resistance at the ends of feeders reduces the degree of asymmetry.
9. If flux diminishes the armature current will decrease.
10. The initial value of short circuit current in 3-phase fault is always symmetrical.
11. The speed of circuit breaker is specified in cycles of power frequency.
12. The circuit breakers will clear the fault in steady state short circuit period.

Answers

1. True	4. False	7. False	10. False
2. True	5. True	8. True	11. True
3. False	6. False	9. True	12. False

II. Fill in the blanks

1. A ..... in a circuit is any failure which interferes with the normal current.
2. Based on the nature fault currents the faults are classified into ..... and ..... faults.
3. The ..... faults are due to short circuits.
4. The ..... faults are due to open conductors.
5. In ..... fault the fault currents are equal in all the phases.
6. In ..... fault the fault currents are not equal in all the phase.
7. The ..... faults are the most severe faults.
8. The ..... faults are the most common and frequently occuring faults.
9. The presence of ..... current makes the short circuit current asymmetrical.
10. The rms value of the current determined by the intercept of the short circuit current envelope with zero time is called ..... .
11. The ..... reactance is the ratio of induced emf and the steady state rms current.
12. The ..... reactance can be used to estimate the initial value of fault current.
13. The ..... reactance is used to estimate the interrupting current rating of circuit breaker.
14. The circuit breakers will open its contacts in the transient period of short circuit.
15. The momentary current rating of circuit breaker is obtained by multiplying the initial symmetrical rms current by ..... .

Answers

1. fault	6. unsymmetrical	11. synchronous	
2. symmetrical, unsymmetrical	7. 3-phase	12. subtransient	
3. shunt	8. single line-to-ground	13. transient	
4. series	9. dc-offset	14. transient	
5. symmetrical	10. subtransient	15. 1.6	

M. Unsolved problems

1. For the system shows in fig E3.1, find the fault current in all the branches where a 3-phase fault occurs at bus-3. All reactance are in pu. Neglect prefault current.

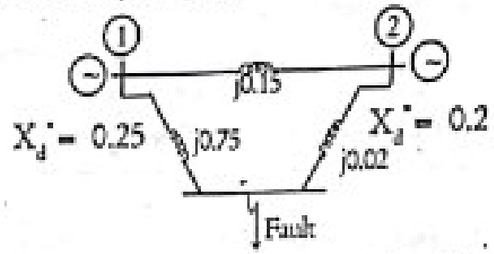


Fig E 3.1

2. The system shown in fig E3.2 is initially at no-load. Calculate the subtransient fault current for a 3-phase fault at point F. Take base values as 120 MVA and 69kV.

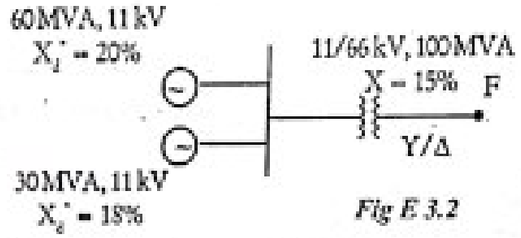


Fig E 3.2

3. A 3-phase transmission line operating at 110kV and having an impedance 5+j20Ω is connected to the generating station through 15,000 kVA step-up transformer. Two alternator are connected to the bus bars. The ratings of the alternator are 10 MVA, 10%, 16kV and 5MVA, 7.5%, 16kV. Calculate the short circuit MVA for a symmetrical fault at the load end of transmission line. Take transformer reactance as 8%.

4. A generator rated at 500 kVA, 440V, with 12% subtransient reactance is supplying a passive load of 400 kW at 0.8 pf lagging. Calculate the initial rms current for a 3ϕ fault at the generator terminals.

5. The bus impedance matrix of a system is given by

$$Z_{bus} = j\begin{bmatrix} 0.14 & 0.06 & 0.08 & 0.09 \\ 0.06 & 0.15 & 0.07 & 0.09 \\ 0.08 & 0.07 & 0.13 & 0.10 \\ 0.09 & 0.09 & 0.10 & 0.15 \end{bmatrix}$$

Generator are connected to buses 1 and 3 and their subtransient reactances were included when finding bus impedance matrix. Find the fault current for a 3-phase fault in bus-3. Also calculate the voltages in all the buses. Find the subtransient fault current in generator at bus-1 if its subtransient reactance is 0.20 pu.

6. Repeat the exercise problem-5 for a 3-phase fault in bus-2.

7. A generator is connected through a circuit breaker to a transformer. The rating of the generator is 40MVA, 12.4kV. Given that $X_d'' = 14\%$, $X_d' = 22\%$ and $X_0 = 90\%$. The transformer is rated at 40 MVA, 13/110kV, with a reactance of 10%. For a three phase fault on the HT side of the transformer determine the ratings of circuit breaker.

UNSYMMETRICAL FAULT ANALYSIS



4.1 INTRODUCTION

The unsymmetrical faults are the faults in which the fault currents in the three phases are unequal. The different types of unsymmetrical faults are

i) Single line-to-ground fault
ii) Line-to-line fault
iii) Double line-to-ground fault
iv) One or two open conductor faults..

Since any unsymmetrical fault causes unbalanced currents to flow in the system, the unsymmetrical faults are analyzed using symmetrical components. The concepts of symmetrical components and sequence networks developed in chapter-1 will be used for the analysis of unsymmetrical faults.

Review of symmetrical components of unbalanced voltages and currents

Let V_a, V_b, V_c be a set of three phase unbalanced voltage vectors.

Let V_{a1}, V_{b1}, V_{c1} = Positive sequence components of voltages
V_{a2}, V_{b2}, V_{c2} = Negative sequence components of voltages
V_{a0}, V_{b0}, V_{c0} = Zero sequence components of voltages

Now the unbalanced voltage vectors are related to sequence component voltages by the following equations.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} \quad \text{.....(4.1)}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad \text{.....(4.2)}$$

$$V_{a0} = V_{b0} = V_{c0} \quad \text{.....(4.3)}$$

$$V_{b1} = a^2 V_{a1} \quad \& \quad V_{c1} = a V_{a1} \quad \text{.....(4.4)}$$

$$V_{b2} = a V_{a2} \quad \& \quad V_{c2} = a^2 V_{a2} \quad \text{.....(4.5)}$$

$$a = 1\angle 120° \quad \& \quad a^2 = 1\angle 240° \quad \text{.....(4.6)}$$

Let I_a, I_b, I_c be a set of three phase unbalanced current vectors.

Let I_{a1}, I_{b1}, I_{c1} = Positive sequence components of currents

I_{a2}, I_{b2}, I_{c2} = Negative sequence components of currents

I_{a0}, I_{b0}, I_{c0} = Zero sequence components of currents

Now the unbalanced current vectors are related to sequence component currents by the following equations.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.7)}$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.8)}$$

$$I_{a2} = I_{b2} = I_{c2} \quad \text{.....(4.9)}$$

$$I_{b1} = a^2 I_{a1} \quad \& \quad I_{c1} = a I_{a1} \quad \text{.....(4.10)}$$

$$I_{b2} = a I_{a2} \quad \& \quad I_{c2} = a^2 I_{a2} \quad \text{.....(4.11)}$$

$$a = 1\angle 120° \quad \& \quad a^2 = 1\angle 240° \quad \text{.....(4.12)}$$

Review of sequence networks of a generator

Let Z_1 = Positive sequence impedance of generator.

Z_2 = Negative sequence impedance of generator.

Z_{g0} = Zero sequence impedance of generator.

Z_0 = Zero sequence impedance of generator including neutral reactance.

The positive, negative and zero sequence networks of a generator with neutral grounded through an impedance Z_n are shown in fig 4.1., 4.2. and 4.3 respectively.

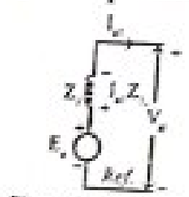


Fig 4.1 : Positive sequence network of a generator

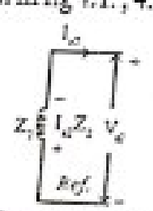


Fig 4.2 : Negative sequence network of a generator

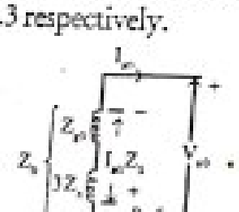


Fig 4.3 : Zero sequence network of a generator

From the sequence networks shown above, using kirchoff's laws we can write the following equations

$$V_{a0} = -I_{a0} Z_0 \quad \text{.....(4.13)}$$

$$V_{a1} = E_a - I_{a1} Z_1 \quad \text{.....(4.14)}$$

$$V_{a2} = -I_{a2} Z_2 \quad \text{.....(4.15)}$$

The equations (4.13), (4.14) and (4.15) can be arranged in the form of matrix equation as shown below.

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.16)}$$

The matrix equation(4.16) will be useful in the analysis of unsymmetrical faults on the generator.

4.2 SINGLE LINE-TO-GROUND FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a single line-to-ground fault on an unloaded Y-connected generator with its neutral grounded through a reactance is shown in fig 4.4. Here it is assumed that phase-a is shorted to ground. Now the fault current, $I_f = I_a$. Since the generator is unloaded the current in other phases are zero.

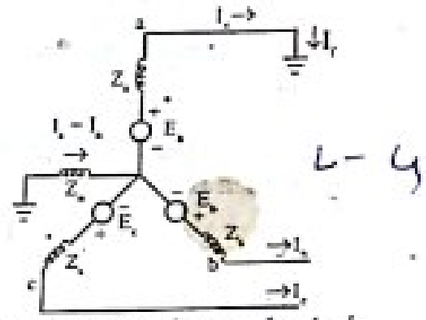


Fig 4.4 : Circuit diagram for single line-to-ground fault on phase-a of an unloaded generator

The condition at the fault is expressed by the following equations

$$I_b = 0 \; ; \; I_c = 0 \; ; \; V_a = 0 \quad \text{.....(4.17)}$$

The symmetrical components of the currents are given by

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.18)}$$

On substituting the conditions $I_b = I_c = 0$ in the symmetrical components of currents (equ 4.18) we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} \quad \text{.....(4.19)}$$

On multiplying the above matrix equation (4.19) we get,

$$I_{a0} = I_{a1} = I_{a2} = \frac{I_a}{3} \quad \text{.....(4.20)}$$

From the sequence networks of the generator we get the following matrix equation [Refer equ(4.16)].

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.21)}$$

On substituting $I_{a0} = I_{a1}$ and $I_{a2} = I_{a1}$ in equ(4.21) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a1} \\ I_{a1} \\ I_{a1} \end{bmatrix} \quad \text{.....(4.22)}$$

On multiplying equ (4.22) we get,

$$V_{a0} = -Z_0 I_{a1} \quad \text{.....(4.23)}$$

$$V_{a1} = E_a - Z_1 I_{a1} \quad \text{.....(4.24)}$$

$$V_{a2} = -Z_2 I_{a1} \quad \text{.....(4.25)}$$

On adding equations (4.23), (4.24) and (4.25) we get,

$$V_{a0} + V_{a1} + V_{a2} = -I_{a1} Z_0 + E_a - I_{a1} Z_1 - I_{a1} Z_2 \quad \text{.....(4.26)}$$

Since phase a is shorted to ground, $V_a = V_{a0} + V_{a1} + V_{a2} = 0$ and so the equ(4.26) can be written as

$$-I_{a1} Z_0 + E_a - I_{a1} Z_1 - I_{a1} Z_2 = 0$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0} \quad \text{.....(4.27)}$$

Using equ (4.27) the equivalent circuit of generator during single line-to-ground fault is drawn as shown in fig 4.5. Here the positive, negative and zero sequence networks of the generator are connected in series.

If the neutral of the generator is not grounded, the zero sequence network is open-circuited and Z_0 is infinite. Under this condition I_{a1} is zero and so I_{a2} & I_{a0} must be zero. Therefore no path exists for the flow of current in the fault unless the generator neutral is grounded.

Here the fault current is given by,

$$I_f = I_a = 3 I_{a1} \quad \text{.....(4.28)}$$

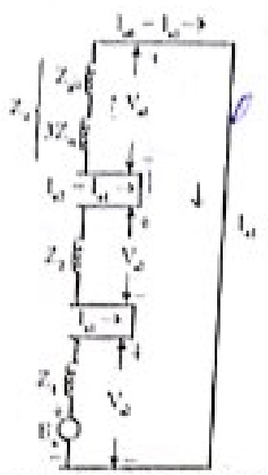


Fig 4.5 : Connection of sequence network of an unloaded generator for line-to-ground fault on phase -a

4.3 LINE-TO-LINE FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a line to line fault on an unloaded, Y-connected generator is shown in fig 4.6a. Here it is assumed that phase-b and phase c are shorted. Now the fault current $I_b = -I_c$. Since the generator is unloaded the current in phase a is zero.

The conditions at the fault are expressed by the following equations

$$V_b = V_c \; ; \; I_a = 0 \; ;$$

$$I_b + I_c = 0 \Rightarrow I_b = -I_c \quad \text{.....(4.29)}$$

With $V_c = V_b$, the symmetrical components of voltage are given by

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_b \end{bmatrix} \quad \text{.....(4.30)}$$

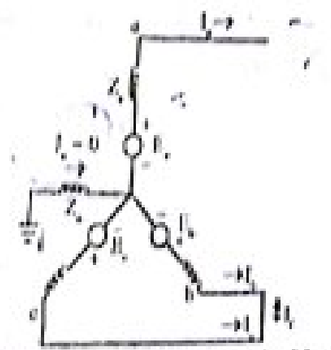


Fig 4.6 : Circuit diagram for double line to ground fault between phase b & c in an unloaded generator

On multiplying row 2 & 3 of equ(4.30) we get,

$$V_{a1} = \frac{1}{3} (V_a + aV_b + a^2 V_b) \quad \text{.....(4.31)}$$

$$V_{a2} = \frac{1}{3} (V_a + a^2 V_b + aV_b) \quad \text{.....(4.32)}$$

From equations (4.31) and (4.32) we can write

$$V_{a1} = V_{a2} \quad \text{.....(4.33)}$$

The symmetrical components of current are given by

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.34)}$$

On substituting $I_b = -I_c$ and $I_a = 0$ in equ(4.34) we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 0 \\ -I_c \\ I_c \end{bmatrix} \quad \text{.....(4.35)}$$

On multiplying equ(4.35) we get,

$$I_{a0} = \frac{1}{3} [-I_c + I_c] = 0 \quad \text{.....(4.36)}$$

$$I_{a1} = \frac{1}{3} [-a I_c + a^2 I_c]$$

$$I_{a2} = \frac{1}{3}\left[-a^2 I_a + a I_c\right] = -\frac{1}{3}\left[-a I_a + a^2 I_c\right] \quad \text{....(4.37)}$$

From equations (4.36) and (4.37) we can write

$$I_{a2} = -I_{a1} \quad \text{....(4.38)}$$

From the sequence networks of the generator we get the following matrix equations. [Refer equ(4.16)].

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{....(4.39)}$$

On substituting $I_{a0} = 0$ and $I_{a2} = -I_{a1}$ in equ (4.39) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} 0 \\ I_{a1} \\ -I_{a1} \end{bmatrix} \quad \text{....(4.40)}$$

On multiplying equ (4.40) we get,

$$V_{a0} = 0$$

$$V_{a1} = E_a - Z_1 I_{a1} \quad \text{....(4.41)}$$

$$V_{a2} = Z_2 I_{a1} \quad \text{....(4.42)}$$

From equ(4.33) we know that $V_{a1} = V_{a2}$. Hence, on equating (4.41) and (4.42) we get,

$$E_a - Z_1 I_{a1} = Z_2 I_{a1}$$

(or) $$I_{a1}(Z_1 + Z_2) = E_a$$

$$\therefore I_{a1} = \frac{E_a}{Z_1 + Z_2} \quad \text{....(4.43)}$$

Using equ (4.43), the equivalent circuit of generator during single line to ground fault is drawn as shown in fig (4.7).

Since $V_{a1} = V_{a2}$, here the positive and negative sequence networks of the generator must be in parallel. Since $V_{a0} = 0$, the zero sequence network is shorted and so it need not be considered. (or Since Z_n does not enter into the equations, the zero-sequence network is not used). Since this type of fault does not involve ground, the neutral current $I_n = 0$. Hence the presence or absence of a grounded neutral at the generator does not affect the fault current.

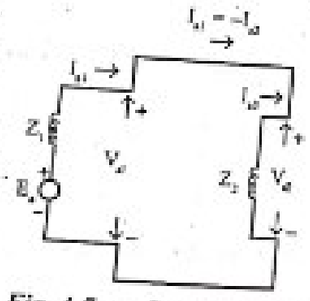


Fig 4.7 : Connection of the sequence networks of an unloaded generator for a line-to-line fault between phases b and c

Here the fault current, $I_f = I_b = -I_c$

The unbalanced currents I_a, I_b and I_c are related to the symmetrical components of currents by the following matrix equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{....(4.44)}$$

From equation (4.44) we get,

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2} \quad \text{....(4.45)}$$

On substituting the conditions $I_{a0} = 0$ and $I_{a2} = -I_{a1}$ which exists during line-to-line fault in equ(4.45) we get,

$$I_b = a^2 I_{a1} - a I_{a1} = I_{a1}(a^2 - a)$$

$$\therefore \text{Fault current, } I_f = I_b = I_{a1}(a^2 - a) \quad \text{....(4.46)}$$

4.4 DOUBLE LINE-TO -GROUND FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a double line-to-ground fault on an unloaded Y connected generator having a grounded neutral is shown in fig 4.8. The faulted phases are b and c. Now the fault current, $I_f = I_b + I_c$. Since the generator is unloaded the current in phase-a is zero.

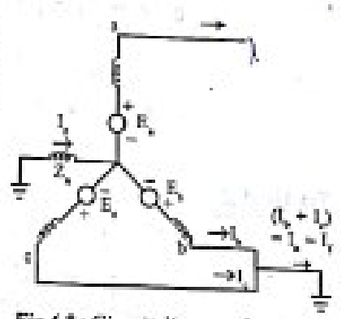


Fig 4.8 : Circuit diagram for double line-to-ground fault between phase b & c in an unloaded generator

The conditions at the fault are expressed by the following equations

$$V_b = 0, \quad V_c = 0, \quad I_a = 0.$$

With $V_b = V_c = 0$, the symmetrical components of voltage are given by

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ 0 \\ 0 \end{bmatrix} \quad \text{....(4.47)}$$

On multiplying equ(4.47) we get,

$$V_{a0} = V_{a1} = V_{a2} = \frac{V_a}{3} \quad \text{....(4.48)}$$

e networks of the generator we get the following matrix equation.

$$-\begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \qquad \text{.....(4.49)}$$

row-2 of equ(4.49) we get,

$$Z_1 \qquad \text{.....(4.50)}$$

et $V_{a0} = V_{a1} = V_{a2}$ and from equ(4.50) we get $V_{a1} = E_a - I_{a1} Z_1$. On s in equ(4.49) we get,

$$\begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \qquad \text{.....(4.51)}$$

u(4.51) we get,

$$\begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} E_a - I_{a1} Z_1 \\ E_a - I_{a1} Z_1 \\ E_a - I_{a1} Z_1 \end{bmatrix} \qquad \text{.....(4.52)}$$

$$\therefore Z^{-1} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix}$$

$$\frac{Z}{fZ} = \frac{Z_{cof}^{T}}{\Delta Z}$$

$$Z_0 Z_1 Z_2$$

$$\begin{bmatrix} 0 \\ 0 \\ Z_0 Z_1 \end{bmatrix}^T = \begin{bmatrix} Z_1 Z_2 & 0 & 0 \\ 0 & Z_0 Z_2 & 0 \\ 0 & 0 & Z_0 Z_1 \end{bmatrix}$$

$$\begin{bmatrix} Z_2 & 0 & 0 \\ 0 & Z_0 Z_2 & 0 \\ 0 & 0 & Z_0 Z_1 \end{bmatrix} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix}$$

On premultiplying the equ (4.52) by Z^{-1} we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix}\begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} 1/Z_0 & 0 \\ 0 & 1/Z \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{E_a}{Z_1} \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{E_a - I_{a1}Z_1}{Z_0} \\ \frac{E_a - I_{a1}Z_1}{Z_1} \\ \frac{E_a - I_{a1}Z_1}{Z_2} \end{bmatrix}$$

On multiplying equ(4.53) we get

$$I_{a0} = -\left(\frac{E_a - I_{a1}Z_1}{Z_0}\right) = -\frac{E_a}{Z_0} + \frac{I_{a1}Z_1}{Z_0}$$

$$I_{a1} = \frac{E_a}{Z_1} - \left(\frac{E_a - I_{a1}Z_1}{Z_1}\right) = \frac{E_a}{Z_1} - \frac{E_a}{Z_1} + \frac{I_{a1}Z_1}{Z_1} = I_{a1}$$

$$I_{a2} = -\left(\frac{E_a - I_{a1}Z_1}{Z_2}\right) = -\frac{E_a}{Z_2} + \frac{I_{a1}Z_1}{Z_2}$$

Here $I_a = 0$, On expressing I_a by its symmetrical compo

$$I_a = I_{a0} + I_{a1} + I_{a2} = 0$$

On substituting for I_{a0}, I_{a1} and I_{a2} from equations (4.54), (4.5

$$-\frac{E_a}{Z_0} + \frac{I_{a1}Z_1}{Z_0} + I_{a1} - \frac{E_a}{Z_2} + \frac{I_{a1}Z_1}{Z_2} = 0$$

$$I_{a1}\left(\frac{Z_1}{Z_0} + 1 + \frac{Z_1}{Z_2}\right) = \frac{E_a}{Z_0} + \frac{E_a}{Z_2}$$

$$I_{a1}\left(1 + Z_1\left(\frac{1}{Z_0} + \frac{1}{Z_2}\right)\right) = E_a\left(\frac{1}{Z_0} + \frac{1}{Z_2}\right)$$

$$I_{a1}\left(1 + Z_1\left(\frac{Z_0 + Z_2}{Z_0 Z_2}\right)\right) = E_a\left(\frac{Z_0 + Z_2}{Z_0 Z_2}\right)$$

On multiplying the equ(4.58) throughout by $\frac{Z_0 Z_2}{Z_0 + Z_2}$ we g

$$I_{a1}\left(\frac{Z_0 Z_2}{Z_0 + Z_2} + Z_1\right) = E_a$$

$$\therefore I_{a1} = \frac{E_a}{Z_1 + \frac{Z_0 Z_2}{Z_0 + Z_2}} \qquad \text{....(4.59)}$$

Using equ(4.59) the equivalent circuit of generator under double line-to-ground fault can be drawn as shown in fig 4.9.

Under this fault conditon the sequence networks should be connected in parallel since the positive, negative, and zero-sequence voltages are equal during this fault. In the absence of a ground connection at the generator no current can flow into the ground at the fault. In this case Z_0 would be infinite and I_{a0} would be zero and so the fault will be similar to line-to-line fault.

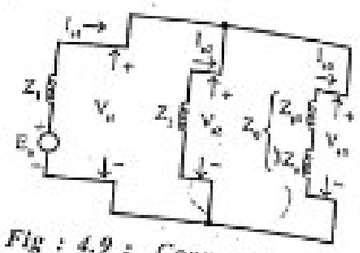


Fig : 4.9 : Connection of the sequence networks of an unloaded generator for a double line-to-ground fault on phase b and c.

Here the fault current, $I_f = I_b + I_c$ —(4.60)

To determine the fault current, first the symmetrical components I_{a1}, I_{a0} and I_{a2} are calculated using equations (4.59), (4.54) and (4.56). These equations are presented here for convenience

$$I_{a1} = \frac{E_a}{Z_1 + \frac{Z_0 Z_2}{Z_0 + Z_2}} \quad ; \quad I_{a0} = -\frac{E_a}{Z_0} + \frac{I_{a1} Z_2}{Z_0} \quad ; \quad I_{a2} = -\frac{E_a}{Z_2} + \frac{I_{a1} Z_1}{Z_2}$$

Alternatively the I_{a1} can be calculated using equ(4.59) and then I_{a2} & I_{a0} can be calculated using current division rule as shown below

$$I_{a2} = -I_{a1} \times \frac{Z_0}{Z_0 + Z_2} \;\&\; I_{a0} = -(I_{a1} + I_{a2})$$

The unbalanced currents I_a, I_b and I_c are related to the symmetrical components of currents by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \qquad \text{....(4.61)}$$

From equation (4.61) we get,

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$\therefore \text{Fault current, } I_f = I_b + I_c = I_{a0} + a^2 I_{a1} + a I_{a2} + I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$= 2I_{a0} + (a + a^2) I_{a1} + (a + a^2) I_{a2}$$

$$= 2I_{a0} + (a + a^2)(I_{a1} + I_{a2}) \qquad \text{....(4.62)}$$

4.5 UNSYMMETRICAL FAULTS ON POWER SYSTEMS

The unsymmetrical faults on the power system are analyzed using thevenin's equivalent of positive, negative and zero sequence networks of the power system. The thevenin's equivalent is obtained with respect to fault point and reference.

In power system the prefault voltages and currents are assumed to be balanced. Hence the prefault voltage at the fault point is a balanced voltage and so it has only positive sequence component. The negative and zero sequence component of prefault voltage at the fault point is zero. Therefore the prefault voltage at the fault point is the thevenin's voltage of positive sequence network.

Let Z_1 = Thevenin's impedance of positive sequence network.
Z_2 = Thevenin's impedance of negative sequence network.
Z_0 = Thevenin's impedance of zero sequence network.
V_{pf} = Prefault voltage at the fault point.

Thevenin's equivalent of positive, negative and zero sequence networks of the power system with respect to fault point will be as shown in fig (4.10), (4.11) and (4.12) respectively.

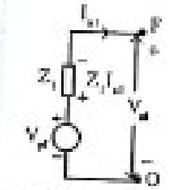


Fig 4.10 : Thevenin's equivalent of positive sequence network

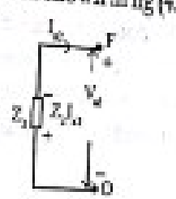


Fig 4.11 : Thevenin's equivalent of negative sequence network

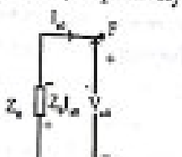


Fig 4.12 : Thevenin's equivalent of zero sequence network

From fig (4.10), (4.11) and (4.12) using kirchoff's voltage law we get,

$$V_{a0} = -Z_0 I_{a0} \qquad \text{....(4.63)}$$

$$V_{a1} = V_{pf} - Z_1 I_{a1} \qquad \text{....(4.64)}$$

$$V_{a2} = -Z_2 I_{a2} \qquad \text{....(4.65)}$$

On arranging the equations in the matrix form we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ V_{pf} \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \qquad \text{....(4.66)}$$

The matrix equation (4.66) will be useful in the analysis of unsymmetrical faults on the power system.

Single line - to Ground Fault

A single line-to-ground fault at point F in a power system through a fault impedance Z_f can be represented by connecting three stubs as shown in fig 4.19. The fault is on phase a.

At the fault point F, the following relations exist

$$I_b = 0; \quad I_c = 0; \quad V_a = Z_f I_a \qquad \text{.....(4.76)}$$

The symmetrical components of the fault current are

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore I_{a1} = I_{a2} = I_{a0} = \frac{1}{3}I_a \qquad \text{..... (4.77)}$$

(or) $I_a = 3I_{a1}$ ..... (4.78)

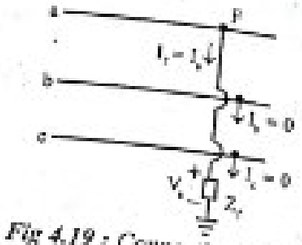


Fig 4.19 : Connection diagram of stubs for a single line-to ground fault through an impedance

Expressing the voltage V_a in terms of symmetrical components, we have

$$V_a = V_{a1} + V_{a2} + V_{a0} \qquad \text{....(4.79)}$$

Using equations (4.76) ,(4.78) and (4.79) we can write

$$V_{a1} + V_{a2} + V_{a0} = Z_f I_a = 3Z_f I_{a1} \qquad \text{.....(4.80)}$$

From equations (4.77) and (4.80) we can say that all sequence currents are equal and the sum of sequence voltages equals $3Z_f I_{a1}$. Therefore, these equations suggest a series connection of sequence networks through an impedance $3Z_f$ as shown in fig 4.20.

From fig 4.20 we can write

$$I_{a1} = \frac{V_{pf}}{(Z_1 + Z_2 + Z_0) + 3Z_f} \qquad \text{.....(4.81)}$$

Fault current, $I_f = I_a = 3I_{a1} = \dfrac{3V_{pf}}{(Z_1 + Z_2 + Z_0) + 3Z_f}$ ....(4.82)

The symmetrical components of phase-a voltage are calculated by writing kirchoff's voltage law equations in the sequence networks of fig 4.20 as shown below.

$$V_{a1} = V_{pf} - Z_1 I_{a1}$$
$$V_{a2} = -Z_2 I_{a1}$$
$$V_{a0} = -Z_0 I_{a1}$$

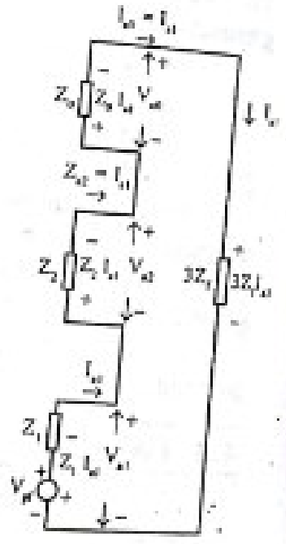


Fig 4.20 Connection of sequence networks for a single line to ground fault through an impedance Z_f

From the symmetrical components of phase-a voltage, the phase voltages V_a, V_b and V_c are calculated using the following matrix equation.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

Line - to - Line Fault

A line-to-line fault at point F in a power system between phases b and c through a fault impedance Z_f can be represented by connecting three stubs as shown in fig 4.21. The currents and voltages at the fault can be expressed as

$$I_a = 0 \text{ and } I_b = -I_c \qquad \text{.....(4.83)}$$

Also $V_b - V_c = I_b Z_f$; $\therefore V_c = V_b - I_b Z_f$ .....(4.84)

The symmetrical components of the currents after substituting $I_a = 0$ and $I_c = -I_b$ we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 0 \\ I_b \\ -I_b \end{bmatrix} \qquad \text{.....(4.85)}$$

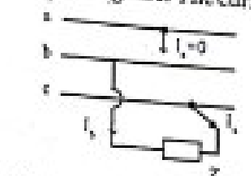


Fig 4.21 : Connection diagram of stubs for a line-to-line fault through an impedance

On multiplying equ(4.85) we get,

$$I_{a0} = \frac{1}{3}(I_b - I_b) = 0$$

$$I_{a1} = \frac{1}{3}(aI_b - a^2 I_b)$$

$$I_{a2} = \frac{1}{3}(a^2 I_b - aI_b) = -I_{a1} \qquad \text{.....(4.86)}$$

The line currents can be obtained from the following matrix equation, after substituting $I_{a0} = 0$ and $I_{a2} = -I_{a1}$

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} 0 \\ I_{a1} \\ -I_{a1} \end{bmatrix} \qquad \text{.....(4.87)}$$

From row -2 of matrix equ(4.87) we get,

$$I_b = a^2 I_{a1} - aI_{a1} = I_{a1}(a^2 - a) = I_{a1}\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} + \frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = -j\sqrt{3}\,I_{a1} \qquad \text{.....(4.88)}$$

The symmetrical components of phase voltages are

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} \quad \text{—(4.89)}$$

On substituting $V_0 = V_a - I_0 Z_f$ in equ (4.89) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_a - I_a Z_f \end{bmatrix} \quad \text{—(4.90)}$$

From row-2 of matrix equ (4.90) we get,

$$V_{a1} = \frac{1}{3}[V_a - aV_b - a^2 V_a + a^2 I_a Z_f]$$

$$= V_a - V_b(a - a^2) - a^2 Z_f I_a \quad \text{—(4.91)}$$

From row-3 of matrix equ (4.90) we get,

$$V_{a2} = \frac{1}{3}[V_a - a^2 V_b - aV_a + aI_a Z_f]$$

$$= V_a - V_b(a^2 - a) + aZ_f I_a \quad \text{—(4.92)}$$

On subtracting equ (4.92) from equ (4.91) we get,

$$V_{a1} - V_{a2} = V_b - V_b(a - a^2) - a^2 Z_f I_a - V_b - V_b(a^2 - a) + aZ_f I_a$$

$$= (-a^2 - a)Z_f I_a$$

$$= \left(\frac{1}{2} - j\frac{\sqrt{3}}{2} - \frac{1}{2} - j\frac{\sqrt{3}}{2}\right) Z_f I_a = -j\sqrt{3} Z_f I_a$$

$$\therefore V_{a1} - V_{a2} = \frac{1}{3}(-j\sqrt{3} Z_f I_a) = -j\frac{1}{\sqrt{3}} Z_f I_a \quad \text{—(4.93)}$$

On substituting for I_a from equ (4.88) in equ(4.93) we get,

$$V_{a1} - V_{a2} = -j\frac{1}{\sqrt{3}} Z_f(-j\sqrt{3} I_{a1}) = Z_f I_{a1}$$

$$\therefore V_{a1} - V_{a2} = Z_f I_{a1} \quad \text{—(4.94)}$$

The equations (4.86) and (4.94) suggest parallel connection of positive and negative sequence networks through a series impedance Z_f. Therefore the equivalent circuit of power system under line to line fault can be drawn as shown in fig 4.22.

From the fig 4.22 we get

$$I_{a1} = \frac{V_{ff}}{Z_1 + Z_2 + Z_f} \quad \text{—(4.95)}$$

The fault current, $I_f = I_b = -j\sqrt{3}\, I_{a1}$ —(4.96)
(Refer equation 4.88)

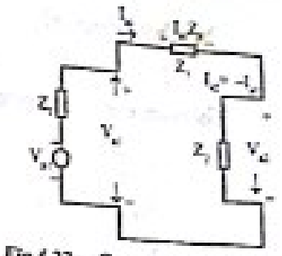


Fig 4.22 : Connection of sequence networks for a line-to-line fault through an impedance Z_f

Double Line - to - Ground Fault

A double line-to-ground fault at point F in a power system, through a fault impedance Z_f can be represented by connecting three stubs as shown in fig 4.23.

The current and voltage conditions at the fault are

$$I_a = 0 \quad \text{—(4.97)}$$

$$V_b = V_c = Z_f(I_b + I_c) \quad \text{—(4.98)}$$

The line currents are given by

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{—(4.99)}$$

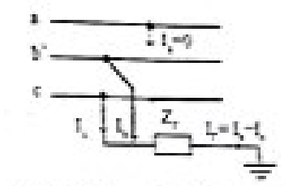


Fig 4.23 :Connection diagram of stubs for a double line-to-ground fault through an impedance

From row-1 of equ (4.99) we get,

$$I_a = I_{a0} + I_{a1} + I_{a2}$$

From equ (4.97) we get $I_a = 0$

$$\therefore I_{a0} + I_{a1} + I_{a2} = 0 \quad \text{(or)} \quad I_{a0} = -(I_{a1} + I_{a2}) \quad \text{—(4.100)}$$

From row-2 and 3 of equ (4.99) we get,

$$I_b = I_{a0} + a^2 I_{a1} + aI_{a2}$$

$$I_c = I_{a0} + aI_{a1} + a^2 I_{a2}$$

$$I_b + I_c = I_{a0} + a^2 I_{a1} + aI_{a2} + I_{a0} + aI_{a1} + a^2 I_{a2} \quad \text{—(4.101)}$$

$$\therefore \quad = 2I_{a0} + (a^2 + a)I_{a1} + (a^2 + a)I_{a2}$$

We know that, $1 + a + a^2 = 0$, $\therefore a + a^2 = -1$ —(4.102)

From equ(4.101) and (4.102) we get,

$$I_b + I_c = 2I_{a0} - I_{a1} - I_{a2} \quad \text{—(4.103)}$$

$$= 2I_{a0} - (I_{a1} + I_{a2})$$

On substituting for $I_{a1} + I_{a2}$ from equ (4.100) in equ (4.103) we get,

$$I_b + I_c = 2I_{a0} - (-I_{a0}) = 3I_{a0} \quad \text{.....(4.104)}$$

The symmetrical components of voltages after subsituting $V_c = V_b$ are given by,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_b \end{bmatrix} \quad \text{.....(4.105)}$$

From row-1 of equ 4.105 we get,

$$V_{a0} = \frac{1}{3}[V_a + V_b + V_b] = \frac{1}{3}(V_a + 2V_b)$$

From row-2 and 3 of equ(4.105) we get, ... (4.106)

$$V_{a1} = V_{a2} = \frac{1}{3}[V_a + aV_b + a^2V_b] \qquad (\because a + a^2 = -1)$$

$$= \frac{1}{3}(V_a + (a + a^2)V_b) = \frac{1}{3}(V_a - V_b) \quad \text{.....(4.107)}$$

From equation (4.106) and (4.107) we can write,

$$V_{a0} - V_{a1} = \frac{1}{3}(V_a + 2V_b) - \frac{1}{3}(V_a - V_b)$$

$$= \frac{1}{3}(V_a + 2V_b - V_a + V_b) = V_b$$

$$\therefore V_{a0} - V_{a1} = V_b$$

On substituting for V_b from equ(4.98) in equ(4.108) we get, —(4.108)

$$V_{a2} - V_{a1} = Z_f(I_b - I_c)$$

On substituting $(I_b + I_c)$ from equ (4.104) in equ(4.109) we get, ...(4.109)

$$V_{a2} - V_{a1} = Z_f 3I_{a2}$$

$$\therefore V_{a2} = V_{a1} + 3Z_f I_{a2} \quad \text{—(4.100)} \qquad (\because V_{a1} = V_{a2})$$

Also, $V_{a0} = V_{a2} + 3Z_f I_{a0}$ —(4.101)

Equations (4.100) and (4.110) suggest the connection of sequence networks as shown in fig 4.24 for a double line-to-ground fault.

From fig 4.24 we can write,

$$I_{a1} = \frac{V_{af}}{Z_1 + \dfrac{Z_2(Z_0 + 3Z_f)}{Z_2 + Z_0 + 3Z_f}} \quad \text{.....(4.112)}$$

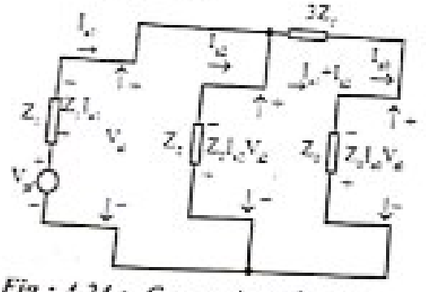


Fig : 4.24 : Connection of sequence networks for a double line-to-ground fault through an impedance

$$V_{a1} = V_{af} - Z_1 I_{a1}$$

$$V_{a2} = V_{a1} \quad ; \quad I_{a2} = \frac{V_{a2}}{Z_2} \quad ; \quad I_{a0} = -(I_{a1} + I_{a2}) \quad \text{.....(4.113)}$$

From fig (4.23) and equ(4.104) we get, ...(4.114)

Fault current, $I_f = I_b + I_c = 3I_{a0}$ .....(4.115)

EXAMPLE 4.1

Two 11 kV, 20 MVA, three-phase, star connected generators operate in parallel as shown in figure 4.1.1.

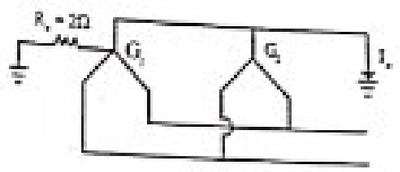


Fig 4.1.1.

The positive, negative and zero sequence reactances of each being respectively, j0.18, j0.15, j0.10 p.u. The star point of one of the generator is isolated and that of the other is earthed through a 2.0 Ω resistor. A single line-to-ground fault occurs at the terminals of one of the generators. Estimate (i) fault current, (ii) current in grounded resistor and (iii) voltage across grounding resistor.

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 20MVA \quad ; \quad kV_b = 11\ kV$$

$$\text{Base impedance, } Z_b = \frac{kV_b^2}{MVA_b} = \frac{11^2}{20} = 6.05\ \Omega$$

$$\text{p.u. value of neutral resistance} = \frac{\text{Actual value}}{\text{Base impedance}} = \frac{2}{6.05} = 0.3306\ \text{p.u.}$$

The single line diagram and the sequence networks of the given power system are shown in fig (4.1.2) to (4.1.5)

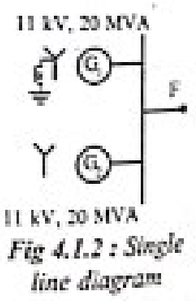


Fig 4.1.2 : Single line diagram

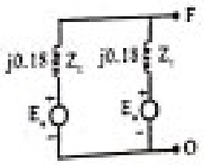


Fig 4.1.3 : Positive sequence network



Fig 4.1.4 : Negative sequence network

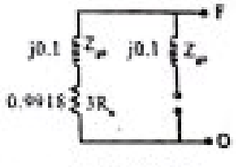


Fig 4.1.5 : Zero sequence network

The thevenin's equivalent of the sequence networks are shown in fig (4.1.6) to (4.1.8)

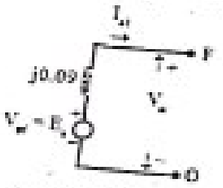


Fig 4.1.6 : Thevenin's equivalent of p.s. network

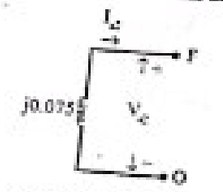


Fig 4.1.7 : Thevenin's equivalent of n.s. network

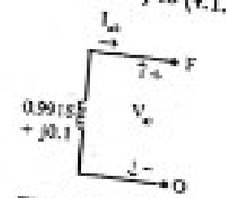


Fig 4.1.8 : Thevenin's equivalent of z.s. network

For a single line-to-ground fault,

$$I_{a1} = I_{a2} = I_{a0} \qquad \text{and} \quad I_f = I_a = 3\,I_{a1}$$

Hence the thevenin's equivalent of sequence networks are connected in series as shown in fig (4.1.9). The fault current is calculated by taking the prefault voltage $V_{pf} = 1$ p.u.

From fig 4.1.9. we get,

$$I_{a1} = \frac{1\angle 0^\circ}{j0.09 + j0.075 + 0.9918 + j0.1}$$

$$= \frac{1}{0.9918 + j0.265} = \frac{1}{1.0266\angle 15^\circ} = 0.9741\angle -15^\circ \text{ p.u.}$$

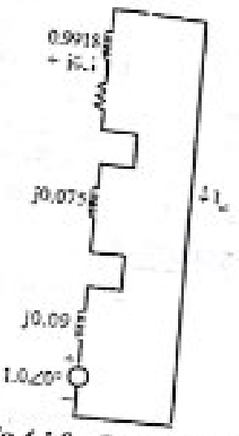


Fig 4.1.9 : Connection of sequence networks

i. To find fault current

Fault current, $I_f = I_a = 3\,I_{a1}$

$= 3 \times 0.9741\angle -15^\circ$

$= 2.9223\angle -15^\circ$ p.u.

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}\,kV_b} = \dfrac{20 \times 1000}{\sqrt{3} \times 11} = 1049.7$ A

∴ Actual value of fault current = p.u value of fault current × Base current

$= 2.9223\angle -15^\circ \times 1049.7 = 3067.5\angle -15^\circ$ A $= 3.0675\angle -15^\circ$ kA

ii. To find the current through neutral resistor

The current through the neutral resistor is same as that of fault current.

∴ Current through neutral resistor = $2.9223\angle -15^\circ$ p.u. or $3.0675\angle -15^\circ$ kA.

iii. To find the voltage across grounding resistor

From the thevenin's equivalent of zero sequence network we get,

The voltage across grounding resistor = $3\,R_n\,I_{a0} = 3\,R_n\,I_{a1}$

$= 3 \times 0.3306 \times 0.9741\angle -15^\circ = 0.9661\angle -15^\circ$ p.u.

Actual value of voltage across grounding resistor = p.u. value of voltage $\times \dfrac{kV_b}{\sqrt{3}}$ $\left(\because kV_b \text{ is } \begin{matrix}\text{line}\\\text{value}\end{matrix}\right)$

$= 0.9661\angle -15^\circ \times \dfrac{11}{\sqrt{3}} = 6.1356\angle -15^\circ$ kV

RESULT

i. Fault current = $2.9223\angle -15^\circ$ p.u. or $3.0675\angle -15^\circ$ kA.

ii. Neutral current = $2.9223\angle -15^\circ$ p.u. or $3.0675\angle -15^\circ$ kA.

iii. Voltage across neutral resistance = $0.9661\angle -15^\circ$ p.u. or $6.1356\angle -15^\circ$ kV.

EXAMPLE 4.2

A salient - pole generator without dampers is rated 20 MVA, 13.8 kV and has a direct axis subtransient reactance of 0.25 per unit. The negative and zero sequence reactances are 0.35 and 0.10 per unit respectively. The neutral of the generator is solidly grounded. Determine the sub-transient current in the generator and the line-to-line voltages for subtransient conditions when a single line-to-ground fault occurs at the generator terminals with generator operating unloaded at rated voltage. Neglect resistance.

SOLUTION

Let us choose the generator ratings as base values.

$\therefore MVA_b = 20MVA$; $kV_b = 13.8$ kV

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}\,kV_b} = \dfrac{20 \times 1000}{\sqrt{3} \times 13.8} = 836.7$ A

Given that, $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a single line-to-ground fault in an unloaded generator the sequence networks are connected in series as shown in fig 4.2.1. The following relations exists for a single line-to-ground fault.

$$I_{a0} = I_{a1} = I_{a2} \quad ; \quad I_f = I_a = 3I_{a1}$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0}$$

Here E_a is phase value of induced emf. Let it be 1 p.u. (i.e. $E_a = 1\angle 0^\circ$ p.u.)

$$\therefore I_{a1} = \frac{1}{j0.25 + j0.35 + j0.1} = \frac{1}{j0.7} = -j1.43 = 1.43\angle -90^\circ \text{ p.u.}$$

Subtransient fault current in p.u., $I_f'' = |I_a| = 3\,|I_{a1}| = 3 \times 1.43$

$= 4.29$ p.u.

Actual value of subtransient fault current = p.u.value of fault × Base current

$= 4.29 \times 836.7 = 3589.4$ A $= 3.5894$ kA

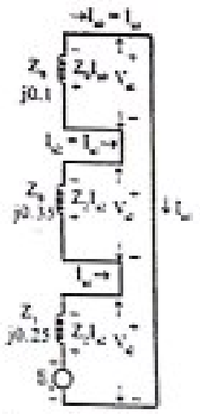


Fig 4.2.1

From the sequence networks of the generator the symmetrical components of phase -a voltage are calculated as shown below.

$V_{a1} = E_a - I_{a1}Z_1 = 1.0 - (-j1.43)\,(j0.25) = 1.0 - 0.357 = 0.643$ p.u.

$V_{a2} = -I_{a2}Z_2 = -(-j1.43)\,(j0.35) = -0.50$ p.u.

$V_{a0} = -I_{a0}Z_0 = -(-j1.43)\,(j0.1) = -0.143$ p.u.

The phase voltages are given by,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$

and $a^2 = 1\angle 240° = -0.5 - j0.866$

$V_a = V_{a0} + V_{a1} + V_{a2} = 0.643 - 0.50 - 0.143 = 0$

$V_b = V_{a0} + a^2 V_{a1} + aV_{a2}$

$= -0.143 + 0.643\,(-0.5 - j0.866) - 0.50(-0.5 + j0.866)$

$= -0.143 - 0.322 - j0.557 + 0.25 - j0.433 = -0.215 - j0.990$ p.u.

$V_c = V_{a0} + aV_{a1} + a^2 V_{a2}$

$= -0.143 + 0.643(-0.5 + j0.866) - 0.5(-0.5 - j0.866)$

$= -0.143 - 0.322 + j0.557 + 0.25 + j0.433 = -0.215 + j0.99$ p.u.

Line-to-line voltages are

$V_{ab} = V_a - V_b = 0.215 + j0.990 = 1.01\angle 77.7°$ p.u.

$V_{bc} = V_b - V_c = -0.215 - j0.99 + 0.215 - j0.99 = -j1.980 = 1.980\angle 270°$ p.u.

$V_{ca} = V_c - V_a = -0.215 + j0.99 = 1.01\angle 102.3°$ p.u.

Since the generated voltage to neutral E_a was taken as 1.0 p.u., the above line-to-line voltages are expressed in p.u. on the base of phase voltage. The postfault line voltages expressed in volts are obtained by multiplying the p.u. value with base value of phase voltage.

Line value of base voltage = 13.8 kV

Phase value of base voltage $= \frac{13.8}{\sqrt{3}}$ kV = 7.97 kV

$V_{ab} = 1.01 \times 7.97\angle 77.7° = 8.05\angle 77.7°$ kV

$V_{bc} = 1.98 \times 7.97\angle 270° = 15.78\angle 270°$ kV

$V_{ca} = 1.01 \times 7.97\angle 102.3° = 8.05\angle 102.3°$ kV

EXAMPLE 4.3

Find the subtransient currents and line-to-line voltages at the fault under subtransient conditions when a line-to-line fault between phase b and c occurs at the terminals of the generator described in example 4.2. Assume that the generator is unloaded and operating at rated terminal voltage when the fault occurs. Neglect resistance.

SOLUTION

Let us choose the generator ratings as base values.

$\therefore MVA_b = 20MVA$; $kV_b = 13.8$ kV

Base current, $I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{20 \times 1000}{\sqrt{3} \times 13.8}$

$= 836.7$ A ≈ 837 A

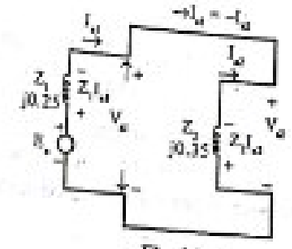


Fig 4.3.1.

Given that $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a line-to-line fault the positive and negative sequence networks are connected in parallel as shown in fig 4.3.1.

The following relations exists for a single line-to-line fault.

$I_a = 0$; $I_b = -I_c$; $I_{a0} = 0$; $I_{a2} = -I_{a1}$; $V_b = V_c$; $V_{bc} = 0$

From fig 4.3. we get,

$$I_{a1} = \frac{E_a}{Z_1 + Z_2}$$

Here E_a is phase value of induced emf. Let it be 1 p.u. (i.e. $E_a = 1\angle 0°$ p.u.)

$\therefore I_{a1} = \frac{1}{j0.25 + j0.35} = \frac{1}{j0.6} = -j1.667 = 1.667\angle -90°$ p.u.

$I_{a2} = -I_{a1} = j1.667$ p.u.

$I_{a0} = 0$

The line currents are related to symmetrical components by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$

and $a^2 = 1\angle 240° = -0.5 - j0.866$

$I_a = I_{a0} + I_{a1} + I_{a2} = -j\,1.667 + j1.667 = 0$

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$= -j1.667(-0.5 - j0.866) + j1.667(-0.5 + j0.866)$$

$$= j0.833 - 1.443 - j0.833 - 1.443 = -2.886 \text{ p.u.}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2} = -a I_{a2} - a^2 I_{a1} = -(a^2 I_{a1} + a I_{a2}) = -I_b = 2.886 \text{ p.u.}$$

Actual values of line currents are obtained by multiplying the p.u. values with base current.

$$I_a = 0$$

$$I_b = -2.886 \times 837 = 2416\angle 180° \text{ A}$$

$$I_c = 2.886 \times 837 = 2416\angle 0° \text{ A}$$

The fault current, $I_f = |I_b| = 2416 \text{ A} = 2.416 \text{ kA}$

From the sequence networks of the generator the symmetrical components of phase-a voltage are calculated as shown below.

$$V_{a0} = 0 \quad ; \quad V_{a1} = E_a - I_{a1} Z_1 \quad ; \quad V_{a2} = V_{a1}$$

$$\therefore V_{a1} = V_{a2} = 1 - (-j1.667)(j0.25) = 1 - 0.417 = 0.583 \text{ p.u.}$$

The phase voltages are given by,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$V_a = V_{a0} + V_{a1} + V_{a2} = 0.583 + 0.583 = 1.166\angle 0° \text{ p.u.}$$

$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2}$$

$$V_b = 0.583(-0.5 - j0.866) + 0.583(-0.5 + j0.866) = -0.583 \text{ p.u.}$$

$$V_c = V_b = -0.583 \text{ p.u.}$$

Line voltages are

$$V_{ab} = V_a - V_b = 1.166 + 0.583 = 1.749\angle 0° \text{ p.u.}$$

$$V_{bc} = V_b - V_c = -0.583 + 0.583 = 0 \text{ p.u.}$$

$$V_{ca} = V_c - V_a = -0.583 - 1.166 = 1.749\angle 180° \text{ p.u.}$$

Since the generated phase voltage E_a is taken as 1 p.u., the above line voltages are expressed in p.u. on the base of phase voltage. The post fault line voltages expressed in volts are obtained by multiplying the p.u. value with the base value of phase voltage.

Line value of base voltage = 13.8 kV

$$\text{Phase value of base voltage} = \frac{13.8}{\sqrt{3}} = 7.9 \text{ kV}$$

$$V_{ab} = 1.749\angle 0° \times 7.97 = 13.94\angle 0° \text{ kV} \quad ; \quad V_{bc} = 0 \text{ kV}$$

$$V_{ca} = 1.749\angle 180° \times 7.97 = 13.94\angle 180° \text{ kV}$$

EXAMPLE 4.4

Find the subtransient currents and the line-to-line voltages at the fault under subtransient conditions when a double line-to-ground fault involving phase b and c occurs at the terminals of the generator described in example 4.2. Assume that the generator is unloaded and operating at rated voltage when the fault occurs. Neglect resistance.

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore \text{MVA}_b = 20\text{MVA} \quad ; \quad \text{kV}_b = 13.8 \text{ kV}$$

$$\text{Base current, } I_b = \frac{\text{kVA}_b}{\sqrt{3}\,\text{kV}_b} = \frac{20 \times 10^3}{\sqrt{3} \times 13.8} = 836.7\text{A} \approx 837\text{A}$$

Given that $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a double line-to-ground fault the positive, negative and zero sequence networks are connected in parallel as shown in fig 4.4.1.

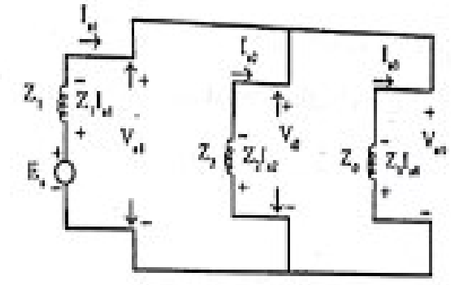


Fig 4.4.1

The following relations exists for a double line-to-ground fault.

$$I_{a1} = -(I_{a2} + I_{a0}) \quad ; \quad V_{a0} = V_{a1} = V_{a2} \quad ; \quad V_b = V_c = 0$$

From fig 4.4.1 we can calculate the symmetrical components of phase-a voltage and current as shown below.

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 Z_0 / (Z_2 + Z_0)} = \frac{1.0 + j0}{j0.25 + (j0.35 \times j0.10)/(j0.35 + j0.10)}$$

$$= \frac{1.0}{j0.25 + j0.0778} = \frac{1.0}{j0.3278} = -j3.05 \text{ p.u.}$$

$$V_{a1} = V_{a0} = V_{a2} = E_a - I_{a1} Z_1 = 1 - (-j3.05)(j0.25) = 1.0 - 0.763 = 0.237 \text{ p.u.}$$

$$I_{a2} = \frac{-V_{a2}}{Z_2} = -\frac{0.237}{j0.35} = j0.68 \text{ p.u}$$

$$I_{a0} = \frac{-V_{a0}}{Z_0} = \frac{-0.237}{j0.10} = j2.37 \text{ p.u}$$

The line currents are related to symmetrical components by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

$I_a = I_{a0} + I_{a1} + I_{a2} = j2.37 - j3.05 + j0.68 = 0$

$I_b = I_{a0} + a^2 I_{a1} + aI_{a2}$

$= j2.37 + (-0.5 - j0.866)(-j3.05) + (-0.5 + j0.866)(j0.68)$

$= j1.525 - 2.641 - j0.34 - 0.589 + j2.37 = -3.230 + j3.555 = 4.80\angle 132.3°$ p.u.

$I_c = I_{a0} + aI_{a1} + a^2 I_{a2}$

$= j2.37 + (-0.5 + j0.866)(-j3.05) + (-0.5 - j0.866)(j0.68)$

$= j1.525 + 2.641 - j0.34 + 0.589 + j2.37 = 3.230 + j3.555$

$= 4.80\angle 47.7°$ p.u.

Fault current, $I_f = I_b + I_c = -3.230 + j3.555 + 3.230 + j3.555 = j7.11$ p.u.

The phase and line voltages are calculated as shown below

$V_a = V_{a1} + V_{a2} + V_{a0} = 3V_{a1} = 3 \times 0.237 = 0.711$ p.u.

$V_b = V_c = 0$

$V_{ab} = V_a - V_b = 0.711$ p.u.

$V_{bc} = 0$

$V_{ca} = V_c - V_a = -0.711$ p.u.

The actual values of line currents and voltages are calculated as shown below.

$I_a = 0$

$I_b = 837 \times 4.8\angle 132.3° = 4017\angle 132.3°$ A

$I_c = 837 \times 4.8\angle 47.7° = 4017\angle 47.7°$ A

$I_f = 837 \times 7.11\angle 90° = 5951\angle 90°$ A

$V_{ab} = 0.711 \times 7.97 = 5.66\angle 0°$ kV

$V_{bc} = 0$

$V_{ca} = -0.711 \times 7.97 = 5.66\angle 180°$ kV

EXAMPLE 4.5

A generator of negligible resistance having 1 p.u. voltage behind transient reactance is subjected to different types of faults.

Type of fault	Resulting fault current in pu
3 - phase	3.33
L - L	2.23
L - G	3.01

Calculate the per unit value of 3 sequence reactances.

SOLUTION

Case(I) : 3-phase fault

The circuit model used to calculate the transient fault current for a 3-phase fault on a generator is shown in fig 4.5.1

Given that, $E_g' = 1$ p.u., $I' = 3.33$ p.u.

When load current is neglected, $E_g'' = E_g' = E_g = E_a$

$\therefore X_d'' = \frac{|E_g|}{|I'|} = \frac{1}{3.33} = 0.3$ p.u.

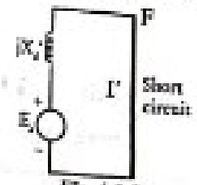


Fig 4.5.1

We know that the reactance during symmetrical fault is positive sequence reactance.

$\therefore$ positive sequence reactance of generator, $X_1 = X_d'' = 0.3$ p.u.

Case (ii) : L - L fault

For a line-to-line fault on a generator the positive and negative sequence networks can be connected in parallel as shown in fig 4.5.2

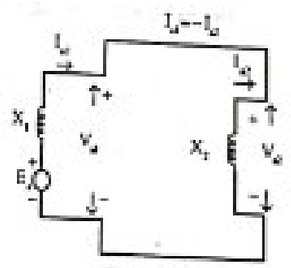


Fig : 4.5.2

From fig 4.5.2 we get

$$I_{a1} = \frac{E_g}{jX_1 + jX_2}$$

$$\therefore |I_{a1}| = \frac{E_g}{X_1 + X_2} \quad \text{....(4.5.1)}$$

Let us assume that the line-to-line fault is due to a short between phase-b and phase-c. Hence the fault current is I_b.

From the symmetrical components of the current we can write

$I_b = I_{a0} + a^2 I_{a1} + aI_{a2}$

For a line-to-line fault, $I_{a0} = 0$ and $I_{a2} = -I_{a1}$

$$\therefore I_b = 0 + a^2 I_{a1} - a I_{a1} = I_{a1}(a^2 - a)$$

$$= I_{a1}(-0.5 - j0.866 - (-0.5 + j0.866)) = I_{a1}(-j1.732) = -j1.732\, I_{a1}$$

Fault current, $I_f = |I_b| = 1.732\, |I_{a1}|$

$$\therefore |I_{a1}| = \frac{I_f}{1.732}$$

Given that $I_f = 2.23$ p.u.

$$\therefore |I_{a1}| = \frac{2.23}{1.732} = 1.2875 \text{ p.u.} \qquad \text{.....(4.5.2)}$$

From equation (4.5.1) & (4.5.2) we can write,

$$1.2875 = \frac{E_a}{X_1 + X_2}$$

$$\therefore X_1 + X_2 = \frac{E_a}{1.2875} \quad ; \quad \therefore X_2 = \frac{E_a}{1.2875} - X_1 = \frac{1}{1.2875} - 0.3 = 0.48 \text{ p.u.}$$

Case (iii) L-G fault

For a single line-to-ground fault, on a generator the sequence networks are connected in series as shown in fig 4.5.3. For a single line-to-ground fault, the fault current is $|I_a|$ and it is equal to $3|I_{a0}|$.

Given that, $I_f = 3.01$ p.u.

$$\therefore 3|I_{a0}| = I_f = 3.01 \quad \text{or} \quad |I_{a0}| = \frac{3.01}{3} = 1.0033 \text{ p.u.}$$

From fig 4.5.3 we can write

$$I_{a0} = \frac{E_a}{jX_1 + jX_2 + jX_0}$$

$$\therefore |I_{a0}| = \frac{E_a}{X_1 + X_2 + X_0}$$

$$X_1 + X_2 + X_0 = \frac{E_a}{|I_{a0}|}$$

$$X_0 = \frac{E_a}{|I_{a0}|} - X_1 - X_2 = \frac{1}{1.0033} - 0.3 - 0.48 = 0.22 \text{ p.u.}$$

RESULT

Positive sequence reactance of generator = 0.3 p.u.

Negative sequence reactance of generator = 0.48 p.u.

Zero sequence reactance of generator = 0.22 p.u.

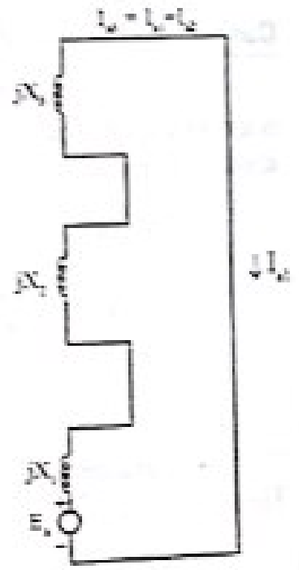


Fig 4.5.3

4.7 SHORT QUESTIONS AND ANSWERS

1. *Name the fault in which positive, negative and zero sequence component currents are equal.*

   In single line-to-ground fault the positive, negative and zero sequence component currents are equal.

2. *Name the fault in which positive and negative sequence component currents together is equal to zero sequence current in magnitude.*

   Double line-to-ground fault

3. *Name the various unsymmetrical faults in a power system.*

   The unsymmetrical faults in a power systems are
   (i) Single line-to-ground fault
   (ii) Line-to-line fault
   (iii) Double line-to-ground fault
   (iv) Open conductor fault

4. *Define negative sequence impedance?*

   The negative sequence impedance of an equipment is the impedance offered by the equipment to the flow of negative sequence current.

5. *Name the faults which do not have zero sequence currents flowing.*

   In line to line faults the zero sequence currents do not flow.

6. *Name the faults involving ground*

   The faults involving ground are
   (i) Single line to ground fault
   (ii) Double line to ground fault
   (ii) Three phase fault

7. *Define positive sequence impedance.*

   The positive sequence impedance of an equipment is the impedance offered by the equipment to the flow of positive sequence currents.

8. *In what type of fault the positive sequence component of current is equal in magnitude but opposite in phase to negative sequence components of current?*

   Line-to-line fault

9. *In which fault the negative and zero sequence currents are absent?*

   In three phase fault the negative and zero sequence currents are absent.

10. *Write the boundary condition in single line-to-ground fault?*

    The boundary conditions are

    $$V_a = 0 \quad ; \quad I_1 = I_2 = 0$$

11. What are the boundary condition in line-to-line fault?

$$I_a = 0 \; ; \; I_b + I_c = 0 \; ; \; V_b = V_c$$

12. Write down the boundary condition in double line-to-ground fault?

$$I_a = 0 \; ; \; V_b = 0 \; ; \; V_c = 0$$

13. Give the boundary condition for the 3-phase fault.

$$I_a + I_b = I_c = 0 \; ; \; V_a = V_b = V_c = 0$$

14. Draw the connection of sequence networks for a single line-to-ground fault at the terminals of an unloaded generator.

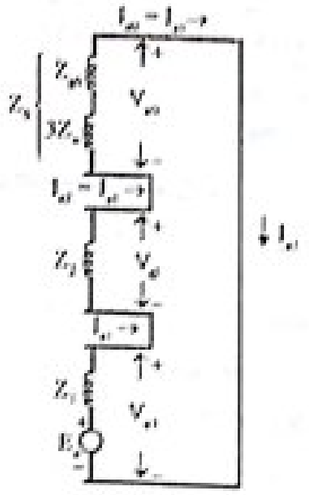


Fig Q14 : Connection of sequence network of an unloaded generator for line-to-ground fault on phase -a

15. Draw the connection of sequence networks for a line-to-line fault at the terminals of an unloaded generator.

16. Draw the connection of sequence networks for a double line-to-ground fault at the terminals of an unloaded generator.

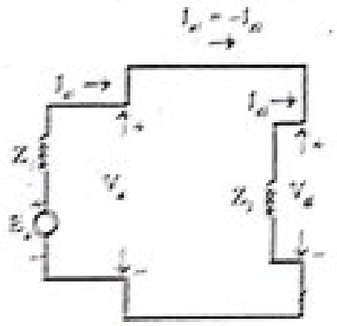


Fig Q15 : Connection of the sequence networks of an unloaded generator for a line-to-line fault between phases b and c

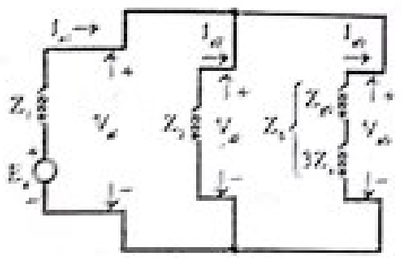


Fig Q16 : Connection of the sequence networks of an unloaded generator for a double line-to-ground fault on phase b and c.

17. Draw the connection of seqence networks for line-to-line fault without fault impedance.

18. Draw the connection of sequence networks for single line-to-ground fault without fault impedance?

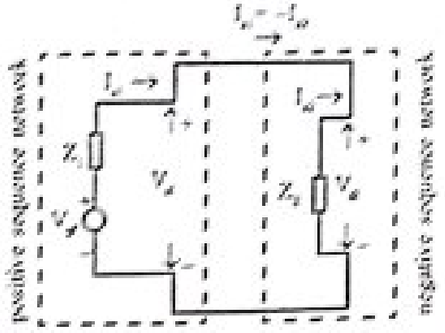


Fig Q17 : Connection of the sequence networks for a line-to-line fault in a power system.

19. Draw the connection of seqence networks for double line-to-ground fault without fault impedance.

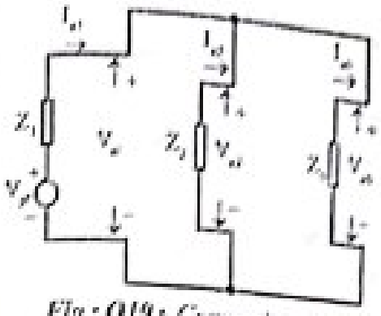


Fig : Q19 : Connection of the sequence networks for a double line-to-ground fault in a power system.

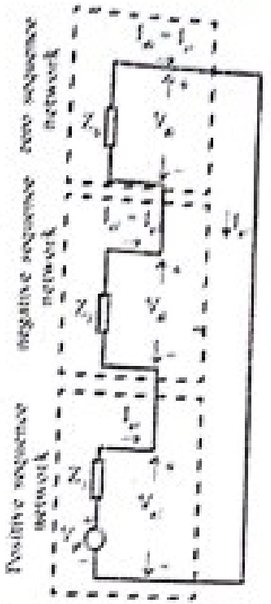


Fig Q18 : Connection of sequence networks for a single line-to-ground fault in a power system

20. Draw the connection of sequence networks for single line-to-ground fault through an impedance Z_f

21. Draw the connection of sequence networks for line-to-line fault through an impedance Z_f

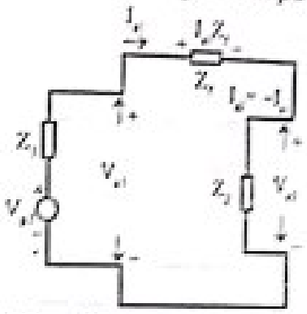


Fig Q21 : Connection of sequence networks for a line-to-line fault through an impedance Z_f

22. Draw the connection of sequence networks for double line-to-ground fault through an impedance Z_f

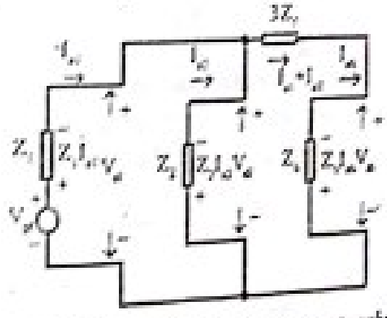


Fig Q22 : Connection of sequence networks for a double line-to-ground fault through an impedance

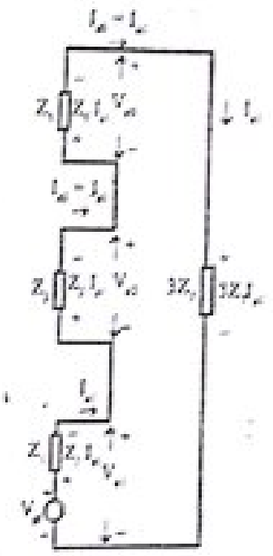


Fig Q20 Connection of sequence networks for a single line to ground fault through an impedance Z_f

4.8 EXERCISES

1. A 25 MVA, 11 kV alternator with $X_0 = 0.05$ p.u., $X_1 = X_2 = 0.15$ p.u. is earthed through a reactance of 0.333 Ω. Calculate the line current for a single line-to-ground fault in amperes.
2. A single line-to-ground fault of 0.05 Ω resistance occur in a 3φ system supplied by a synchronous generator with a generator emf of 11 kV between lines. The positive, negative and zero sequance reactances of the generators and network upto the fault are 0.5 Ω, 0.2 Ω &0.1 Ω respectively. Find the fault current.
3. Two similar generators each having emfs of 2.2 kV per phase and impedences $Z_1 = 0 + j1$, $Z_2 = 0 + j.8$ and $Z_0 = 0 + j.3$ Ω per phase are connected in parallel. Both generators are solidly grounded. Determine (i) the total current to ground (ii) phase voltage when a double line-to-ground fault occurs at the terminals.
4. A 20 MVA, 11 kV alternator solidly grounded neutral has a subtransient reactances of 0.25 p.u. The negative and zero sequence reactances are j0.148, j0.05 respectively. A single Line-ground fault occurs at the terminals of an unloaded generator. Determine the fault currents & line-to-line voltages. Neglect resistances.
5. A generator without dampers is rated 30 MVA, 12.8 kV and has a direct axis subtransient reactance is 0.3 per unit. The negative and zero sequance reactances is 0.25, 0.1 p.u. The neutral of the generator is solidly ground. Determine the subtransient currents and line-to-line voltages at the faults under subtransient conditions, when a line-to-line fault occurs at the generator terminals. Assume that the generator is unloaded and operating at rated terminal voltage when fault occurs. Neglect resistance.
6. A double-line-to-ground fault occurs on line b and c at point F in the system of fig E4.6. Find the subtransient current in each phase c of generator 1, assuming prefault currents to be zero. Both synchronous machines are rated 3000 kVA, 900 V with reactances of $X'' = X_2 = 20\%$ and $X_0 = 0\%$. Each three phase transformer is rated 3000 kVA, 900/3,300 V with leakage reactance of 10%. The reactances of the transmission line are $X_1 = X_2 = 40\%$ and $X_0 = 20\%$ on a base of 3000 kVA, 3,300 V. The reactance of the neutral grounding reactors are 10% on the kVA base of the machines.

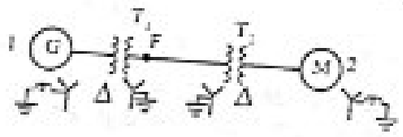


Fig E4.6

7. A synchronous generator is rated 25 MVA, 11 kV. It is star connected with the neutral points solidly grounded. The generator is operating at no load at rated voltage. It's reactances are $X_1 = X_2 = 20$ and $X_0 = 0.02$ p.u. Calculate the symmetrical subtransient line currents for
   (i) Single line-to-ground fault
   (ii) Double line fault
   (iii) Double line-to-ground fault
   (iv) Symmetrical 3φ fault

STABILITY ANALYSIS



5.1 INTRODUCTION

The term stability refers to stable operation of the synchronous machines connected to a power system when they are subjected to sudden disturbances. Hence we can say that the stability is the ability of power system to return to stable operation when it is subjected to a disturbance. The instability refers to the loss of synchronism (or falling out of step) of synchronous machines in the power system. The stability studies are classified into three types depending upon the nature of the disturbance. They are Transient, Dynamic and Steady state stability.

The study of steady state stability is concerned with the determination of upper limit of loading synchronous machines before loosing synchronism. In these studies it is assumed that the machine is loaded gradually.

In a power system, small disturbances will continuously occur due to variations in load, changes in the speed of prime mover, fault in certain part of system, etc.,. These small disturbances may excite the system into a state of natural oscillations.

In dynamically stable systems the amplitude of the oscillations will be small and they die out quickly. In dynamically unstable systems the amplitude of the oscillations is very large and they exist for a long time. This type of unstable behaviour may create serious problems in power system operation. (i.e., it may lead to unnecessary frequent load shedding).

Large disturbances may occur in a power system due to switching heavy loads, switching long transmission lines, major faults, etc., The sudden large disturbances are characterized by large changes in the speed of the rotor of synchronous machine, large changes in power angle and fast change in power transfer. These characteristic may lead to loss of synchronism. This type of instability is called transient instability. Hence the aim of transient stability is to determine whether the system will remain in synchronism or not following a sudden or major disturbances, such as transmission line fault, sudden heavy loads, tripping of generating units or switching lines with heavy loads.

The concepts of stability can be better understood by considering a single synchronous machine connected to infinite bus. It is equivalent to two machine system. A practical power system may have a number of machines running in synchronism.

For theoretical analysis the multimachine system can be converted to an equivalent single machine system connected to infinite bus. Alternatively, the stability study of a multimachine system can be carried on a computer.

The dynamic behaviour of a power system have the following basic features.

1. A synchronous machine has a maximum limit of power transfer when remains in synchronism. When the power transfer exceeds the limit it cannot stay in synchronism.
2. The synchronous machine connected to an infinite bus is basically a spring-inertia oscillatory system. The inertia is due to the mechanical part (rotor) and the spring action is due to the synchronous tie with infinite bus.
3. If δ is the angular displacement of the rotor in electrical radians then the power transfer is proportional to sin δ. Hence the mathematical equation governing the system dynamics is nonlinear.

In order to simplify the computational task in stability studies, the following three assumptions are made in all stability studies.

1. The dc-offset currents and harmonic components are neglected. The currents and voltages are assumed to have fundamental component alone.
2. The symmetrical components are used for the representation of unbalanced faults.
3. It is assumed that the machine speed variations will not affect the generated voltage.

5.2 DYNAMICS OF SYNCHRONOUS MACHINE ROTOR

List of terms used for the analysis of the dynamics of a synchronous machine.

- E_{ke} = Kinetic energy of the rotor in MJ (Mega Joules)
- J = Moment of inertia of the rotor in kg - m^2
- ω_{sm} = Synchronous angular speed of the rotor in mech.rad/sec.
- ω_s = Synchronous angular speed of rotor in elect.rad/sec.
- p = Number of poles in synchronous machine
- M = Moment of inertia of rotor in MJ-s/elec. rad or MJ-s/mech rad.
- S = Power rating of machine in MVA
- H = Inertia constant in MJ/MVA or MW-s/MVA.
- f = Frequency in cycles/sec or Hz.
- δ_m = Angular displacement of rotor with respect to synchronously rotating reference frame in mech.rad.
- δ = Angular displacement of rotor with respect to synchronously rotating reference frame in elect rad.

- θ_m = Angular displacement of rotor with respect to a stationary axis in mech.rad.
- θ = Angular displacement of rotor with respect to a stationary axis in elect.rad.
- t = Time in seconds.
- T_m = Mechanical torque at the shaft of rotor (supplied by prime mover) in N-m.
- T_e = Net electromagnetic torque in N-m.
- T_a = Net accelerating torque in N-m.
- P_m = Mechanical power input in p.u.
- P_e = Electrical power output in p.u.

Note : Here the mechanical losses in the rotating system and electrical losses in the stator of synchronous machine are neglected.

The kinetic energy (in MJ) of the rotor of a synchronous machine is given by equ (5.1).

$$E_{ke} = \frac{1}{2} J \omega_{sm}^2 \times 10^{-6} \quad \text{---(5.1)}$$

The mechanical and electrical angular speeds are related to the number of poles in synchronous machine as shown in equ(5.2).

$$\omega_s = \frac{p}{2} \omega_{sm} \quad \text{(or)} \quad \omega_{sm} = \frac{2}{p} \omega_s \quad \text{.....(5.2)}$$

On substituting for ω_{sm} from equ (5.2) in equ (5.1) we get,

$$E_{ke} = \frac{1}{2} J \left(\frac{2}{p}\right)^2 \omega_s^2 \times 10^{-6}$$

Let, $E_{ke} = \frac{1}{2} M \omega_s$ .....(5.3)

where, $M = J \left(\frac{2}{p}\right)^2 \omega_s \times 10^{-6}$ .....(5.4)

Here M is the moment of inertia in MJ-s/elec.rad. This moment of inertia is used popularly in stability studies.

Another useful constant which is popularly used in stability studies is the inertia constant, H. It is defined as the ratio of stored kinetic energy in megajoules at synchronous speed to the machine rating in megavoltampere.

$$H = \frac{\text{Store kinetic energy in MJ at synchronous speed}}{\text{Machine rating in MVA}}$$

$$\therefore H = \frac{E_{ke}}{S} \quad \text{.....(5.5)}$$

On substituting for E_{ke} from equ (5.1) in equ(5.5) we get,

$$H = \frac{\frac{1}{2} J \omega_{sm}^2}{S} \quad \text{.....(5.6)}$$

On substituting for E_k from equ(5.3) in equ(5.5) we get,

$$H = \frac{\frac{1}{2} M \omega_s}{S}$$

$$\therefore M = \frac{2HS}{\omega_s} \quad \text{.....(5.7)}$$

We know that, $\omega_s = 2\pi f$ .....(5.8)

$$\therefore M = \frac{2HS}{2\pi f} = \frac{HS}{\pi f} \text{ (in MJ - s/ elect. rad)}$$

In equ(5.9) the moment of inertia, M is obtained in the units of MJ-s/elect.rad. Sometimes it is required in MJ-s/elect.degree. In this case the π radians in equ(5.9) should be replaced by 180° as shown in equ(5.10). .....(5.9)

$$\therefore M = \frac{HS}{180\, f} \text{ (in MJ - s/ elect.deg)} \quad \text{.....(5.10)}$$

The value of M can be expressed in per unit by selecting a base for MVA.

Let, S_b = Base MVA

Now p.u value of $M = M_{p.u} = \frac{SH/\pi f}{S_b}$ or $\frac{SH/180f}{S_b}$

If the machine rating S is chosen as base value, then $S = S_b$, in this case the equation 5.11 can be written as shown in equ (5.12) .....(5.11)

p.u value of M with machine rating as base MVA $\Big\}$ $M_{p.u} = \frac{H}{\pi f}$ or $\frac{H}{180f}$ .....(5.12)

Note : *Many authors treat M and H as two different inertia constants in different units.*

5.3 SWING EQUATION

The rotor of a synchronous machine is subjected to two torques, T_e and T_m, which are acting in opposite directions as shown in fig 5.1.

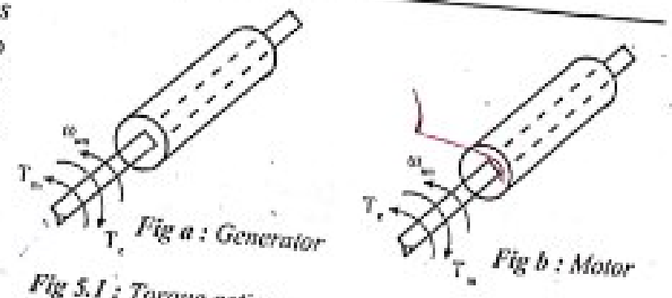


Fig a : Generator *Fig b : Motor*

Fig 5.1 : Torque acting on rotor of synchronous machine

where, T_a = Net electrical or electromechanical torque in N-m.

T_m = Mechanical or shaft torque supplied by the prime mover in N-m.

Under steady state operating condition the T_e and T_m are equal and the machine runs at constant speed, which is called synchronous speed. If there is a difference between the two torques then the rotor will have an accelerating or deaccelerating torque, denoted as T_a.

$$\therefore T_a = T_m - T_e \quad \text{.....(5.13)}$$

Here T_m & T_e are positive for generators and T_m & T_e are negative for motors.

Let θ_m = Angular displacement of rotor with respect to stationary reference axis.

δ_m = Angular displacement of rotor with respect to synchronously rotating reference axis.

By Newton's second law of motion, we can say that the accelerating torque, T_a is directly proportional to angular acceleration and the constant of proportionality is the moment of inertia, J.

$$\therefore T_a \propto \frac{d^2\theta_m}{dt^2} \quad \text{(or)} \quad T_a = J \frac{d^2\theta_m}{dt^2} \quad \text{.....(5.14)}$$

On substituting for T_a from equ(5.14) in (5.13) we get,

$$J \frac{d^2\theta_m}{dt^2} = T_m - T_e \quad \text{.....(5.15)}$$

The angular displacements θ_m and δ_m are related to synchronous speed by the following equation [equ(5.16)]

$$\theta_m = \omega_{sm} t + \delta_m \quad \text{.....(5.16)}$$

On differentiating the equ(5.16) with respect to time, t we get,

$$\frac{d\theta_m}{dt} = \omega_m = \omega_{sm} + \frac{d\delta_m}{dt} \quad \text{.....(5.17)}$$

On differentiating the equ(5.17) with respect to time, t we get,

$$\frac{d^2\theta_m}{dt^2} = \frac{d^2\delta_m}{dt^2} \quad \text{.....(5.18)}$$

From equ(5.17) we can say that the rotor angular velocity $d\theta_m/dt$ is constant and equal to ω_{sm} (synchronous speed) only if $d\delta_m/dt$ is zero. Hence $d\delta_m/dt$ represents the deviation of the rotor speed from synchronism.

From equation (5.15) and (5.18) we can write, $J \frac{d^2\delta_m}{dt^2} = T_m - T_e$

Let, $P_{m,act}$ = Shaft power input to the machine neglecting los

$P_{e,act}$ = Electrical power developed in rotor (in MW

We know that, power, $P = \frac{2\pi NT}{60} = \omega T$

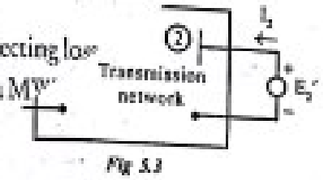


Fig 5.2

The node basis matrix equation of 2-bus system of fig5.3. can be written as shown below

$$I = Y_{bus} V \qquad \text{—(5.30)}$$

In the expanded form the equation (5.30) can be written as,

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} E_1' \\ E_2' \end{bmatrix} \qquad \text{—(5.31)}$$

where, I_1 & I_2 are the currents injected by the sources E_1' & E_2' respectively to the system.

From the row-1 of matrix equation (5.31) we get,

$$I_1 = Y_{11} E_1' + Y_{12} E_2' \qquad \text{—(5.32)}$$

The complex power S_1 injected to bus-1 is given by

$$S_1 = P_1 + jQ_1 = E_1' I_1^* \qquad \text{—(5.33)}$$

On substituting for I_1 from equ(5.32) in equ(5.33) we get,

$$\begin{aligned} P_1 + jQ_1 &= E_1'(Y_{11} E_1' + Y_{12} E_2')^* \\ &= E_1'(Y_{11}^* (E_1')^* + Y_{12}^* (E_2')^*) \\ &= (E_1')(E_1')^* Y_{11}^* + Y_{12}^* E_1'(E_2')^* \\ &= |E_1'|^2 Y_{11}^* + Y_{12}^* E_1'(E_2')^* \end{aligned} \qquad \text{—(5.34)}$$

The quantities E_1', E_2', Y_{11} and Y_{12} in equation (5.39) are complex quantities and so they can be expressed in polar coordinates (or in rectangular coordinates) as shown below.

Let $E_1' = |E_1'| \angle \delta_1$ $\qquad Y_{11} = G_{11} + jB_{11} = |Y_{11}| \angle \theta_{11}$

$E_2' = |E_2'| \angle \delta_2$ $\qquad Y_{12} = G_{12} + jB_{12} = |Y_{12}| \angle \theta_{12}$

Using the polar or rectangular coordinates of the complex quantities involved in equation (5.34) it can be written as shown below.

$$\begin{aligned} P_1 + jQ_1 &= |E_1'|^2 (G_{11} + jB_{11})^* + (Y_{12} \angle \theta_{12})^* |E_1'| \angle \delta_1 (|E_2'| \angle \delta_2)^* \\ &= |E_1'|^2 (G_{11} - jB_{11}) + |Y_{12}| \angle -\theta_{12}) |E_1'| \angle \delta_1 |E_2'| \angle -\delta_2 \\ &= |E_1'|^2 (G_{11} - jB_{11}) + |E_1'||E_2'||Y_{12}| \angle (\delta_1 - \delta_2 - \theta_{12}) \\ &= |E_1'|^2 (G_{11} - jB_{11}) + |E_1'||E_2'||Y_{12}| \cos(\delta_1 - \delta_2 - \theta_{12}) \\ &\quad + j|E_1'||E_2'||Y_{12}| \sin(\delta_1 - \delta_2 - \theta_{12}) \end{aligned} \qquad \text{.....(5.35)}$$

On equating the real and imaginary part of equation (5.35) we get,

$$P_1 = |E_1'|^2 G_{11} + |E_1'||E_2'||Y_{12}| \cos(\delta_1 - \delta_2 - \theta_{12}) \qquad \text{.....(5.36)}$$

$$Q_1 = -|E_1'|^2 B_{11} + |E_1'||E_2'||Y_{12}| \sin(\delta_1 - \delta_2 - \theta_{12}) \qquad \text{.....(5.37)}$$

The equ(5.36) and (5.37) are equations of real and reactive power injected by a generator at bus-1.

Let, $\delta = \delta_1 - \delta_2$

$\gamma = \theta_{12} - \pi/2$ $\qquad$ (and $\theta_{12} = \gamma + \pi/2$)

$P_c = |E_1'|^2 G_{12}$

$P_{max} = |E_1'||E_2'||Y_{12}|$

$P_1 = P_e$

By using the terms defined above, the real power P_1 (i.e., equ 5.36) can be written as shown below.

$$P_e = P_c + P_{max} \cos(\delta - \gamma - \pi/2)$$

$$\therefore P_e = P_c + P_{max} \sin(\delta - \gamma) \qquad \text{—(5.38)}$$

The equation (5.38) is called power angle equation. In this equation the term P_c represents power loss in the system and P_{max} represents the maximum real power that can be delivered by the generator to an infinite bus.

Note : *Here the bus-2 is considered as infinite bus and so $\delta_2 = 0$. Therefore $\delta = \delta_1$*

The power angle equation can be further simplified by considering the network as purely reactive network (by neglecting the resistances).

When resistances are neglected $G_{12} = 0$, $\theta_{12} = \pi/2$ and so $\gamma = 0$. On substituting these conditions in equ(5.38) we get,

$$P_e = P_{max} \sin\delta$$

$$\text{where } P_{max} = \frac{|E_1'||E_2'|}{X_{12}} \qquad \text{—(5.39)}$$

and X_{12} = Transfer reactance between bus-1 & 2.

The equation (5.39) is called simplified power angle equation. The graph of (or plot of) P_e as a function δ is called the power angle curve. A typical power angle curve is shown in fig (5.4).

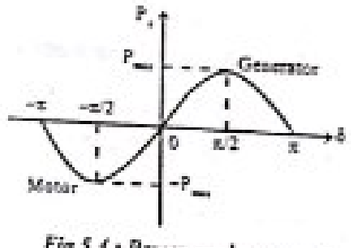


Fig 5.4 : Power angle curve of synchronous machine

On substituting the expression for P_e from equ(5.39) in the swing equation (i.e., equ(5.29), we get.

$$\frac{H}{\pi f} \frac{d^2\delta}{dt^2} = P_m - P_{max} \sin\delta \qquad \text{.....(5.40)}$$

The equation (5.40) is the swing equation in which the electrical power is expressed as a function of δ.

5.5 STEADY STATE STABILITY

In steady state every synchronous machine has a limit for power transfer to a receiving system. The steady state limit of a machine or transmitting system is defined as the maximum power that can be transmitted to the receiving system without loss of synchronism.

Let us consider a single synchronous machine delivering power to a large system through a transmitting system. The power angle equation of such a system have been developed in section 5.4, and it can be used for the analysis of steady state stability if the transient emfs are replaced by steady state emfs.

Let $|E|$ = Magnitude of steady state internal emf of synchronous machine.
$|V|$ = Magnitude of voltage of receiving system.
X = Transfer reactance between the synchronous machine and receiving system.

$$\text{Now, } P_{max} = \frac{|E|\,|V|}{X} \qquad \text{.....(5.41)}$$

$\therefore$ Real power injectd by machine to system, $P_e = P_{max} \sin\delta$ .....(5.42)

Let the system be operating with steady power transfer with a torque angle δ_0. In this operating condition, let the electrical power output be P_{e0}. Now, the mechanical power input P_m is equal to P_{e0} under ideal conditions.

With the power input (P_m) remaining same let us assume that the electrical power output increases by a small amount ΔP. Now the torque angle change by a small amount $\Delta\delta$. Therefore the new value of torque angle is $(\delta_0 + \Delta\delta)$. The electrical power output for this new torque angle can be obtained from equ (5.42)

$$\left.\begin{array}{l}\text{The electrical power output}\\ \text{for a torque angle of } (\delta_0+\Delta\delta)\end{array}\right\} = P_{e0} + \Delta P = P_{max}\sin(\delta_0+\Delta\delta)$$

$$= P_{max}[\sin\delta_0\cos\Delta\delta + \cos\delta_0\sin\Delta\delta] \qquad \text{.....(5.43)}$$

Since $\Delta\delta$ is a small incremental displacement from δ_0,

$$\sin\Delta\delta \approx \Delta\delta \quad \text{and} \quad \cos\Delta\delta \approx 1 \qquad \text{.....(5.44)}$$

$$\therefore P_{e0} + \Delta P = P_{max}\sin\delta_0 + (P_{max}\cos\delta_0)\,\Delta\delta \qquad \text{.....(5.45)}$$

From equation (5.42) when $\delta = \delta_0$ we get,

$$P_e = P_{e0} = P_m \sin\delta_0 \qquad \text{.....(5.46)}$$

From equations (5.45) and (5.46) we can write,

$$\Delta P = (P_{max}\cos\delta_0)\,\Delta\delta \qquad \text{.....(5.47)}$$

The swing equation for the system discussed here, have been developed in section (5.3) [The swing equation is given by equ 5.29]. The nonlinear swing equation can be linearized about the operating point for steady state stability analysis. The nonlinear swing equation is

$$\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \qquad \text{.....(5.48)}$$

We know that the p.u. value of $M = H/\pi f$. Hence the equation (5.48) cane be written as

$$M\frac{d^2\delta}{dt^2} = P_m - P_e \qquad \text{.....(5.49)}$$

For a torque of $\delta = (\delta_0 + \Delta\delta)$, The electrical power P_e in the equ(5.49) should be replaced by $(P_{e0} + \Delta P)$.

$$\therefore M\frac{d^2(\delta_0+\Delta\delta)}{dt^2} = P_m - (P_{e0} + \Delta P) \qquad \text{.....(5.50)}$$

Since δ_0 is constant and $P_m = P_{e0}$, the equation (5.51) can be written as,

$$M\frac{d^2\Delta\delta}{dt^2} = -\Delta P \qquad \text{.....(5.51)}$$

On substituting for ΔP from equ(5.47) in equ(5.50) we get,

$$M\frac{d^2\Delta\delta}{dt^2} = -(P_{max}\cos\delta_0)\,\Delta\delta$$

$$M\frac{d^2}{dt^2}\Delta\delta + P_{max}\cos\delta_0\,\Delta\delta = 0 \qquad \text{.....(5.52)}$$

Let $\frac{d^2}{dt^2} = x^2$ and $P_{max}\cos\delta_0 = C$

$$\therefore M x^2\,\Delta\delta + C\,\Delta\delta = 0$$

$$(Mx^2 + C)\,\Delta\delta = 0 \qquad \text{.....(5.53)}$$

In equ(5.53) since $\Delta\delta \neq 0$, $M x^2 + C = 0$ .....(5.54)

The equation (5.54) is the characteristic equation of the system for small changes. The stability is determined by the roots of the characteristic equation.

$$\therefore x^2 = \frac{-C}{M} \quad (\text{or}) \quad x = \pm\sqrt{\frac{-C}{M}} \qquad \text{.....(5.55)}$$

Case-I : When C is positive , (i.e, $P_{max}\cos\delta_0 > 0$)

In this case the roots are purely imaginary and conjugate. Hence the system behaviour is oscillatory about δ_0. In this analysis the resistances in the system have been neglected. If we include the resistances in the analysis then the roots will be complex conjugate and the response will be damped oscillatory. Therefore in a practical system the system is stable for small increment in power provided, $P_{max}\cos\delta_0 > 0$.

Case-ii : When C is negative, (i.e, $P_{max} \cos \delta_0 < 0$)

In this case the roots are real and equal in magnitude. One of the root is positive and the other one is negative. Due to the positive root the torque angle increases without bound when there is a small increment in power and the machine will loose synchronism. Hence the machine becomes unstable for small changes in power provided $P_{max} \cos\delta_0 < 0$.

Steady state limit

The term $P_{max} \cos \delta_0$ decides the steady state stability of the system and so it is called synchronizing coefficient or stiffness of synchronous machine. From the power angle curve for generator action shown in fig 5.5, we get the range of δ_0 as 0 to π. (The fig 5.5 is drawn by using the equation $P_e = P_{max} \sin \delta$.)

When $0 \le \delta_0 \le \pi/2$; $P_{max} \cos \delta_0$ and P_e are positive

When $\delta_0 = \pi/2$; $P_{max} \cos \delta_0 = 0$ and $P_e = P_{max}$

When $\pi/2 < \delta_0 \le \pi$; $P_{max} \cos \delta_0$ is negative and P_e is positive

From the above discussion we can say that the synchronizing co-efficient ($P_{max} \cos \delta_0$) and real power injected to the system (P_e) are positive when δ is in the range of 0 to π/2. Therefore the maximum power that can be transmitted without loss of stability occurs for $\delta = \delta_0 = \pi/2 = 90°$. The maximum power transmitted is P_{max}.

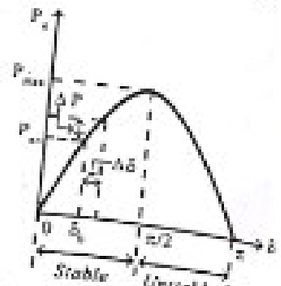


Fig 5.5 : Power angle curve for steady state operation of generator

where, $P_{max} = \frac{|E||V|}{X}$

The fig 5.5 shows the stable and unstable steady state operating regions of a generator. The concepts developed in this section is also applicable for power transfer from one system to another system if the transmitting system is represented by single equivalent generator.

In stable operating region of the system the damping should be sufficient to reduce the oscillations developed due to small changes in loads. If the oscillations exists for a long time then it may pose a problem to system security.

Practically, the system has to be operated below the steady state stability limit. This limit can be improved by reducing the reactance X or by increasing the voltages at sending end and/or at receiving end. The reactance can be reduced by introducing series capacitors in the transmission line. Alternatively the line reactance can be reduced by using two parallel transmission lines. (The two parallel lines improves the reliability of the system).

5.6 TRANSIENT STABILITY

The transient stability of a system is concerned with the study of system behaviour for large disturbances. The short circuits (fault conditions) and switching heavy loads can be treated

The dynamics of the system under transient state are governed by the nonlinear swing equation developed in section 5.3 [equation 5.29]. Since the changes in δ is very large in the transient state, the swing equation cannot be linearized for a general solution and so solution has to be obtained by using any one of the numerical techniques. The most popular methods available for the solution of swing equation are point-by-point method and Runga-Kutta method.

The transient stability of a single machine connected to infinite bus bar can be easily determined by a simple criterion called equal area criterion.

The computational task involved in transient stability studies can be understood by considering the transient state of a practical system. Consider a single machine system shown in fig 5.6, feeding energy through a transmission line to an infinite bus. Let the circuit breakers shown in the figure 5.6 be autoreclosure type. In this type the circuit breaker will open its contact upon sensing a fault and after a small time it will close it contacts, if the fault still exist then again it will open its contact to permanently disconnect the faulty part.

This feature is useful in clearing transient faults. Because the transient fault exists for a small time and it gets cleared when the circuit is opened. Then the circuit can be closed for normal operation.

Most of the autoreclosure circuit breakers will open and close the contacts twice before permanently disconnecting the circuit. In the majority of faults the first reclosure will be successful. Hence the system stability is improved by using autoreclosure circuit breakers.

Let a fault occurs at the sending end of the transmission line in the system shown in fig 5.6. Let the fault be transient in nature and cleared in the transient state. Hence the circuit breaker will close the circuit automatically after the fault is cleared. The transient stability study for this situation involves the following steps.



Fig 5.6 : A single generator feeding energy to infinite bus.

1. Calculate the transient internal emf and torque angle δ_0 using the prefault load currents. (Here δ_0 is the torque angle corresponding to prefault operating point).
2. Determine an equation for power during the fault condition. Let this equation be $P_e(\delta)$. If the fault is 3-phase fault then power transferred to infinite bus is zero and the entire power goes to fault.
3. Calculate $\delta(t)$ for various time instant by solving the swing equation using a numerical technique. The initial value of δ for this solution in δ_0. (Here $\delta(t)$ is the torque angle expressed as a function of time).
4. Let us assume that the fault is cleared when the circuit breaker open its contact the first time. Now $P_e(\delta) = 0$. Continue calculating $\delta(t)$ by taking previous step value as initial condition.

5. Let us assume that the circuit breaker close its contract and power feeding to infinite bus is resumed. For this situation find $P_a(\delta)$ and continue to calculate $\ddot{\delta}(t)$.
6. Now examine the variations of $\delta(t)$. If $\delta(t)$ goes through a maximum value and starts to reduce then the system is a stable system. On the otherhand if $\delta(t)$ remains increasing for a specified length of time then the system is considered unstable.

5.7 EQUAL AREA CRITERION

The transient stability analysis of simple system can be performed by using a simple criterion called equal area criterion.

During the transient state of a power system we may come across the following two situations for changes in δ (torque angle) with respect to time.

i) The δ may increase to a maximum value and then decrease to a stable value. The system is considered as stable.

ii) The δ may keep on increasing indefinitely. In this case the system is unstable.

The above facts can be stated as a stability criterion as given below.

i) The system is stable if, $\frac{d\delta}{dt} = 0$ at some time instant.

ii) The system is unstable if, $\frac{d\delta}{dt} > 0$ for a sufficiently long time (typically 1 second or more)

For a single machine-infinite bus bar system, the stability criterion stated above can be converted to a simple condition as shown below.

Consider the swing equation of a generator connected to infinite bus (which is derived in section 5.3, equ 5.29).

$$\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \qquad \text{.....(5.56)}$$

Let there be a change in P_e due to a large disturbance, with P_m remaining constant.

Now, $P_m - P_e = P_a$ .....(5.57)

where P_a is the accelerating power.

Also we know that p.u value of $M = H/\pi f$. Hence the equ(5.56) can be written as,

$$M\frac{d^2\delta}{dt^2} = P_a$$

$$\therefore \frac{d^2\delta}{dt^2} = \frac{P_a}{M} \qquad \text{.....(5.58)}$$

On multiplying the equ(5.58) by $2\frac{d\delta}{dt}$ we get,

$$2\frac{d\delta}{dt}\frac{d^2\delta}{dt^2} = 2\frac{d\delta}{dt}\frac{P_a}{M}$$

$$2\frac{d\delta}{dt}\frac{d}{dt}\frac{d\delta}{dt} = \frac{2}{M}P_a\frac{d\delta}{dt}$$

$$\frac{d}{dt}\left(\frac{d\delta}{dt}\right)^2 = \frac{2}{M}P_a\frac{d\delta}{dt}$$

$$d\left(\frac{d\delta}{dt}\right)^2 = \frac{2}{M}P_a\,d\delta$$

On integrating the equation (5.59) we get

$$\left(\frac{d\delta}{dt}\right)^2 = \frac{2}{M}\int_{\delta_0}^{\delta} P_a\,d\delta \qquad \text{.....(5.60)}$$

where δ_0 is the initial value of torque angle or rotor angle

On taking square root of equ(5.60) we get,

$$\frac{d\delta}{dt} = \sqrt{\frac{2}{M}\int_{\delta_0}^{\delta} P_a\,d\delta} \qquad \text{.....(5.61)}$$

For a stable system $\frac{d\delta}{dt} = 0$, at a particular time instance. There fore for a stable system the equation (5.61) can be written as

$$\sqrt{\frac{2}{M}\int_{\delta_0}^{\delta} P_a\,d\delta} = 0 \qquad \text{.....(5.62)}$$

The equation (5.62) is zero if the integral of P_a is zero. Hence we can say that,

for $\frac{d\delta}{dt} = 0$, the term $\int_{\delta_0}^{\delta} P_a\,d\delta = 0$ .....(5.63)

The physical meaning of integration is the estimation of the area under the curve. Hence the integral of P_a equal to zero refers to zero area. Hence the condition of stability can be stated as : the system is stable if the area under $P_a - \delta$ curve reduces to zero at some value of δ. This is possible only if the positive (accelerating) area under $P_a - \delta$ curve is equal to the negative (decelerating) area under $P_a - \delta$ curve for a finite change in δ. Hence this stability criterion is called equal area criterion of stability.

The equal area criterion of stability can be applied to any type of disturbances that may occur in a single machine-infinite bus bar system. The transient stability study for a sudden change in mechanical input using equal area criterion in presented in this section.

Transient stability analysis for a sudden change in mechanical input

Consider a single generator feeding energy to infinite bus as shown in fig 5.7. The electrical power transmitted by the generator is given by

$$P_e = \frac{|E'||V|}{X} \sin\delta = P_{max} \sin\delta \qquad \text{—(5.64)}$$



Fig 5.7.

where, $P_{max} = \frac{|E'||V|}{X}$

Let the generator be operating in steady state with a torque angle δ_0. At this condition the mechanical power input is P_{m0} and electrical power output is P_{e0}. Under ideal conditions $P_{m0} = P_{e0}$. Hence from equ(5.64) for this operating condition we can write,

$$P_{m0} = P_{e0} = P_{max} \sin\delta_0 \qquad \text{—(5.65)}$$

The power angle curve of the generator is shown in fig 5.8. In this the steady state operating point as described by equ(5.65) is the point-a.

Let the mechanical input to the generator rotor be suddenly increased to P_{m1} (by some adjustment in prime mover). Since the mechanical power is more than electrical power, the generator will have an accelerating power, P_a given by

$$P_a = P_{m1} - P_e \qquad \text{—(5.66)}$$

where $P_e = P_{max} \sin\delta$

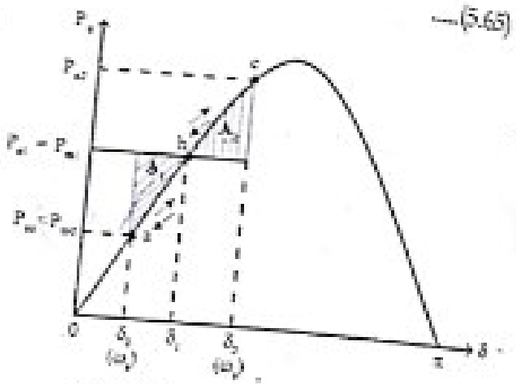


Fig 5.8 : Power angle curve for sudden increase in mechanical input to generator

Due to accelerating power the rotor speed increases and so the rotor angle also increases. This results in increased electrical power generation. Therefore the operating point will move upwards along the power angle curve. At point-b again the mechanical power P_{m1} equals the electrical power P_{e1}, where P_{e1} is the electrical power output corresponding to torque angle δ_1.

Now the rotor angle cannot stay at this point, because the inertia of the rotor will make the rotor to oscillate with respect to point-b. Hence the torque angle will continue to increase till point-c, when the operating point moves from b to c, the electrical power is more than mechanical power. Therefore the power, P_a given by equ(5.66) is negative and it is called decelerating power.

In this region (i.e., from point-b to c) the rotor angle δ increases but the rotor speed decreases due to decelerating power. The point-c is decided by the damping of the system (because the oscillations in the rotor are reduced by damping). At point-c the speed of rotor will be equal to synchronous speed.

Note : *At point-a the speed is synchronous speed (ω_s). From point a to b the speed increases and then from point b to c the speed decreases. Once again at point-c the speed is equal to synchronous speed (ω_s). Thus the rotor oscillates between point-a and point-c before settling to point-b.*

In fig.5.8, the area A_1 is the acceleration area and area A_2 is the deceleration area. The equal area criterion says that, the system is stable if,

$$\int_{\delta_0}^{\delta_2} P_a \, d\delta = 0 \qquad \text{—(5.67)}$$

To satisfy equ(5.67) the acceleration area A_1, should be equal to deceleration area A_2. When the oscillations die out the system will settle to a new state. In this new steady state, $P_{m1} = P_{e1}$.

$$\therefore P_{m1} = P_{e1} = P_{max} \sin\delta_1$$

The areas A_1 and A_2 can be evaluated as shown below.

$$A_1 = \int_{\delta_0}^{\delta_1} (P_{m1} - P_e) \, d\delta \qquad \text{—(5.68)}$$

$$A_2 = \int_{\delta_1}^{\delta_2} (P_e - P_{m1}) \, d\delta \qquad \text{—(5.69)}$$

where, $P_e = P_{max} \sin\delta$ —(5.70)

From the above discussions we can say that there is a upper limit for increase in mechanical power input P_{m1}. As the mechanical power is increased, a limiting condition is finally reached at a point where the area A_1 equals the area above the $P_{e1,max}$ line as shown in fig 5.9. The corresponding δ can be $\delta_{1,max}$.

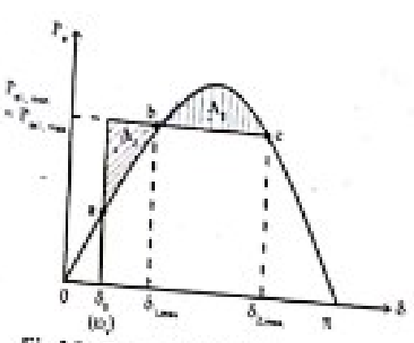


Fig 5.9 : Transient stability limit for sudden change in mechanical input

Under this condition δ_2 takes a maximum value of $\delta_{2,max}$.

Here, $\delta_{2,max} = \pi - \delta_{1,max}$ —(5.71)

From equ(5.64) we can write,

$$P_{e1,max} = P_{m1,max} = P_{max} \sin\delta_{1,max}$$

$$\therefore \sin\delta_{1,max} = \frac{P_{m1,max}}{P_{max}} \quad \text{(or)} \quad \delta_{1,max} = \sin^{-1}\left(\frac{P_{m1,max}}{P_{max}}\right) \qquad \text{—(5.72)}$$

From equations(5.71) and (5.72) we can write,

$$\delta_{2,max} = \pi - \sin^{-1}\left(\frac{P_{m1,max}}{P_{max}}\right)$$

From fig 5.9 we can say that, any further increase in P_{accn} will make the area A_2 less than the area A_1. This means that the acceleration power is more than the deceleration power. Hence the system will have an excess kinetic energy which causes δ to increase beyond point-c. If the δ increases beyond point-c the deceleration power changes to acceleration power and so the system become unstable.

Note : *It is important to note that the system will remain stable even though the rotor may oscillate beyond δ = 90°, as long as the equal area criterion is met. Hence the condition of δ = 90° for stability is applicable only for steady state stability and not for transient stability.*

Clearing time and clearing angle

Consider a single machine system shown in fig 5.10. Let the mechanical input be P_m and the machine is operating in steady state with torque angle, δ_0. The power angle curve for this system is shown in fig 5.11. The operating point is shown as point-a.

Let a three phase fault occur at point F in the system. Now $P_e = 0$ and the operating point drops to b. This means that the power transferred to infinite bus is zero and the entire power generated is flowing through fault. Now the operating point moves along bc.

Let the fault be transient in nature and so the fault be cleared by opening of the circuit breaker at point-c, where $\delta = \delta_c$ and the corresponding time be t_c. Here t_c is called clearing time and δ_c is called clearing angle.

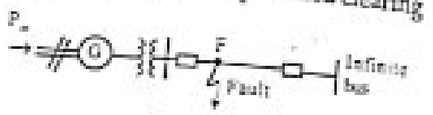


Fig 5.10.

It is assumed that the circuit breaker closes its contact immediately after opening. Hence the normal operation is restored. Now the operating point shifts to point-d. The rotor now decelerates and the operating point moves along de.

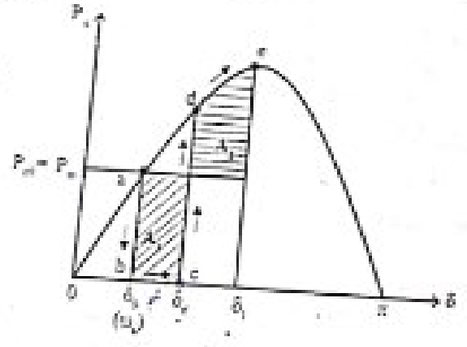


Fig 5.11

For this transient state, if an angle δ_1, can be found such that, $A_2 = A_1$, then the system is found to be stable. The stable system may finally settles down to the steady operating point-a in an oscillatory manner due to damping in the system.

In the above discussion it is assumed that the fault is cleared at δ_c, but if the fault clearing is delayed then the angle δ_c continue to increase to an upper limit, δ_{max}. This corresponds to a point where equal areas for A_1 and A_2 can be found for a given P_m as shown in fig 5.12.

For this situation the fault would have been cleared at an angle δ_{cr}, as shown in fig 5.12. This angle δ_{cr} is called critical clearing angle. The time corresponding to this angle is called critical clearing time, t_{cr}.

If the fault is not cleared within critical time, then δ_1 would increase to a value greater than δ_{max}. In such a situation the area A_2 will be less than the area A_1 and so the system would be unstable.

For a 3-phase fault in simple systems, the equations for δ_{cr} and t_{cr} can be obtained as shown below.

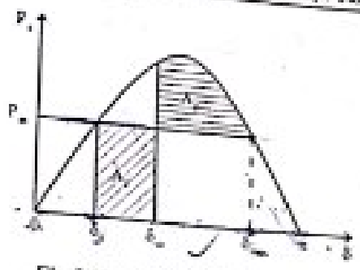


Fig 5.12 : Critical clearing angle

From fig 5.12, we get

$$\delta_{max} = \pi - \delta_c$$

Under steady state, for a given δ_0, $P_m = P_e$ and it is constant .....(5.73)

$$\therefore P_m = P_e = P_{max} \sin \delta_0$$

The acceleration power, $P_a = P_m - P_e$ .....(5.74)

When a three phase fault occurs, $P_e = 0$. .....(5.75)

$$\therefore P_a = P_m = \text{constant}$$

The acceleration area A_1 can be evaluated by integrating P_a of equ(5.76) from $\delta = \delta_0$ to $\delta = \delta_{cr}$. .....(5.76)

$$\therefore A_1 = \int_{\delta_0}^{\delta_{cr}} P_m \, d\delta = P_m [\delta]_{\delta_0}^{\delta_{cr}} = P_m (\delta_{cr} - \delta_0)$$ .....(5.77)

When the power feeding is resumed after the fault, $P_e = P_{max} \sin \delta$.

Now, $P_a = P_e - P_m = P_{max} \sin \delta - P_m$ .....(5.78)

The deacceleration area A_2 can be evaluated by integrating P_a of equ(5.78) from $\delta = \delta_{cr}$ to $\delta = \delta_{max}$.

$$\therefore A_2 = \int_{\delta_{cr}}^{\delta_{max}} (P_{max} \sin\delta - P_m) \, d\delta = [-P_{max} \cos\delta - P_m \delta]_{\delta_{cr}}^{\delta_{max}}$$

$$= [-P_{max} \cos\delta_{max} - P_m \delta_{max} + P_{max} \cos\delta_{cr} + P_m \delta_{cr}]$$

$$= P_{max} (\cos\delta_{cr} - \cos\delta_{max}) - P_m (\delta_{max} - \delta_{cr})$$ .....(5.79)

For a stable system $A_1 = A_2$. Hence the equations (5.77) and (5.79) can be equated to solve δ_{cr}.

On equating (5.77) and (5.79) we get

$$P_{max} (\cos\delta_{cr} - \cos\delta_{max}) - P_m (\delta_{max} - \delta_{cr}) = P_m (\delta_{cr} - \delta_0)$$

$$\therefore P_{max} \cos\delta_{cr} = P_m \delta_{cr} - P_m \delta_0 + P_{max} \cos\delta_{max} + P_m \delta_{max} - P_m \delta_{cr}$$

$$P_{max} \cos \delta_{cc} = P_m (\delta_{max} - \delta_0) + P_{max} \cos \delta_{max}$$

$$\therefore \cos \delta_{cc} = \frac{P_m}{P_{max}} (\delta_{max} - \delta_0) + \cos \delta_{max}$$

$$\text{(or)} \quad \delta_{cc} = \cos^{-1} \left[\frac{P_m}{P_{max}} (\delta_{max} - \delta_0) + \cos \delta_{max} \right] \qquad \text{---(5.80)}$$

The equation (5.80) can be used to estimate the value of critical clearing angle, δ_{cc}.

Consider the swing equation of single machine system,

$$\frac{H}{\pi f} \frac{d^2\delta}{dt^2} = P_m - P_e$$

During a three phase fault, $P_e = 0$ --- (5.81)

$$\therefore \frac{H}{\pi f} \frac{d^2\delta}{dt^2} = P_m$$

$$\frac{d^2\delta}{dt^2} = \frac{\pi f P_m}{H}$$

On integrating equ(5.82) twice we get .....(5.82)

$$\delta = \frac{\pi f}{2H} P_m t^2 + \delta_0$$

where δ_0 is the integral constant .....(5.83)

In equ(5.83) when $\delta = \delta_{cc}, t = t_{cc}$.

$$\therefore \text{at} \quad t = t_{cc}, \qquad \delta_{cc} = \frac{\pi f}{2H} P_m t_{cc}^2 + \delta_0$$

$$\left.\begin{aligned} \frac{\pi f}{2H} P_m t_{cc}^2 &= \delta_{cc} - \delta_0 \\ \therefore t_{cc} &= \sqrt{\frac{2H(\delta_{cc} - \delta_0)}{\pi f P_m}} \end{aligned}\right\} \qquad \text{---(5.84)}$$

The equation (5.84) can be used to estimate the value of critical clearing time, t_{cc}.

Note : *Since $P_e = 0$, it is possible to find an explicit solution for δ by integrating the swing equation.*

5.8 SOLUTION OF SWING EQUATION BY POINT-BY-POINT METHOD

Consider the swing equation of a power system.

$$\frac{H}{\pi f} \frac{d^2\delta}{dt^2} = P_m - P_e$$

We know that, $M = \frac{H}{\pi f}$ ---(5.85)

$$P_e = P_{max} \sin \delta \qquad \text{---(5.86)}$$

$$P_a = P_m - P_e = P_m - P_{max} \sin \delta \qquad \text{---(5.87)}$$

$$\therefore M \frac{d^2\delta}{dt} = P_a \qquad \text{---(5.88)}$$

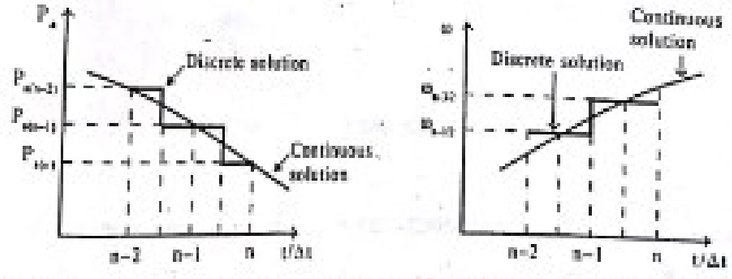
$$\frac{d^2\delta}{dt} = \frac{P_a}{M}$$

The equation (5.89) is a nonlinear equation. During transient state the δ is a function of time, t and so it can be denoted as $\delta(t)$. In point-by-point method, the solution of $\delta(t)$ is obtained by dividing the time into small equal values of Δt. (i.e., the entire time (range) of interest is divided into number of small equal interval Δt). .....(5.89)

The accelerating power and the change in speed are also continuous function of time. They are discretized as follows,

1. *The accelerating power P_a computed at the beginning of an interval is assumed to remain constant from the middle of the preceding interval to the middle of the interval being considered.*
2. *The angular velocity is assumed constant throughout any interval. This constant value is the value corresponding to the midpoint of concerned interval.*

The discretization of P_a and ω are shown in Fig 5.13.



Note : *The n represents discretized time instant and it pertains to the end of the interval.*

Fig 5.13 : *Discretization of P_a and ω.*

For each discrete interval the values of P_a, ω and δ are calculated as shown below.

The solution starts from the initial condition values, which corresponds to a stable operating point. Let δ_0 be the angle corresponding to initial operating point.

Let us assume that, the values ω and δ for $(n-1)^{th}$ interval are known

δ_{n-1} = The value of δ at the end of $(n-1)^{th}$ interval

$\omega_{n-1/2}$ = The value of ω at the end of $(n-1)^{th}$ interval

$P_{a(n-1)}$ = Value of P_a at the end of $(n-1)^{th}$ interval.

Now from equ(5.89) we get,

$$P_{a(n-1)} = P_m - P_{max} \sin \delta_{n-1}$$

The equ (5.89) can be written in the modified form as shown in equ(5.91)

$$\frac{d\omega}{dt} = \frac{P_a}{M} \quad \text{....(5.90)}$$

where, $\omega = \frac{d\delta}{dt}$ and $\frac{d\omega}{dt} = \frac{d^2\delta}{dt^2}$ ....(5.91)

For small changes in δ, the equation (5.91) can be linearized as shown below

$$\frac{\Delta\omega}{\Delta t} = \frac{P_a}{M}$$

$$\therefore \Delta\omega = \frac{\Delta t}{M} P_a$$

Let $\omega_{n-1/2}$ = The value of ω at the end of nth interval. ....(5.92)

For calculating n^{th} interval value of ω, put, $\Delta\omega = \omega_{n-1/2} - \omega_{n-3/2}$ and $P_a = P_{a(n-1)}$ in equ(5.92).

$$\therefore \omega_{n-1/2} - \omega_{n-3/2} = \frac{\Delta t}{M} P_{a(n-1)}$$

$$\therefore \omega_{n-3/2} = \omega_{n-1/2} - \frac{\Delta t}{M} P_{a(n-1)}$$

For small changes in δ, we can write

$$\omega = \frac{\Delta\delta}{\Delta t} \quad \text{....(5.93)}$$

To solve for change in δ in $(n-1)^{th}$ interval put $\Delta\delta = \Delta\delta_{n-1}$ and $\omega = \omega_{n-3/2}$ in equ(5.94)

$$\therefore \Delta\delta_{n-1} = \Delta t\, \omega_{n-3/2} \quad \text{....(5.94)}$$

Similarly for change in δ in n^{th} interval put $\Delta\delta = \Delta\delta_n$ and $\omega = \omega_{n-1/2}$ in equ(5.94)

$$\therefore \Delta\delta_n = \Delta t\, \omega_{n-1/2} \quad \text{....(5.95)}$$

Let δ_n = The value of δ at the end of nth interval. ....(5.

Now, $\delta_n = \delta_{n-1} + \Delta\delta_n$ ....(5.97)

The above process of computation is repeated to obtain $P_{a(n)}$, $\Delta\delta_{(n+1)}$ and δ_{n+1}. The solution of $\delta(t)$ is thus obtained in discrete form over the desired length of time. The normal desired length of time is 0.5 sec.

The continuous form of solution is obtained by drawing a smooth curve through discrete values as shown in fig.5.14.



Fig 5.14 : Solution of swing equation by point-by-point method

5.9 METHODS OF IMPROVING TRANSIENT STABILITY

The transient stability depends on the type and location of a fault. Let us consider the most severe type of fault, i.e., 3-phase fault and the following discussions are applicable for transient conditions due to three phase fault.

An increase in the inertia constant M of a single machine connected to infinite bus, reduces the angle through which the rotor swings in a given time interval. Hence stability can be improved by increasing M.

However this cannot be employed in practice because of economic reasons. Also, increasing M will have an undesirable effect of slowing down the response of the speed governor loops.

It can be proved that for a given clearing angle, the acceleration area decreases but the deceleration area increases as the maximum power limit of a power angle curve is raised which improves the transient stability limit of the system. The above conclusion suggest the following methods of improving the transient stability limit of a power system.

1. Increase of system voltages, use of AVR (Automatic Voltage Regulators)
2. Use of high speed excitation systems.
3. Reduction in system transfer reactance.
4. Use of high speed reclosing breakers.

During fault, the voltage reduction of all buses at the generator terminals are sensed by the automatic voltage regulators which help to restore the generator voltage. Modern exciter systems having solid state controls which quickly respond to bus voltage reduction.

The transfer reactance is reduced to improve the stability limit. Incidentally this also raises the system voltage ... reactance of a transmission line can be decreased (i) by ... conductor diameter.

... is an effective and economical ... distances of more than 350 km. ... of load voltages and raise the ... limit.

... the number of parallel lines ... unhealthy lines followed by

Some of the recent methods of improving stability are,

1. **HVDC Links :** *Increased use of HVDC links employing thyristors would deviate stability problem. A dc link is asynchronous. There is no risk of a fault in one system causing loss of stability in the other system.*

2. **Breaking Resistors :** *For improving stability where clearing is delayed or large load is suddenly lost, a resistive load called a breaking resistor is connected at or near the generator bus. This load compensates for some of the reduction of load on the generators.*

3. **Bypass Valving :** *In this method, the stability of a unit is improved by decreasing the mechanical input power to the turbine.*

4. **Full load Rejection Technique :** *Fast valving combined with high-speed clearing time will sufficient to maintain stability in most cases. However, there are still situations where stability is difficult to maintain. To remedy these situations, a full load rejection scheme could be utilised after the unit is spectrated from the system. To do this, the unit has to be equipped with a large steam bypass system. After system has recovered from the shock caused by the fault, the unit could be resynchronised and reloaded. The main disadvantage is the extra cost of the large bypass system.*

EXAMPLE 5.1.

The generator shown in fig 5.1.1. is delivering power to infinite bus. Take $|V_1| = 1.1$ p.u. Find the maximum power that can be transferred when

(i) *The system is healthy*

(ii) *One line is open*

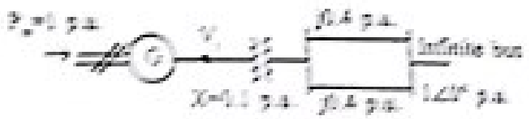


Fig 5.1.1.

SOLUTION

The reactance diagram of the system is shown in fig 5.1.2.

Here, $V_1 = |V_1| \angle\delta$ and $V = 1\angle 0°$ p.u.

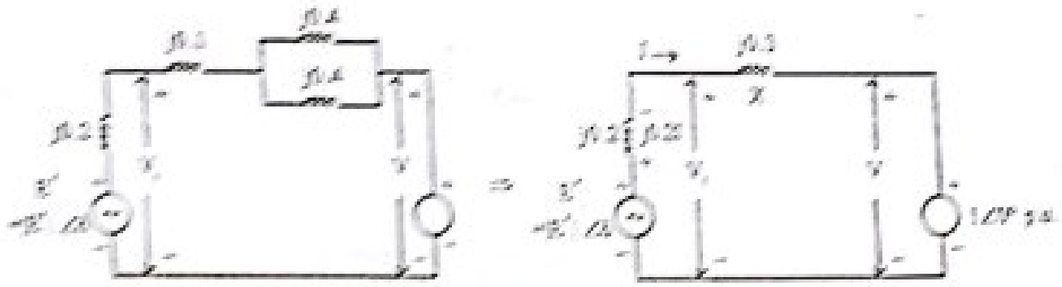


Fig 5.1.2

Let electrical power transferred at steady state be equal to P_m.

$\therefore P_e = P_m = 1$ p.u.

We know that,

$$P_e = \frac{|V_1||V|}{X}\sin\theta$$

$$\therefore \theta = \sin^{-1}\left[\frac{P_e X}{|V_1||V|}\right] = \sin^{-1}\left[\frac{1\times 0.3}{1.1\times 1}\right] = 15.8°$$

$$\therefore V_1 = |V_1|\angle\theta = 1.1\angle 15.8° = 1.058 + j0.3$$

$$\text{Current, } I = \frac{V_1 - V}{jX} = \frac{1.058 + j0.3 - 1}{j0.3} = \frac{0.058 + j0.3}{0.3\angle 90°} = \frac{0.3056\angle 79°}{0.3\angle 90°} = 1.0187\angle -11° \text{ p.u.}$$

In fig 5.1.2. using KVL we can write,

The transient internal voltage, $E' = jX_1 I + V_1$

$$= j0.2 \times 1.0187\angle -11° + 1.1\angle 15.8°$$
$$= 0.2\angle 90° \times 1.0187\angle -11° + 1.058 + j0.3$$
$$= 0.2037\angle 79° + 1.058 + j0.3$$
$$= 0.039 + j0.2 + 1.058 + j0.3$$
$$= 1.097 + j0.5 = 1.2056\angle 24.5° \text{ p.u.}$$

$\therefore |E'| = 1.2056$ p.u. and $\delta = \delta_c = 24.5°$

case (i) Healthy system

Let X_{12} be the reactance between the transient internal emf and the source representing infinite bus.

From fig 5.1.2, $X_{12} = 0.5$ p.u.

$$\therefore P_{max} = \frac{|E'||V|}{X_{12}} = \frac{1.2056\times 1}{0.5} = 2.4112 \text{ p.u.}$$

$$\therefore P_e = P_{max}\sin\delta = 2.4112\sin\delta$$

Case (ii) When one line is open

The reactance diagram when one line is open is shown in fig 5.1.3.

Let X_{12} = Transfer reactance, from fig 5.1.3.

$$X_{12} = 0.2 + 0.1 + 0.4 = 0.7 \text{ p.u.}$$

$$\therefore P_{max} = \frac{|E'||V|}{X_{12}} = \frac{1.2056\times 1}{0.7} = 1.722 \text{ p.u.}$$

$$\therefore P_e = P_{max}\sin\delta = 1.722\sin\delta$$

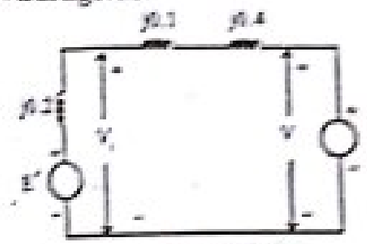


Fig 5.1.3

EXAMPLE 5.2.

Find the steady state power limit of a system consisting of a generator with reactance 0.6 p.u., connected to an infinite bus through a reactance of 0.8 p.u. The terminal of the generator is 1.15 p.u. and the voltage of infinite bus is $1\angle 0°$ p.u.

SOLUTION

The reactance diagram of the system is shown in fig 5.2.1.

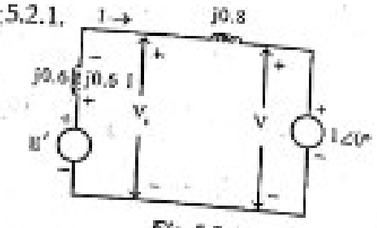


Fig 5.2.1

Given that, $V = 1\angle 0°$ and $|V_t| = 1.15$.

Let, $V_t = 1.15\angle\theta$

From fig 5.2.1. we get,

$$I = \frac{V_t - V}{j0.8}$$

$$E' = j0.6\, I + V_t = j0.6\left(\frac{V_t - V}{j0.8}\right) + V_t = 0.75(V_t - V) + V_t = 1.75\, V_t - 0.75\, V$$

$$= 1.75 \times 1.15\angle\theta - 0.75 \times 1\angle 0° = 2.0125\angle\theta - 0.75 = 2.0125\cos\theta + j2.0125\sin\theta - 0.75$$

$$= (2.0125\cos\theta - 0.75) + j(2.0125\sin\theta)$$

Consider the transient internal emf in polar form. —(5.2.1)

$$\text{i.e., } E' = |E'|\angle\delta = |E'|\cos\delta + j\,|E'|\sin\delta \quad \text{.....(5.2.2)}$$

When the steady state power limit is reached, the value of δ is 90°, when $\delta = 90°$, $\cos\delta = 0$ and so the real part of equ(5.2.2) is zero. Therefore at steady state limit, the real part of equ(5.2.1.) is also zero.

$$\therefore 2.0125\cos\theta - 0.75 = 0$$

$$2.0125\cos\theta = 0.75$$

$$\theta = \cos^{-1}\left(\frac{0.75}{2.0125}\right) = 68.1°$$

$$\therefore V_t = |V_t|\angle\theta = 1.15\angle 68.1° \text{ p.u.}$$

$$\text{Now, } I = \frac{V_t - V}{j0.8} = \frac{1.15\angle 68.1° - 1\angle 0°}{j0.8} = \frac{0.4289 + j1.067 - 1}{0.8\angle 90°}$$

$$= \frac{-0.5711 + j1.067}{0.8\angle 90°} = \frac{1.21\angle 118.2°}{0.8\angle 90°} = 1.5125\angle 28.2° = 1.333 + j0.715 \text{ p.u.}$$

$$E' = j0.6\, I + V_t = 0.6\angle 90° \times 1.5125\angle 28.2° + 1.15\angle 68.1°$$

$$= 0.9075\angle 118.2° + 1.15\angle 68.1° = -0.4288 + j0.8 + 0.4289 + j1.067$$

$$= 0.0001 + j1.867 = 1.867\angle 90° \text{ p.u.}$$

$$\text{Steady state power limit, } P_{max} = \frac{|E'|\,|V|}{0.6 + 0.8} = \frac{1.867 \times 1}{1.4} = 1.3336 \text{ p.u.}$$

EXAMPLE 5.3

A synchronous generator having a reactance of 1 p.u. is connected to an infinite bus through a transmission system with a reactance of 0.7 p.u. The generator is running on no-load with a voltage of 1.1 p.u. Take H = 4.5 MW-s/MVA. The voltage of infinite bus is $1\angle 0°$ p.u. and its frequency is 50 Hz.

Calculate the frequency of natural oscillations if the machine is suddenly loaded to (i) 60% and (ii) 75% of its maximum power limit. Neglect the resistance and machine damping.

SOLUTION

Maximum power limit, $P_{max} = \frac{|E||V|}{X}$

Given that, $|E| = 1.1$ p.u. ; $|V| = 1$ p.u. ; $X = (1 + 0.7)$ p.u.

$$\therefore P_{max} = \frac{1.1 \times 1}{1.7} = 0.6471 \text{ p.u.}$$

The inertia constant, M = H/πf in p.u.

Given that, H = 4.5 MW-s/MVA and f = 50 Hz.

$$\therefore M = \frac{4.5}{\pi \times 50} = 0.0287 \text{ p.u.}$$

Case (i) : 60% loading

The electrical power, $P_e = P_{max}\sin\delta$

For 60% loading, let $\delta = \delta_1$ and $P_e = P_{s1}$

It is given that, $\frac{P_{s1}}{P_{max}} = 60\% = 0.6$

$$\therefore P_{s1} = P_{max}\sin\delta_1 \quad \text{(or)} \quad \sin\delta_1 = \frac{P_{s1}}{P_{max}} = 0.6$$

$$\therefore \delta_1 = \sin^{-1} 0.6 = 36.9° \text{ (elect.deg)}$$

The characteristic equation of the system during small disturbances is given by

$$Mx^2 + C = 0$$

where, $C = P_{max}\cos\delta_1$

The natural frequency of oscillation of the system for small disturbances is given by the roots of characteristic equation.

$$\text{The root of characteristic equation, } x = \pm\sqrt{\frac{-C}{M}} = \pm j\sqrt{\frac{P_{max}\cos\delta_1}{M}}$$

$$= \pm j\sqrt{\frac{0.6471 \times \cos 36.9°}{0.0287}} = \pm j4.25$$

∴ Natural frequency of oscillation = 4.25 rad / sec = $\frac{4.25}{2\pi}$ = 0.6764 Hz.

Case (ii) : 75 % loading

The electrical power, $P_e = P_{max} \sin\delta$

For 85% loading, let $\delta = \delta_2$ and $P_e = P_{e2}$

It is given that, $\frac{P_{e2}}{P_{max}} = 75\% = 0.75$

$$\therefore P_{e2} = P_{max} \sin\delta_2 \quad \text{(or)} \quad \sin\delta_2 = \frac{P_{e2}}{P_{max}} = 0.85$$

$$\therefore \delta_2 = \sin^{-1} 0.75 = 48.6^\circ \text{ (elect. deg)}$$

The characteristic equation of the system during small disturbances is given by

$$Mx^2 + C = 0$$

where, $C = P_{max} \cos\delta_2$

The natural frequency of oscillation of the system for small disturbances is given by the roots of characteristic equation.

The root of characteristic equation, $x = \pm\sqrt{\frac{-C}{M}} = \pm j\sqrt{\frac{P_{max}\cos\delta_2}{M}}$

$$= \pm j\sqrt{\frac{0.6471 \times \cos 48.6^\circ}{0.0287}} = \pm j3.86$$

∴ Natural frequency of oscillation = 3.86 rad / sec = $\frac{3.86}{2\pi}$ = 0.6143 Hz.

EXAMPLE 5.4

Two power stations A and B are located close together. Station A has four identical generator sets each rated 100 MVA and having an inertia constant of 9 MJ/MVA whereas the station B has 3 sets each rated 200 MVA, 4 MJ/MVA. Calculate the inertia constant of a single equivalent machine on a base of 100 MVA.

SOLUTION

Let us assume that the machines are swinging coherently (i.e., working in unison). For two machines swinging coherently the equivalent inertia constant, H_{eq} is given by

$$H_{eq} = \frac{H_{1,mach}\, S_{1,mach}}{S_{sys}} + \frac{H_{2,mach}\, S_{2,mach}}{S_{sys}}$$

where, S_{sys} = MVA rating of system (or specified base)

$S_{1,mach}$ & $H_{1,mach}$ = MVA rating and inertia constant of machine-1.

$S_{2,mach}$ & $H_{2,mach}$ = MVA rating and inertia constant of machine-2.

The equation(5.4.1) can be extended to any number of machines swinging coherently.

Station-A

The station-A has four machines of identical rating

∴ Equivalent inertia constant of station A = $H_A = 4\left(\frac{H_{mach}\, S_{mach}}{S_{sys}}\right)$

Given that, H_{mach} = 9 MJ/MVA ; S_{mach} = 100 MVA and S_{sys} = 100 MVA

$$\therefore H_A = 4\left(\frac{9 \times 100}{100}\right) = 36 \text{ MJ / MVA}$$

Station-B

The station-B has three machines of identical rating

∴ Equivalent inertia constant of station B = $H_B = 3\left(\frac{H_{mach}\, S_{mach}}{S_{sys}}\right)$

Given that, H_{mach} = 4 MJ/MVA ; S_{mach} = 200 MVA and S_{sys} = 100 MVA

$$\therefore H_B = 3\left(\frac{4 \times 200}{100}\right) = 24 \text{ MJ / MVA}$$

H_{eq} of system

The inertia constants of station-A and station-B are calculated on a common system base MVA (of 100 MVA). Hence the equivalent inertia constant of the system is given by the sum of inertia constants of the two stations.

∴ Equivalent inertia constant of the system = $H_{eq} = H_A + H_B = 36 + 24 = 60$ MJ/MVA

EXAMPLE 5.5.

A 2-pole 50 Hz, 11 kV turbo alternator has a ratio of 100 MW, power factor 0.85 lagging. The rotor has a moment of inertia of 10,000 kg m². Calculate H and M.

GIVEN DATA

pole, p = 2	Rated voltage, V = 11 kv
frequency, f = 50 Hz	Rated power, P = 100 MW
power factor = 0.85 lag	Moment of inertia, J = 10,000 kg m²

SOLUTION

Synchronous speed in rps, $n_s = \frac{2f}{p} = \frac{2 \times 50}{2} = 50$ rps

Synchronous speed in rad / sec, $\omega_s = 2\pi n_s = 2\pi \times 50 = 314.16$ elect. rad / sec

(Note : One revolution is equal to 2π radians)

Inertia constant, $M = J\left(\frac{2}{P}\right)^2 \omega_s \times 10^{-6}$ in MJ - s / elect. rad

$\therefore M = 10,000 \times \left(\frac{2}{2}\right)^2 \times 314.16 \times 10^{-6} = 3.146$ MJ - s / elec. rad

MVA rating of machine, $S = \frac{P}{pf} = \frac{100}{0.85} = 117.675$ MVA

Let us choose the rated voltage and MVA as base values.

$\therefore kV_b = 11\,kV$ and $MVA_b = 117.65$ MVA

Inertia constant, M in p.u., $M_{pu} = \frac{M \text{ in MJ - s / elect. rad}}{MVA_b} = \frac{3.1416}{117.65} = 0.0267$ p.u.

The inertia constant, $H = \pi f M_{pu} = \pi \times 50 \times 0.0267 = 4.194$ MW - s / MVA

RESULT

Inertia constant, $M = 3.1416$ MJ-s/elect.rad (or) 0.0267 p.u.

Inertia constant, $H = 4.194$ MW-s/MVA.

5.10 SHORT QUESTIONS AND ANSWERS

1. *Define stability*

The stability of a system is defined as the ability of power system to return to stable (synchronous) operation when it is subjected to a disturbance.

2. *Define steady state stability*

The steady state stability is defined as the ability of a power system to remain stable (i.e, without loosing synchronism) for small disturbances.

3. *Define transient stability*

The transient stability is defined as the ability of a power system to remain stable (i.e., without loosing synchronism) for large disturbances.

4. *What is steady state stability limit?*

The steady state stability limit is the maximum power that can be transmitted by a machine (or transmitting system) to a receiving system without loss of synchronism. In steady state the power transferred by synchronous machine (or power system) is always less than the steady state stability limit.

5. *What is transient stability limit?*

The transient stability limit is the maximum power that can be transmitted by a machine (or transmitting system) to a fault or a receiving system during a transient state without loss of synchronism. The transient stability limit is always less than the steady state stability limit.

6. *How stability studies are classified. What are they?*

The stability studies are classified depending on the nature of the disturbance.

Depending on the nature of disturbance the stability studies can be classified into the following three types.

1. Steady state stability
2. Dynamic stability
3. Transient stability.

The steady state stability study is concerned with the determination of upper limit of loading without loss of synchronism.

The dynamic stability study is concerned with the study of nature of oscillations and its decay for small disturbances.

The transient stability study is concerned with the study of dynamics of a system for large disturbances.

7. *Write down the units of inertia constants M and H and their interrelationship.*

The unit of M is MJ-s/elect.rad or MJ-s/mech.rad.

The unit of H is MJ/MVA or MW-s/MVA

The M and H are related by the equation,

$$M = \frac{HS}{\pi f}$$

where, S = MVA rating of machine

f = Frequency in Hz

8. *If two machines are swinging coherently with inertia M_1 and M_2 what will be the inertia of the equivalent machine?*

The equivalent moment of inertia, $M_{eq} = \frac{M_1 S_1}{S_b} + \frac{M_2 S_2}{S_b}$

where, S_1 & S_2 = MVA rating of machine 1 & 2 respectively

S_b = Base MVA or MVA rating of system.

9. *Give typical values of inertia constant, H for turbo generators, synchronous condensers and water wheel generators.*

Turbo generators	-	4.0 to 9.0 MW-s/MVA
Synchronous condensers	-	1.0 to 1.25 MW-s/MVA
Waterwheel generator	-	2.0 to 4.0 MW-s/MVA

10. *In a 3-machine system having ratings S_1, S_2 and S_3 and inertia constants M_1, M_2 and M_3, what is the inertia constants M and H of the equivalent system.*

$$M_{eq} = \frac{M_1 S_1}{S_b} + \frac{M_2 S_2}{S_b} + \frac{M_3 S_3}{S_b}$$

where, S_1, S_2, S_3 = MVA ratings of machines 1,2,3, respectively

S_b = Base MVA or MVA rating of system.

$$H_{eq} = \frac{\pi f M_{eq}}{S_b}$$

11. *Define synchronizing coefficient. For what value of synchronizing coefficient the system remains stable.*

The term $P_{max} \cos \delta_0$ is called synchronizing coefficient or stiffness at synchronous machine.

The system is stable if the synchronizing coefficient is positive. It is positive when $0 \le \delta_0 \le \pi / 2$.

12. *Given an expression for swing equation. Explaining each term alongwith their units.*

The swing equation is given below.

$$\frac{H}{\pi f} \frac{d^2\delta}{dt^2} = P_m - P_e$$

Since M in p.u. $= H/\pi f$, we can write,

$$M \frac{d^2\delta}{dt^2} = P_m - P_e$$

H = Inertia constant in MW-s / MVA.

f = Frequency in Hz.

M = Inertia constant in p.u.

P_m = Mechanical power input to the system (neglecting mechanical losses) in p.u.

P_e = Electrical power output of the system (neglecting electrical losses) in p.u.

13. *Define swing curve. What is the use of swing curve?*

The swing curve is the plot or graph between the power angle δ and time t.

It is usually plotted for a transient state to study the nature of variation in δ for a sudden large disturbance. From the nature of variations of δ the stability of a system for any disturbance can be determined.

14. *Give the simplified power angle equation and the expression for P_{max}.*

The simplified power angle equation for a generator feeding energy to infinite bus is given by,

$$P_e = P_{max} \sin \delta$$

where, $P_{max} = \frac{|E||V|}{X}$

$|E|$ = Magnitude of internal emf of generator.

$|V|$ = Magnitude of infinite bus voltage.

X = Transfer reactance between generator and infinte bus.

δ = Power angle or torque angle.

15. *Define power angle.*

The power angle (or torque angle) is defined as the angular displacement of the rotor from synchronously rotating reference frame.

16. *For the swing equation $M \frac{d^2\delta}{dt^2} = P_a$ what will be the value of P_a during steady state operation?*

During steady state operation, there is no acceleration or deceleration on the rotor. Therefore, $P_a = 0$. (Under this condition δ is a constant).

17. *Name the two ways by which transient stability study can be made in a system where one machine is swinging with respect to an infinite bus.*

For single machine - infinite bus system, the following two methods can be employed for transient stability studies.

i) Equal area criterion.

ii) Point - by-point method.

18. *Define critical clearing time and critical clearing angle and give equations for both.*

The critical clearing angle, δ_{cr} is the maximum allowable change in the power angle δ before clearing the fault, without loss of synchronism) The time corresponding to this angle is called critical clearing time, t_{cr}.

The critical clearing time, t_{cr} can be defined as the maximum time delay that can be allowed to clear a fault without loss of synchronism)

19. *List the methods of improving the transient stability limit of a power system.*

The following are the methods used to improve the transient stability of a system.

i) Increase of system voltage and use of AVR (Automatic Voltage Regulation)

ii) Use of high speed excitation systems.

iii) Reduction in system transfer reactance.

iv) Use of high speed reclosing breakers.

20. *A system has two stability limits, 100 MW and 150 MW. Which is the steady state stability limit and which is transient stability limit.*

The steady state stability limit = 150 MW

Transient stability limit = 100 MW.

21. *State equal area criterion.*

The equal area criterion for stability states that the system is stable if the area under $P_a - \delta$ curve reduces to zero at some value of δ.

This is possible only if the positive (accelerating) area under $P_a - \delta$ curve is equal to the negative (deceleration) area under $P_a - \delta$ curve for a finite change in δ. Hence this stability criterion is called equal area criterion.

22. *The sending end and the receiving end voltages of a transmission line at 150 MW load are equal at 132 kV. The per phase line impedance is (3+j4)Ω. Calculate the maximum steady state power that can be transmitted over the line.*

SOLUTION

Maximum steady state power, $P_{max} = \frac{|V_S||V_R|}{|Z|} = \frac{132 \times 132}{\sqrt{3^2+4^2}} = 3484.8\ MW$

23. *If the maximum power $P_{max} = 50MW$ find the electrical power output for a torque angle of 60°.*

SOLUTION

Electrical power output, $P_e = P_{max} \sin\delta = 50 \times \sin 60° = 43.3\ MW$

24. *A 50Hz, 4 pole turbogenerator of rating 20 MVA, 13.2 kV has an inertia constant of H = 9 kW - sec/kVA. Find the kinetic energy stored in the rotor at synchronous speed.*

SOLUTION

Kinetic energy, $E_{ke} = \frac{1}{2} M \omega_s$

where, $M = \frac{HS}{\pi f}$ and $\omega_s = 2\pi f$

$\therefore E_{ke} = \frac{1}{2}\frac{HS}{\pi f} 2\pi f = HS = 9 \times 20 = 180\ MJ$ (Mega Joules)

25. *Find the frequency of oscillation for a synchronizing coefficient of 0.6, inertia constant, H = 4 and system frequency of 50 Hz.*

SOLUTION

Frequency of oscillation $= \sqrt{\frac{C}{M}}$

where, C = Synchronizing coefficient ; M = Inertia constant in p.u.

Given that, $C = 0.6$

M in p.u. $= \frac{H}{\pi f} = \frac{4}{\pi \times 50} = 0.0255$ p.u.

$\therefore$ Frequency of oscillation $= \sqrt{\frac{0.6}{0.0255}} = 4.85$ rad / sec $= \frac{4.85}{2\pi} = 0.7719$ Hz

5.11 EXERCISES

I. Fill in the blanks

1. The angle between relative position of rotor axis and the stator magnetic field axis is .....
2. If transmitted power is increased beyond ....., then the synchronism will be lost.
3. The ratio of KE at rated speed to the rated apparent power of the machine is known as .....
4. The swing equation can be solved by ..... and ..... methods.
5. In a two-machine system, ... grounding at the sending end and ..... grounding at the receiving end is preferable.

Answers

1. Load angle or torque angle 2. Steady state stability limit 3. Inertia constant
4. Runge-Kutta, modified Euler's 5. Resistance, reactance

II. State whether the following statements are TRUE/FALSE

1. A system is said to be stable, if and only if $\frac{dP}{d\delta}$ is positive
2. If loading of machine is large, the torque angle will be less.
3. Equal area criterion method is not applicable to a multi-machine system directly.
4. If the actual clearing angle is greater than the critical value, the system is unstable, otherwise it is stable.
5. The series reactance of transmission lines is reduced in order to increase the transient stability limit of the system.
6. In point-by-point method, the computed swing curve approaches the true curve, only when the time interval is increased.
7. During 3-phase fault, the power transfer and stability of the system can be maintained by having resistance and reactance grounding.

Answers

1. True 2. False 3. True 4. True
5. True 6. False 7. False

III. Unsolved Problems

1. A 50 Hz four-pole turbo generator rated 300MVA, 13.2kV has an inertia constant of $H = 6.0$ kW-sec/kVA. Determine the kinetic energy stored in the rotor at synchronous speed. Determine the acceleration if the input less the rotational losses is 250000 HP and the electric power developed is 200000 kW. If the acceleration computed for the generator is constant for a period of 15 cycles, determine the change in torque angle in that period and the rpm at the end of 25 cycles. Assume that the generator is synchronized with a large system and has no accelerating torque before the 25 cycle period begins.

2. Determine the critical fault clearing angle for the network in fig EX.2 when a three-phase fault takes place at P and the breakers at A and B operate simultaneously. The generator is delivering 1 p.u. power before the fault takes place. Assuming the inertia constant $H = 4.0$, (a) determine the critical clearing time, and (b) plot the swing curves both under sustained fault and when the breakers have operated. Use point-by-point method taking time interval of 0.05 sec.



Fig EX.2.

3. Find the critical clearing angle for the system shown in fig EX.3 for a three-phase fault at the point F. The generator is delivering 1.0 p.u power under prefault conditions.



Fig E.X.3.

4. Calculate the steady state power limit consisting of a generator with equivalent reactance 0.8 p.u connected to an infinite bus through a series reactance of 1.0 p.u. The generator emf is held fixed at 1.2 p.u. and the voltage of the infinite bus is 1.0 p.u.

5. A 50 Hz 8 pole generator rated 500MVA, 11kV has an inertia constant of 7MJ/MVA. Find the moment of inertia W.

6. Find the rotor acceleration of a 50Hz, 2 pole turbogenerator rated 33kV having 600MJ of stored energy, if the mechanical input is suddenly raised to 120 MW for an electrical load of 70MW.

7. Calculate the total kinetic energy of rotation of two 50 Hz generating units operating in parallel within the same power plant.

   Gen 1 : 500 MVA , 20kV, H_1 = 9MJ/MVA

   Gen 2 : 1000 MVA , 22kV, H_2 = 3.27 MJ/MVA

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After interchanging column - 3 & column - 4 of Y_{bus} matrix of equ(1.16.2) we get,

$$Y_{bus} = \begin{bmatrix} -j5 & 0 & j1 & j2 \\ 0 & -j6 & j2 & 0 \\ j1 & j2 & -j9 & j5 \\ j2 & 0 & j5 & -j9 \end{bmatrix} \quad \text{.....(1.16.3)}$$

Now the last row and last column [i.e., 4th row and 4th column] of Y_{bus} matrix of equ(1.16.3) can be eliminated.

The elements of new bus admittance matrix after eliminating 4th row and 4th column is given by

$$Y_{jk,new} = Y_{jk} - \frac{Y_{jn}\, Y_{nk}}{Y_{nn}} \quad \text{where } n = 4 \;;\; j = 1,2,3 \text{ and } k = 1,2,3$$

$$Y_{11,new} = Y_{11} - \frac{Y_{14}\, Y_{41}}{Y_{44}} = -j5 - \frac{(j2)\,(j2)}{-j9} = -j4.556$$

$$Y_{12,new} = Y_{12} - \frac{Y_{14}\, Y_{42}}{Y_{44}} = 0 - \frac{j2 \times 0}{-j9} = 0$$

$$Y_{13,new} = Y_{13} - \frac{Y_{14}\, Y_{43}}{Y_{44}} = j1 - \frac{(j2)\,(j5)}{-j9} = j2.111$$

$$Y_{21,new} = Y_{12,new} = 0$$

$$Y_{22,new} = Y_{22} - \frac{Y_{24}\, Y_{42}}{Y_{44}} = -j6 - \frac{0 \times 0}{-j9} = -j6$$

$$Y_{23,new} = Y_{23} - \frac{Y_{24}\, Y_{43}}{Y_{44}} = j2 - \frac{0 \times j5}{-j9} = j2$$

$$Y_{31,new} = Y_{13,new} = j2.111$$

$$Y_{32,new} = Y_{23,new} = j2$$

$$Y_{33,new} = Y_{33} - \frac{Y_{34}\, Y_{43}}{Y_{44}} = -j9 - \frac{j5 \times j5}{-j9} = -j6.222$$

The reduced bus admittance matrix after eliminating bus-3 is given by,

$$Y_{bus} = \begin{bmatrix} -j4.556 & 0 & j2.111 \\ 0 & -j6 & j2 \\ j2.111 & j2 & -j6.222 \end{bmatrix}$$

1.6 BUS IMPEDANCE MATRIX

Consider the node basis matrix equation representing the power system,

$$Y_{bus}\, V = I \quad \text{.....(1.74)}$$

On premultiplying the equ(1.74) by Y_{bus}^{-1} we get,

$$V = Y_{bus}^{-1}\, I \quad \text{.....(1.75)}$$

Now the elements of Y_{bus}^{-1} will be impedances and so the matrix Y_{bus}^{-1} can be represented by an impedance matrix called bus impedance matrix, Z_{bus}.

$$Z_{bus} = Y_{bus}^{-1} \quad \text{.....(1.76)}$$

$$\text{and } V = Z_{bus}\, I \quad \text{.....(1.77)}$$

Since the bus admittance matrix is symmetrical, the bus impedance matrix is also symmetrical around the principal diagonal. In bus impedance matrix, the elements on the main diagonal are called driving point impedance of the buses or nodes and the off-diagonal elements are called the transfer impedances of the buses or nodes. The bus impedance matrix is very useful in fault analysis or calculations.

Note : *The equ(1.77) resembles a mesh basis matrix equation but it is not so, because the matrix V is a node voltage matrix and the matrix I is a current source matrix.*

The bus impedance matrix can be determined by two methods. In one method we can form the bus admittance matrix and then taking its inverse to get the bus impedance matrix. In another method the bus impedance matrix can be directly formed from the reactance diagram and this method requires the knowledge of the modifications of existing bus impedance matrix due to addition of new bus or addition of a new line (or impedance) between existing buses.

Modification of an existing bus impedance matrix

Let us denote the original Z_{bus} of a system with n-number of independent buses as Z_{orig}. When a branch of impedance Z_b is added to the system the Z_{orig} gets modified. The branch impedance Z_b can be added to the original system in the following four different ways.

Case 1 : *Adding a branch of impedance Z_b from a new bus-p to the reference bus.*

Case 2 : *Adding a branch of impedance Z_b from a new bus-p to an existing bus-q.*

Case 3 : *Adding a branch of impedance Z_b from an existing bus-q to the reference bus.*

Case 4 : *Adding a branch of impedance Z_b between two existing buses h and q.*

The modification of Z_{orig} for the above four cases have been presented here without proof.

Case 1 : *Adding Z_b from a new bus-p to the reference bus.*

Consider a n-bus system as shown in fig 1.9. Let us add a bus - p through an impedance Z_b to the reference bus. The addition of a bus will increase the order of the bus impedance matrix by one.

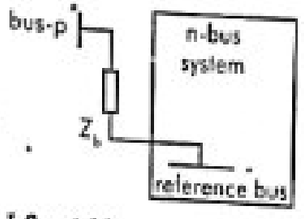


Fig 1.9 : Adding a new bus through an impedance to reference bus.

$$Z_{bus,\,new} = \begin{bmatrix} & & 0 \\ & Z_{orig} & 0 \\ & & \vdots \\ & & 0 \\ 0 & 0 \ldots\ldots 0 & Z_b \end{bmatrix} \qquad \ldots\ldots(1.78)$$

In this case the elements of $(n+1)^{th}$ column and row are all zeros except the diagonal. The diagonal element is the added branch impedance Z_b. The elements of original Z_{bus} matrix are not altered. The new bus impedance matrix will be as shown in equ(1.78).

Case 2 : *Adding Z_b from a new bus-p to existing bus-q.*

Consider a n-bus system as shown in fig 1.10. in which a new bus-p is added through an impedance Z_b to an existing bus-q. The addition of a bus will increase the order of the bus impedance matrix by one.

In this the elements of $(n+1)^{th}$ column are the elements of qth column and elements of $(n+1)^{th}$ row are the elements q^{th} row. The diagonal element is given by sum of Z_{qq} and Z_b. The elements of original Z_{bus} matrix are not altered. The new bus impedance matrix will be as shown in equ(1.79).

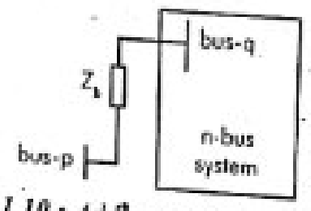


Fig 1.10 : Adding a new bus through an impedance to an existing bus.

$$Z_{bus,\,new} = \begin{bmatrix} & & Z_{1q} \\ & Z_{orig} & Z_{2q} \\ & & \vdots \\ & & Z_{nq} \\ Z_{q1} & Z_{q2} \ldots\ldots Z_{qn} & Z_{qq} + Z_b \end{bmatrix} \qquad \ldots\ldots(1.79)$$

Case 3 : *Adding Z_b from an existing bus-q to the reference bus*

Consider a n-bus system shown in fig 1.11 in which an impedance Z_b is added from an existing bus-q to the reference bus. Let us consider as if the impedance Z_b is connected from a new bus-p and existing bus-q. Now it will be an addition as that of case-2. The new bus impedance matrix order $(n+1)$ can be formed as that of case-2. Then we can short-circuit the bus-q to reference bus. This is equivalent to eliminating $(n+1)^{th}$ bus (i.e., bus-p in this case) and so the bus impedance matrix has to be modified by eliminating

$(n+1)^{th}$ row and $(n+1)^{th}$ column. The reduced bus impedance matrix can be formed by a procedure similar to that of bus elimination in bus admittance matrix, developed in section 1.5. This reduced bus impedance matrix is the actual new bus impedance matrix. Every element of actual new bus impedance matrix can be determined using the equation (1.80).

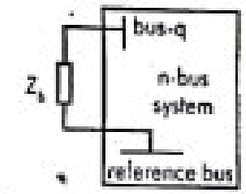


Fig 1.11 : Adding an impedance between existing bus and reference.

$$Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}\, Z_{(n+1)k}}{Z_{(n+1)(n+1)}} \qquad \ldots\ldots(1.80)$$

Note : 1. $Z_{jk,act}$ is the impedance corresponding to row-j and column-k of actual new bus impedance matrix.

2. Z_{jk}, $Z_{(n+1)k}$, $Z_{j(n+1)}$, $Z_{(n+1)(n+1)}$ are impedances of new bus impedance matrix of order (n+1).

3. Since bus impedance matrix is symmetrical

$$Z_{jk,\,act} = Z_{kj,act}$$

Case 4 : *Adding Z_b between two existing buses h and q.*

Consider a n-bus system shown in fig 1.12, in which an impedance Z_b is added between two existing buses h and q.

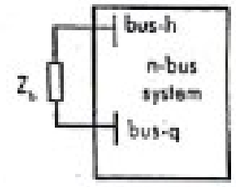


Fig 1.12 : Adding an impedance between bus-h and bus-q.

In this case the bus impedance matrix is formed as shown in equ(1.81):-Here the elements of $(n+1)^{th}$ column is the difference between the elements of column-h and column-q. The elements of $(n+1)^{th}$ row is the difference between the elements of row-h and row-q. The diagonal element is given by equ(1.82).

$$Z_{bus,\,new} = \begin{bmatrix} & & Z_{1h} - Z_{1q} \\ & Z_{orig} & Z_{2h} - Z_{2q} \\ & & \vdots \\ & & Z_{nh} - Z_{nq} \\ Z_{h1} - Z_{q1} & Z_{h2} - Z_{q2} \ldots\ldots & Z_{hn} - Z_{qn} \quad Z_{(n+1)(n+1)} \end{bmatrix} \qquad \ldots\ldots(1.81)$$

$$Z_{(n+1)(n+1)} = Z_b + Z_{hh} + Z_{qq} - 2Z_{hq} \qquad \ldots\ldots(1.82)$$

Since the modification does not involve addition of new bus, the order of new bus impedance matrix has to be reduced to nxn by eliminating the $(n+1)^{th}$ column and $(n+1)^{th}$ row.

This reduced bus impedance matrix is the actual new bus impedance matrix. Every element of this actual new bus impedance matrix can be determined using equ(1.80) which is also given below for reference.

$$Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}\, Z_{(n+1)k}}{Z_{(n+1)(n+1)}}$$

Direct determination of a bus impedance matrix

The bus impedance matrix can be directly obtained from impedance or reactance diagram instead of forming Y_{bus} and then inverting it. In direct determination of Z_{bus} first consider an impedance Z_a connected from bus-1 to reference bus. Now the bus impedance matrix will have a single element as shown below.



$$Z_{bus} = [Z_a]$$

Then add each element of impedance or reactance diagram one by one and modify the Z_{bus} in each step. Each modification of Z_{bus} involve any one of the four cases discussed above.

EXAMPLE 1.17

Determine Z_{bus} for system whose reactance diagram is shown in fig 1.17.1. where the impedance is given in p.u. Preserve all the three nodes.

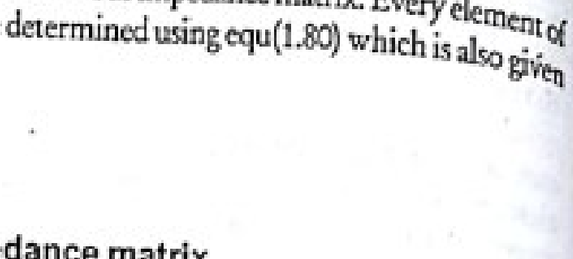


Fig 1.17.1.

SOLUTION

Step 1 : Consider the branch with impedance j1.2 p.u. connected between bus-1 and reference as shown in fig 1.17.2. The system shown in fig 1.17.2. has a single bus and so the order of bus impedance matrix is one, as shown below.

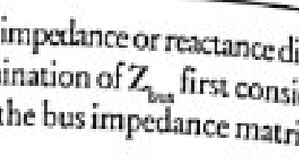


Fig 1.17.2

$$Z_{bus} = [j1.2]$$

Step 2 : Connect bus-2 to bus-1 through an impedance j0.2 as shown in fig 1.17.3. This is case-2 modification and so the order of bus impedance matrix increases by one. In the new bus impedance matrix, the elements of 1st column are copied as elements of 2nd column and the elements of 1st row are copied as elements of 2nd row. The diagonal element is given by $Z_{11} + Z_b$ where $Z_b = j0.2$.

$$\therefore Z_{bus} = \begin{bmatrix} j1.2 & j1.2 \\ j1.2 & j1.2 + j0.2 \end{bmatrix} = \begin{bmatrix} j1.2 & j1.2 \\ j1.2 & j1.4 \end{bmatrix}$$

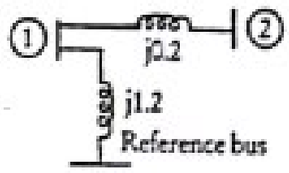


Fig 1.17.3

Step 3 : Connect the bus-3 to bus-2 through an impedance j0.15 as shown in fig 1.17.4. This is case-2 modification and so the order of the bus impedance matrix increases by one. In the new bus impedance matrix the elements of 2nd column are copied as elements of 3rd column and the elements of 2nd row are copied as elements of 3rd row. The diagonal element is given by $Z_{22} + Z_b$ where $Z_b = j0.15$.

$$\therefore Z_{bus} = \begin{bmatrix} j1.2 & j1.2 & j1.2 \\ j1.2 & j1.4 & j1.4 \\ j1.2 & j1.4 & j1.4 + j0.15 \end{bmatrix}$$

$$= \begin{bmatrix} j1.2 & j1.2 & j1.2 \\ j1.2 & j1.4 & j1.4 \\ j1.2 & j1.4 & j1.55 \end{bmatrix}$$

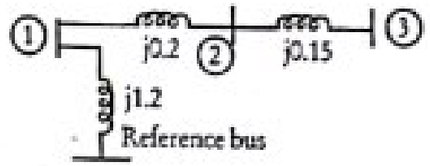


Fig 1.17.4

Step 4 : Connect the impedance j1.5 from bus-3 to reference bus as shown in fig 1.17.5. This is case-3 modification. In case-3 modification the new bus impedance matrix is formed as that of case-2 and then the last row and column are eliminated by node elimination techniques.

In new bus impedance matrix the elements of 3rd column are copied as elements of 4th column and the elements of 3rd row are copied as elements of 4th row. The diagonal element is given by $Z_{33} + Z_b$ where $Z_b = j1.5$.

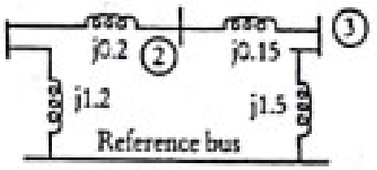


Fig 1.17.5

$$\therefore Z_{bus} = \begin{bmatrix} j1.2 & j1.2 & j1.2 & j1.2 \\ j1.2 & j1.4 & j1.4 & j1.4 \\ j1.2 & j1.4 & j1.55 & j1.55 \\ j1.2 & j1.4 & j1.55 & j1.55 + j1.5 \end{bmatrix} = \begin{bmatrix} j1.2 & j1.2 & j1.2 & j1.2 \\ j1.2 & j1.4 & j1.4 & j.14 \\ j1.2 & j1.4 & j1.55 & j1.55 \\ j1.2 & j1.4 & j1.55 & j3.05 \end{bmatrix}$$

The actual new bus impedance matrix is obtained by eliminating the 4th row and 4th column. The element Z_{jk} of the actual new bus impedance matrix is given by,

$$\therefore Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}\, Z_{(n+1)k}}{Z_{(n+1)(n+1)}} \qquad \text{where } n = 3 \; ; j = 1, 2, 3 \text{ and } k = 1, 2, 3$$

$$Z_{11,act} = Z_{11} - \frac{Z_{14}\, Z_{41}}{Z_{44}} = j1.2 - \frac{j1.2 \times j1.2}{j3.05} = j0.728$$

$$Z_{12,act} = Z_{12} - \frac{Z_{14}Z_{42}}{Z_{44}} = j1.2 - \frac{j1.2 \times j1.4}{j3.05} = j0.649$$

$$Z_{13,act} = Z_{13} - \frac{Z_{14}Z_{43}}{Z_{44}} = j1.2 - \frac{j1.2 \times j1.55}{j3.05} = j0.590$$

$$Z_{21,act} = Z_{12,act} = j0.649$$

$$Z_{22,act} = Z_{22} - \frac{Z_{24}Z_{42}}{Z_{44}} = j1.4 - \frac{j1.4 \times j1.4}{j3.05} = j0.757$$

$$Z_{23,act} = Z_{23} - \frac{Z_{24}Z_{43}}{Z_{44}} = j1.4 - \frac{j1.4 \times j1.55}{j3.05} = j0.689$$

$$Z_{31,act} = Z_{13,act} = j0.590$$

$$Z_{32,act} = Z_{23,act} = j0.689$$

$$Z_{33,act} = Z_{33} - \frac{Z_{34}Z_{43}}{Z_{44}} = j1.55 - \frac{j1.55 \times j1.55}{j3.05} = j0.762$$

$$\therefore Z_{bus} = \begin{bmatrix} j0.728 & j0.649 & j0.590 \\ j0.649 & j0.757 & j0.689 \\ j0.590 & j0.689 & j0.762 \end{bmatrix}$$

Step 5 : Connect the impedance j0.3 between bus-1 and bus-3 as shown in fig 1.17.6. This is case-4 modification.

In new bus impedance matrix, the elements of 4th column are obtained by substracting the elements of 3rd column from 1st column and the elements of 4th row are obtained by substracting the elements of 3rd row from 1st row. The diagonal element Z_{44} is given by the following equation.

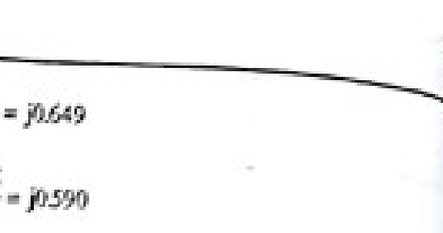


Fig 1.17.6

$$Z_{44} = Z_b + Z_{11} + Z_{33} - 2Z_{13}$$

where $Z_b = j0.3$

$$\therefore Z_{44} = j0.3 + j0.728 + j0.762 - 2(j0.59) = j0.61$$

$$Z_{bus} = \begin{bmatrix} j0.728 & j0.649 & j0.590 & j0.728 - j0.59 \\ j0.649 & j0.757 & j0.689 & j0.649 - j0.689 \\ j0.59 & j0.689 & j0.762 & j0.59 - j0.762 \\ j0.728 - j0.59 & j0.649 - j0.689 & j0.59 - j0.762 & j0.61 \end{bmatrix}$$

$$Z_{bus} = \begin{bmatrix} j0.728 & j0.649 & j0.590 & j0.138 \\ j0.649 & j0.757 & j0.689 & -0.04 \\ j0.59 & j0.689 & j0.762 & -j0.172 \\ j0.138 & -0.04 & -j0.172 & j0.61 \end{bmatrix}$$

Since this modification does not add a new bus, the 4th row and column has to be eliminated using node elimination technique, to determine the actual new bus impedance matrix. The element Z_{jk} of actual new bus impedance matrix is given by,

$$\therefore Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}Z_{(n+1)k}}{Z_{(n+1)(n+1)}} \quad \text{where } n = 3 \; ; j = 1,2,3 \text{ and } k = 1,2,3$$

$$Z_{11,act} = Z_{11} - \frac{Z_{14}Z_{41}}{Z_{44}} = j0.728 - \frac{j0.138 \times j0.138}{j0.61} = j0.697$$

$$Z_{12,act} = Z_{12} - \frac{Z_{14}Z_{42}}{Z_{44}} = j0.649 - \frac{j0.138 \times (-j0.04)}{j0.61} = j0.658$$

$$Z_{13,act} = Z_{13} - \frac{Z_{14}Z_{43}}{Z_{44}} = j0.59 - \frac{j0.138 \times (-j0.172)}{j0.61} = j0.629$$

$$Z_{21,act} = Z_{12,act} = j0.658$$

$$Z_{22,act} = Z_{22} - \frac{Z_{24}Z_{42}}{Z_{44}} = j0.757 - \frac{(-j0.04)(-j0.04)}{j0.61} = j0.754$$

$$Z_{23,act} = Z_{23} - \frac{Z_{24}Z_{43}}{Z_{44}} = j0.689 - \frac{(-j0.04)(-j0.172)}{j0.61} = j0.678$$

$$Z_{31,act} = Z_{13,act} = j0.629$$

$$Z_{32,act} = Z_{23,act} = j0.678$$

$$Z_{33,act} = Z_{33} - \frac{Z_{34}Z_{43}}{Z_{44}} = j0.762 - \frac{(-j0.172)(-j0.172)}{j0.61} = j0.714$$

$$\therefore Z_{bus} = \begin{bmatrix} j0.697 & j0.658 & j0.629 \\ j0.658 & j0.754 & j0.678 \\ j0.629 & j0.678 & j0.714 \end{bmatrix}$$

EXAMPLE 1.18

Find the bus impedance matrix for the system whose reactance diagram is shown in fig 1.18.1. All the impedances are in p.u.

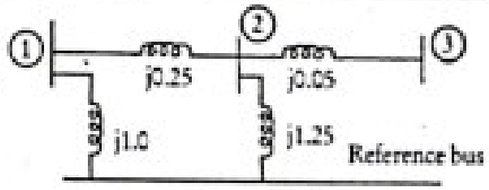


Fig 1.18.1.

SOLUTION

Step 1 : Consider the branch with impedance j1 p.u. connected between bus-1 and reference as shown in fig 1.18.2. The system shown in fig 1.18.2. has a single bus and so the

order of bus impedance matrix is one, as shown below.

$$Z_{bus} = [j1.0]$$

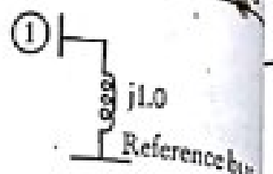


Fig 1.18.2

Step 2 : Connect bus-2 to bus-1 through an impedance j0.25 as shown in fig 1.18.3. This is case-2 modification and so the order of bus impedance matrix increases by one.

In the new bus impedance matrix, the elements of 1st column are copied as elements of 2nd column and the elements of 1st row are copied as elements of 2nd row. The diagonal element is given by $Z_{11} + Z_b$ where $Z_b = j0.25$.

$$\therefore Z_{bus} = \begin{bmatrix} j1.0 & j1.0 \\ j1.0 & j1.0 + j0.25 \end{bmatrix} = \begin{bmatrix} j1.0 & j1.0 \\ j1.0 & j1.25 \end{bmatrix}$$

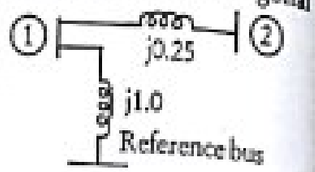


Fig 1.18.3

Step 3 : Connect the impedance j1.25 from bus-2 to reference bus as shown in fig 1.18.4. This is case-3 modification. In case-3 modification the new bus impedance matrix is formed as that of case-2 and then the last row and column are eliminated by node elimination technique.

In the new bus impedance matrix the elements of 2nd column are copied as elements of 3rd column and the elements of 2nd row are copied as elements of 3rd row. The diagonal element is given by $Z_{22} + Z_b$ where $Z_b = j1.25$.

$$\therefore Z_{bus} = \begin{bmatrix} j1.0 & j1.0 & j1.0 \\ j1.0 & j1.25 & j1.25 \\ j1.0 & j1.25 & j1.25 + j1.25 \end{bmatrix} = \begin{bmatrix} j1.0 & j1.0 & j1.0 \\ j1.0 & j1.25 & j1.25 \\ j1.0 & j1.25 & j2.5 \end{bmatrix}$$

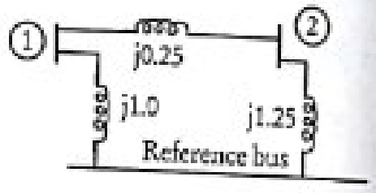


Fig 1.18.4

The actual new bus impedance matrix is obtained by eliminating the 3rd row and 3rd column. The element Z_{jk} of the actual new bus impedance matrix is given by,

$$\therefore Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}\, Z_{(n+1)k}}{Z_{(n+1)(n+1)}} \qquad \text{where } n = 2 \;;\; j = 1, 2, \text{ and } k = 1, 2,$$

$$Z_{11,act} = Z_{11} - \frac{Z_{13} Z_{31}}{Z_{33}} = j1.0 - \frac{j1.0 \times j1.0}{j2.5} = j0.6$$

$$Z_{12,act} = Z_{12} - \frac{Z_{13} Z_{32}}{Z_{33}} = j1.0 - \frac{j1.0 \times j1.25}{j2.5} = j0.5$$

$$Z_{21,act} = Z_{12,act} = j0.5$$

$$Z_{22,act} = Z_{22} - \frac{Z_{23} Z_{32}}{Z_{33}} = j1.25 - \frac{j1.25 \times j1.25}{j2.5} = j0.625$$

$$\therefore Z_{bus} = \begin{bmatrix} j0.6 & j0.5 \\ j0.5 & j0.625 \end{bmatrix}$$

Step 4 : Connect the bus-3 to bus-2 through an impedance j0.05 as shown in fig 1.18.5. This is case-2 modification and so the order of the bus impedance matrix increases by one.

In the new bus impedance matrix, the elements of 2nd column are copied as elements of 3rd column and the elements of 2nd row are copied as elements of 3rd row. The diagonal element is given by $Z_{22} + Z_b$ where $Z_b = j0.5$.

$$Z_{bus} = \begin{bmatrix} j0.6 & j0.5 & j0.5 \\ j0.5 & j0.625 & j0.625 \\ j0.5 & j0.625 & j0.625 + j0.05 \end{bmatrix}$$

$$\therefore Z_{bus} = \begin{bmatrix} j0.6 & j0.5 & j0.5 \\ j0.5 & j0.625 & j0.625 \\ j0.5 & j0.625 & j0.675 \end{bmatrix}$$

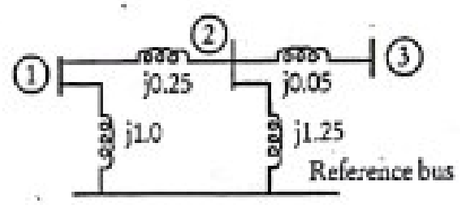


Fig 1.18.5

1.7. SYMMETRICAL COMPONENTS

The analysis of unsymmetrical polyphase network by the method of symmetrical components was introduced by *Dr. C. Fortesque*. He proved that an unbalanced system of *n* related vectors can be resolved into *n* system of balanced vectors called symmetrical components of original vectors. The *n* vectors of each set of components are equal in length and the phase angles between adjacent vectors of the set are equal.

In a three phase system, the three unbalanced vectors [either V_a, V_b & V_c or I_a, I_b & I_c] can be resolved into three balanced system of vectors. The vectors of the balanced system are called symmetrical components of the original system. The symmetrical components of three phase system are,

1. Positive sequence components
2. Negative sequence components
3. Zero sequence components.

The positive sequence components consists of three vectors equal in magnitude, displaced form each other by 120° in phase, and having the same phase sequence as the original vectors